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STUDY OF STAR UML

AIM:

To study the basics of STAR UML

INTRODUCTION:

Star UML is an open-source software modeling tool that supports the UML (Unified Modeling Language) framework for system and software modeling. It is based on UML version 1.4, provides eleven different types of diagrams, and accepts UML 2.0 notation, It actively supports the MDA (Model Driven Architecture) approach by supporting the UML profile concept and allowing to generate code for multiple languages.

SYSTEM REQUIREMENTS:

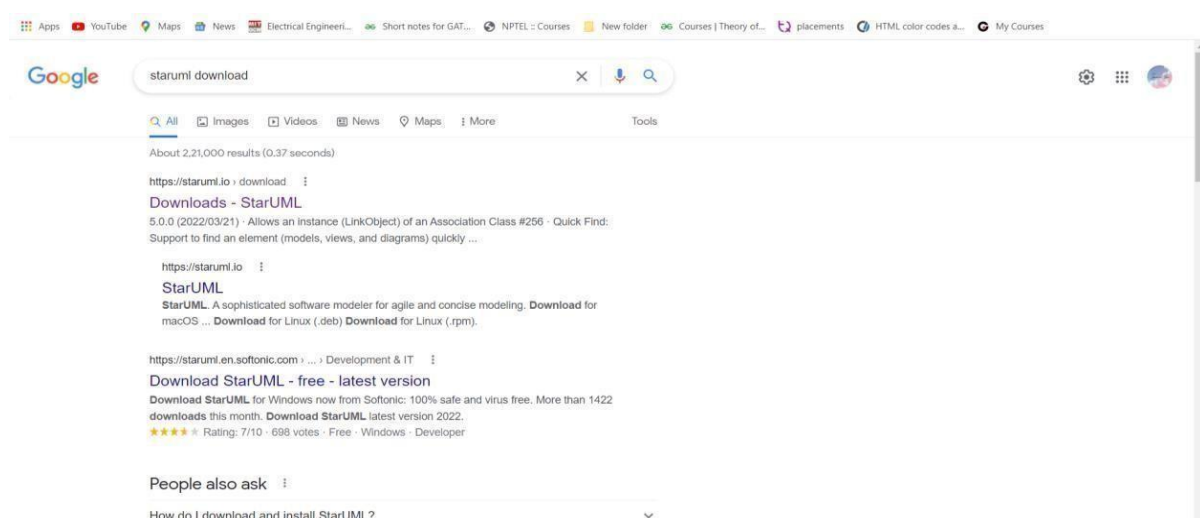
Windows 2000, Windows XP, or higher; Microsoft Internet Explorer 5.0 or higher; 128 MB RAM(256MB recommended); 110MB hard disc space (150MB space recommended)

INSTALLATION

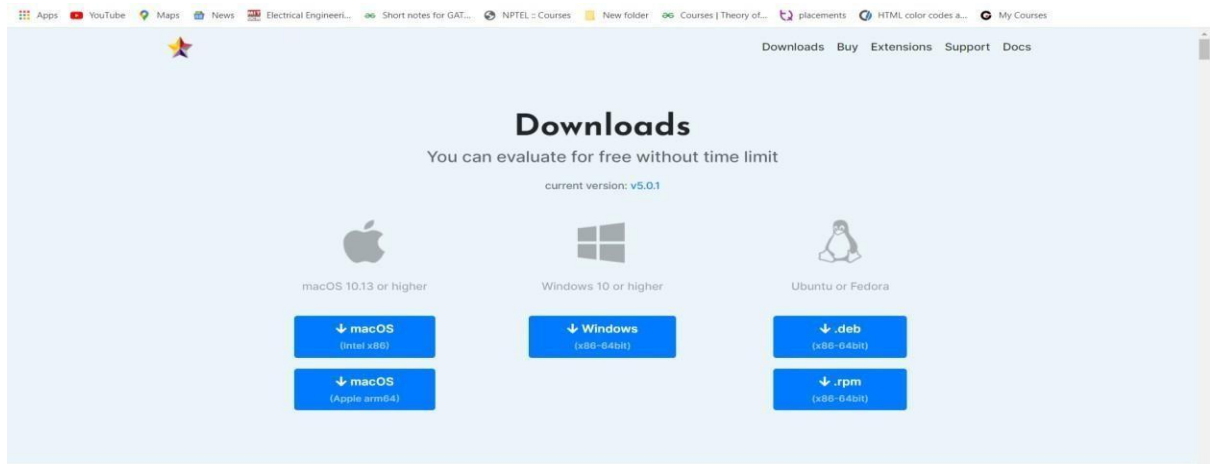
The installer follows the classic Windows install procedure without issues.

STEPS FOR DOWNLOADING STAR UML

STEP 1: Search Star UML in google



STEP 2 : Download for Mac OS/Windows/.deb, .rpm



STEP 3 : Install it



DOCUMENTATION

The same help that could be browsed on the StarUML website is available with the tool on your desktop. Documentation describes the concepts of the tool but on high-level vision. More detailed documentation is available for the diagramming functions. Sample projects are provided with the tool and one of them contains the model of the tool itself, showing that the developers were able to eat their dog food. Besides English, documentation exists in Korean, Japanese, and Russian.

CONFIGURATION

Some general and diagram configuration options are available from the Tools/Options menu. You will find in this window also the configuration switches for the code generation. The interface is also very configurable as you can select what part of the tool you would like to view or not.

FEATURES

When you start a new project, StarUML proposes which approach you want to use: 4+1 (Kruchten), Rational, UML, components (from Cheesman and Daniels book), default or empty. Depending on the approach, profiles and/or frameworks may be included and loaded. If you don't follow a specific approach, the "empty" choice could be used. Although a project can be managed as one file, it may be convenient to divide it into many units and manage them separately if many developers are working on it together.

StarUML makes a clear conceptual distinction between models, views, and diagrams. A Model is an element that contains information for a software model. A View is a visual expression of the information contained in a model, and a Diagram is a collection of view elements that represent the user's specific design thoughts.

StarUML is built as a modular and open tool. It provides frameworks for extending the functionality of the tool. It is designed to allow access to all functions of the model/meta-model and tool through COM Automation, and it provides an extension of menu and option items. Also, users can create their own approaches and frameworks according to their methodologies. The tool can also be integrated with any external tools.

StarUML supports the following Diagram types

- Use Case Diagram
- Class Diagram
- Sequence Diagram Collaboration Diagram
- State chart Diagram
- Activity Diagram
- Component Diagram
- Deployment Diagram
- Composite Structure Diagram

RESULT:

Thus the basics of star UML have been studied successfully.

ONLINE RAILWAY RESERVATION SYSTEM

AIM:

To develop the online railway reservation system using star uml.

PROBLEM ANALYSIS AND PROJECT PLANNING

INTRODUCTION

This document deals with online ticket reservations for the railway. This document is designed in such a way that the reader understands it. The use case description and other documents are described in such a way that the system reaches the people easily.

OBJECTIVES

The purpose of the document is to know about the availability of seats, railway, etc. According to the requirements, the passenger reserves his/her tickets.

SCOPE

This document for online ticket reservations for airlines makes the work easy for the passenger to book tickets.

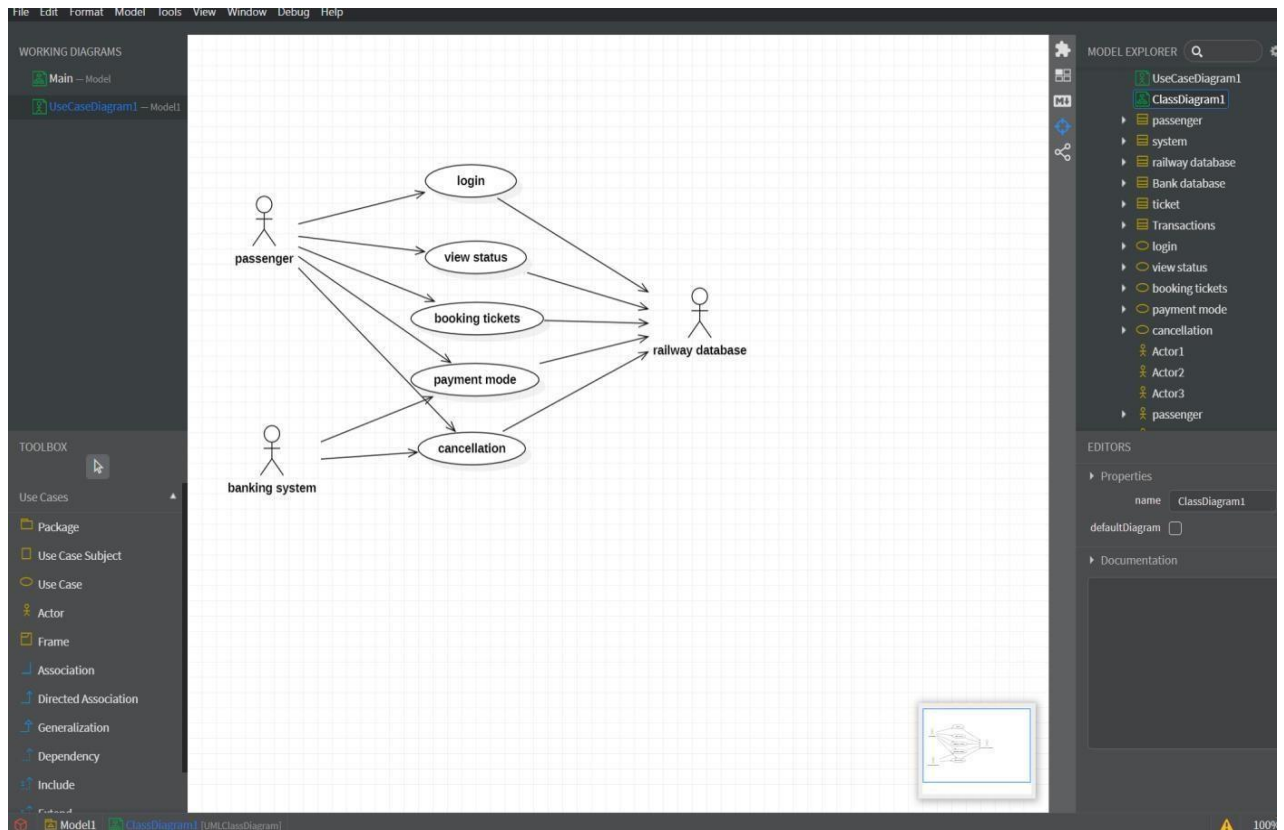
PROBLEM STATEMENT

Computers are an integral part of our day-to-day life. It makes the work easy and faster. Every job is computerized now. So is the ticket reservation, we can book our tickets online. During the reservation of tickets, the passenger has to select the origin and date of the journey. passport number, etc.,

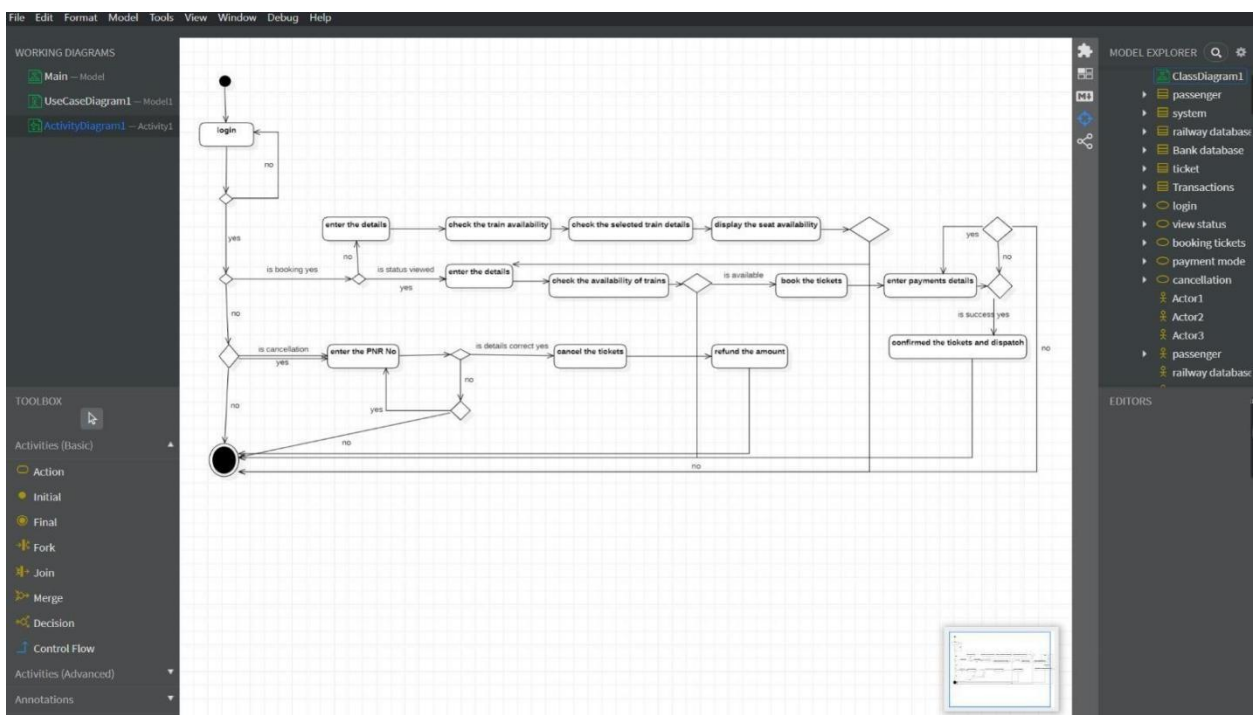
The reservation counter keeps track of the passenger's information. The system will have all the details about the trains and the facilities provided by them. There are various trains provided according to the convenience of the passenger. A database is maintained by the database administrator.

STAR UML DIAGRAMS :

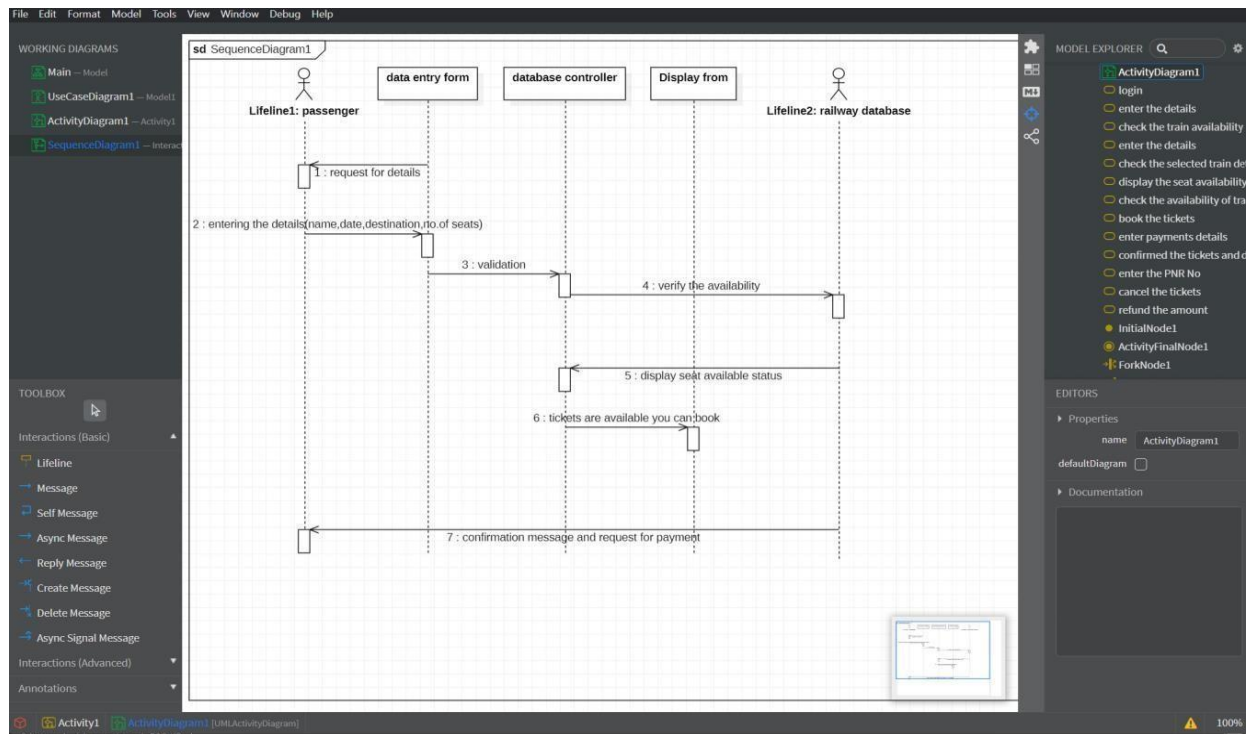
USE CASE DIAGARM



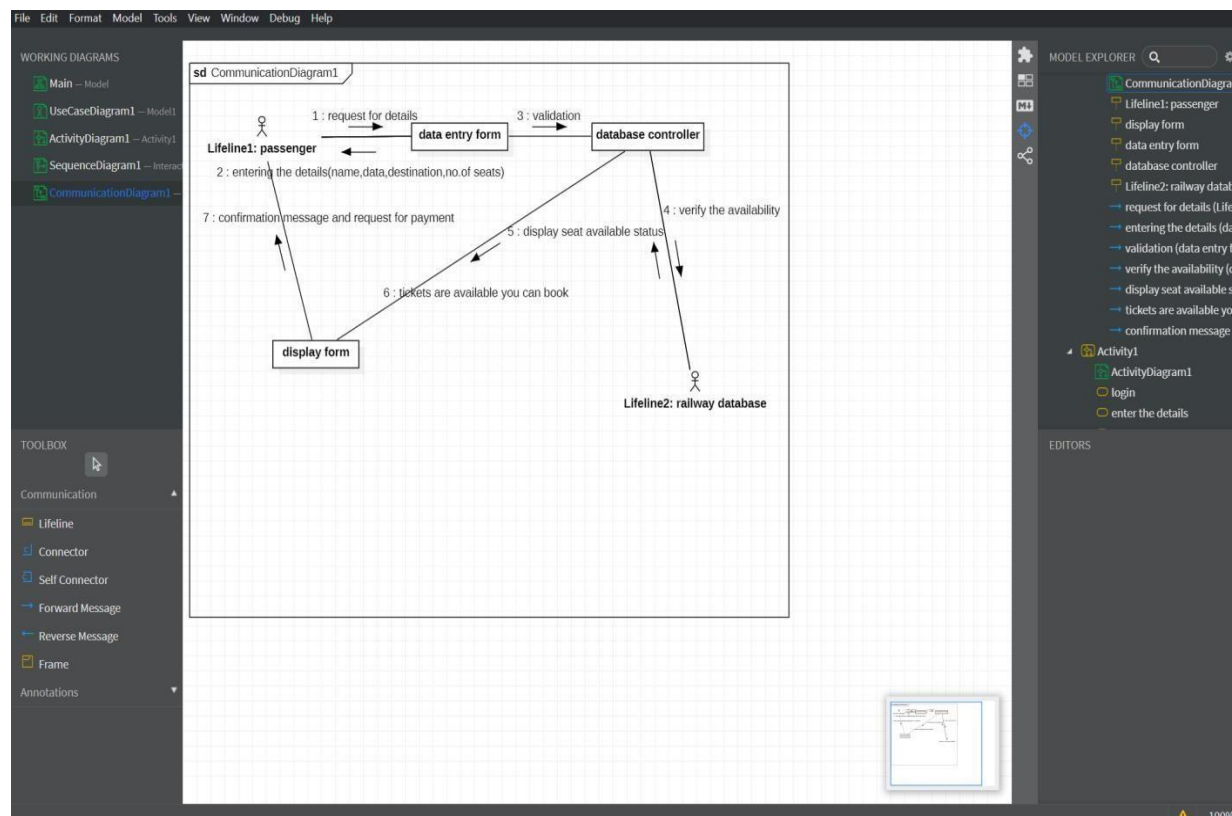
ACTIVITY DIAGRAM



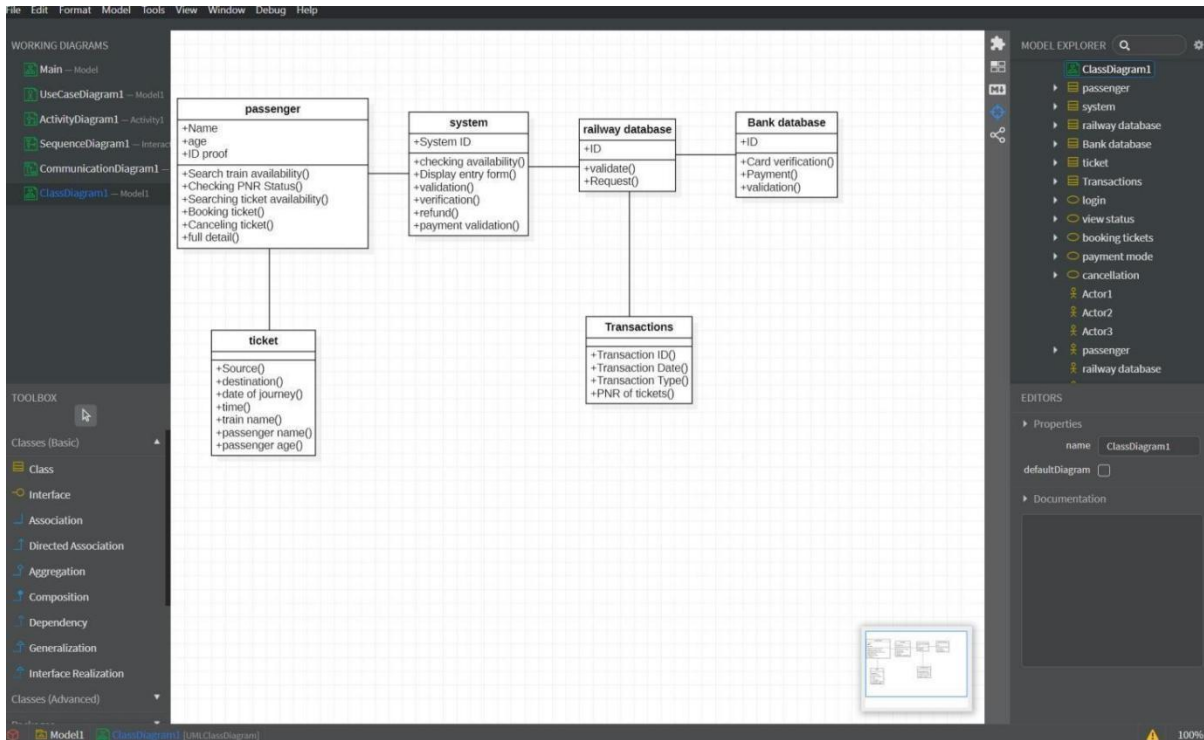
SEQUENCE DIAGRAM



COMMUNICATION DIAGRAM



CLASS DIAGRAM



STEPS TO GENERATE CODE IN STAR UML

- 1) Draw the class diagram and then save it.
- 2) Open the saved file with star uml.
- 3) Then go to tools at right corner.
- 4) Select the code language which we can prefer.
- 5) Then we have the code generator in tools so give access to it.
- 6) Then go to tools and select the language then it will take us to another window.
- 7) There select the class attribute to generate the code.
- 8) Then it will ask for the directory to store the code.
- 9) Select the directory and click on generate code.
- 10) Then code generated successfully.

OUTPUT:

> yoga oose > yoga oose exp2 > Model1

Name	Date modified	Type	Size
Bank database.java	11-03-2023 10:06 AM	JAVA File	1 KB
passenger.java	11-03-2023 10:06 AM	JAVA File	1 KB
railway database.java	11-03-2023 10:06 AM	JAVA File	1 KB
system.java	11-03-2023 10:06 AM	JAVA File	1 KB
ticket.java	11-03-2023 10:06 AM	JAVA File	1 KB
Transactions.java	11-03-2023 10:06 AM	JAVA File	1 KB

BANK DATA BASE

```
> yoga oose > yoga oose exp2 > Model1

Bank database.java

import java.util.*;

/**
 *
 */
public class Bank database {

    /**
     * Default constructor
     */
    public Bank database() {
    }
}
```

PASSENGER

```
> yoga oose > yoga oose exp2 > Model1

passenger.java

import java.util.*;

/**
 *
 */
public class passenger {

    /**
     * Default constructor
     */
    public passenger() {
    }
}
```

RAILWAY DATABASE



```
> yoga oose > yoga oose exp2 > Model1

railway database.java

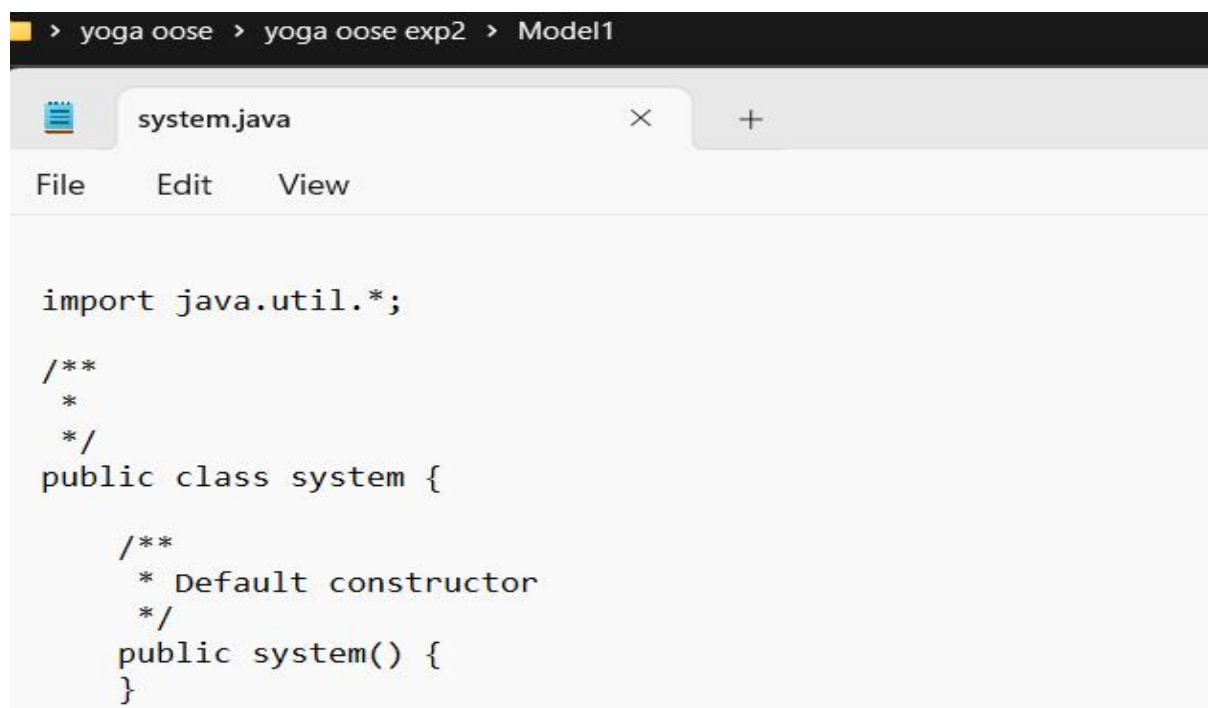
File Edit View

import java.util.*;

/**
 *
 */
public class railway database {

    /**
     * Default constructor
     */
    public railway database() {
    }
```

SYSTEM



```
> yoga oose > yoga oose exp2 > Model1

system.java

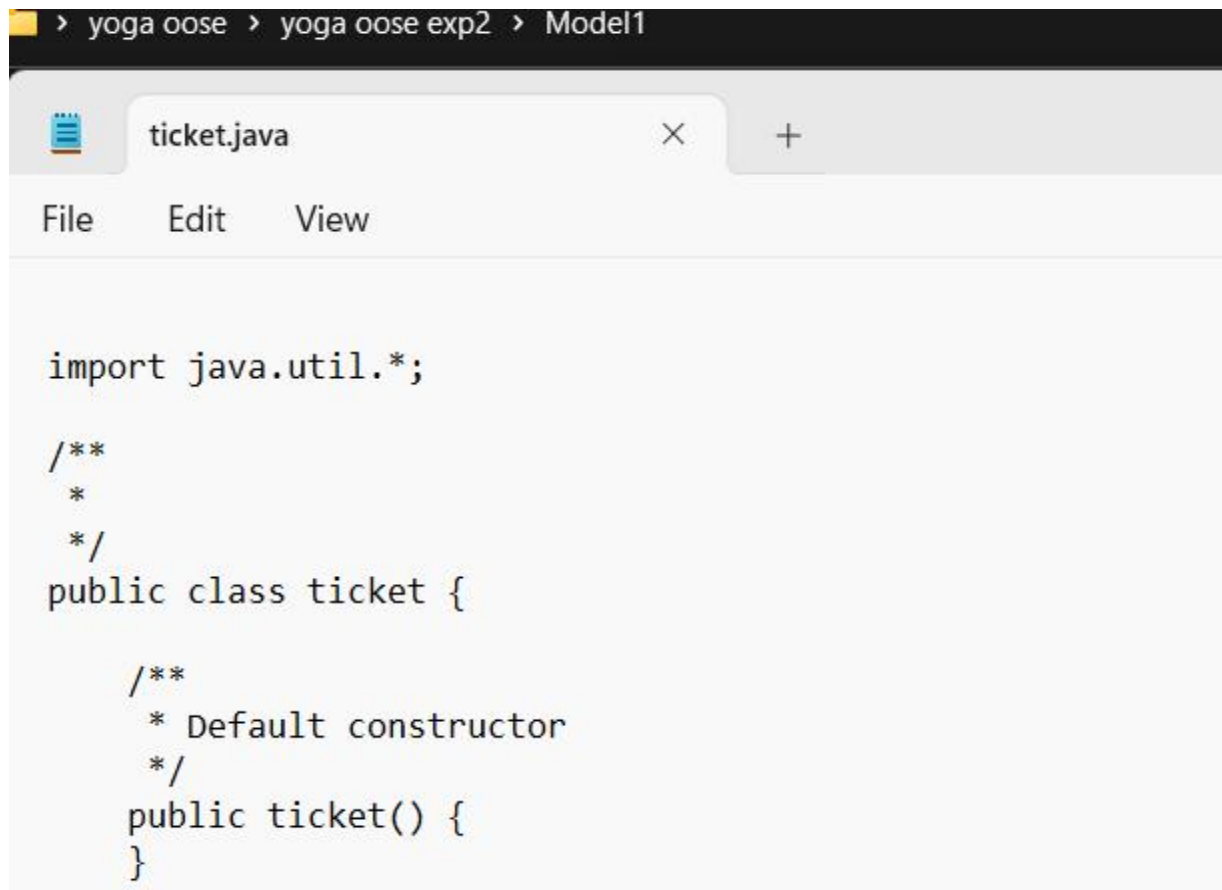
File Edit View

import java.util.*;

/**
 *
 */
public class system {

    /**
     * Default constructor
     */
    public system() {
    }
```

TICKET



The screenshot shows an IDE window with the file path > yoga oose > yoga oose exp2 > Model1. The active file is ticket.java. The code in the editor is as follows:

```
import java.util.*;

/**
 *
 */
public class ticket {

    /**
     * Default constructor
     */
    public ticket() {
    }
```

TRANSACTIONS



The screenshot shows an IDE window with the file path > yoga oose > yoga oose exp2 > Model1. The active file is Transactions.java. The code in the editor is as follows:

```
import java.util.*;

/**
 *
 */
public class Transactions {

    /**
     * Default constructor
     */
    public Transactions() {
    }
```

Result:

Thus, the UML diagrams for the online railway reservation system in StarUML have been designed and a java code template was generated successfully.

STUDENT MARK ANALYSIS SYSTEM

AIM:

To develop the Student Mark Analysis System using StarUML.

PROBLEM ANALYSIS AND PROJECT PLANNING

INTRODUCTION

Student mark analyzing system has been designed to carry out the mark analysis process in an educational institution. The results of respective departments can be efficiently computed without much manual involvement.

OBJECTIVES

The purpose of this project is to define the requirements of the mark analysis system. This system reduces manual work to a great extent. The mark analysis is carried out by the system in an efficient manner.

SCOPE

This system is very essential for every educational institution as it reduces manpower. This system can be used by all kinds of educational institutions to evaluate and analyze the marks and generate reports of specified criteria.

PROBLEM STATEMENT

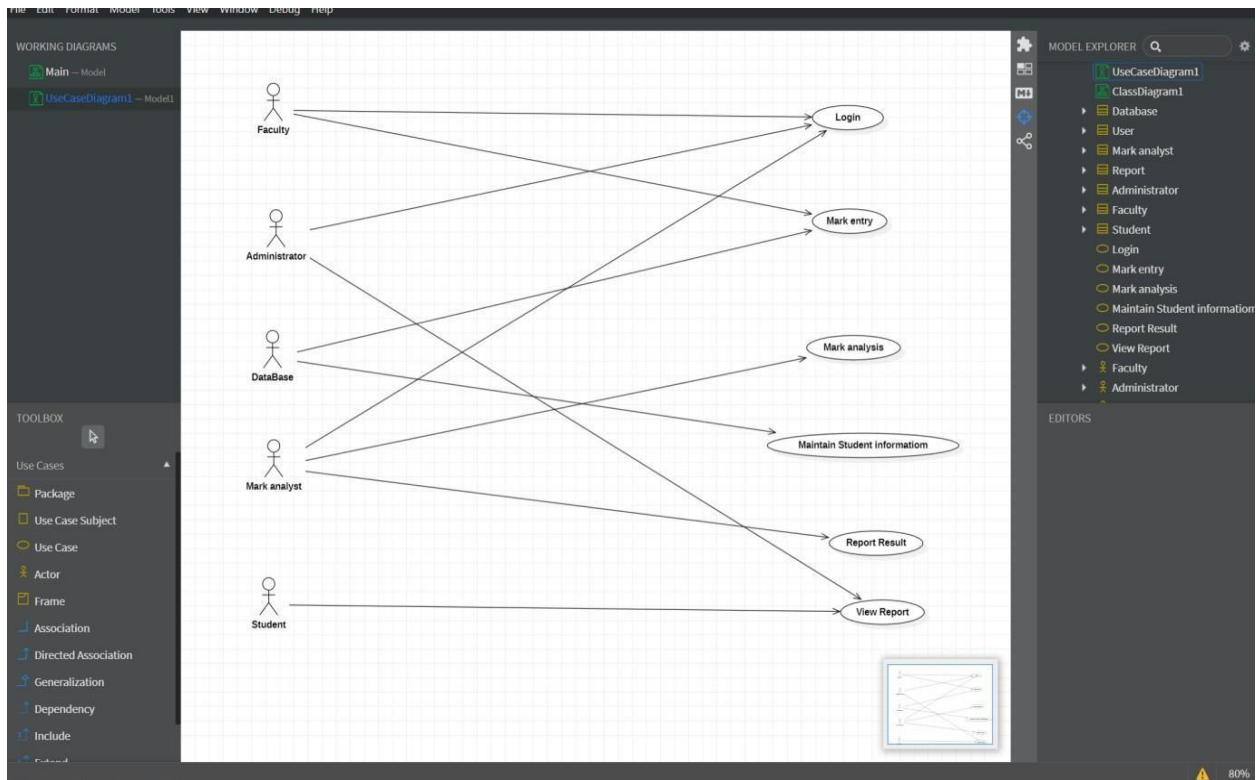
For analyzing the marks obtained by students in an educational institution. We are tasked to build up a student mark analyzing system. This is done to replace the manual entering and processing of marks which are error-prone and tedious. This system also maintains information about students. The system will have a Windows-based desktop interface to allow the faculty to enter marks obtained by the students, update them and generate various reports.

For security reasons, the administrator and faculty only can update the marks and other information. First, the user needs to login into the system for accessing it. The system will retain information on all the students and the institution.

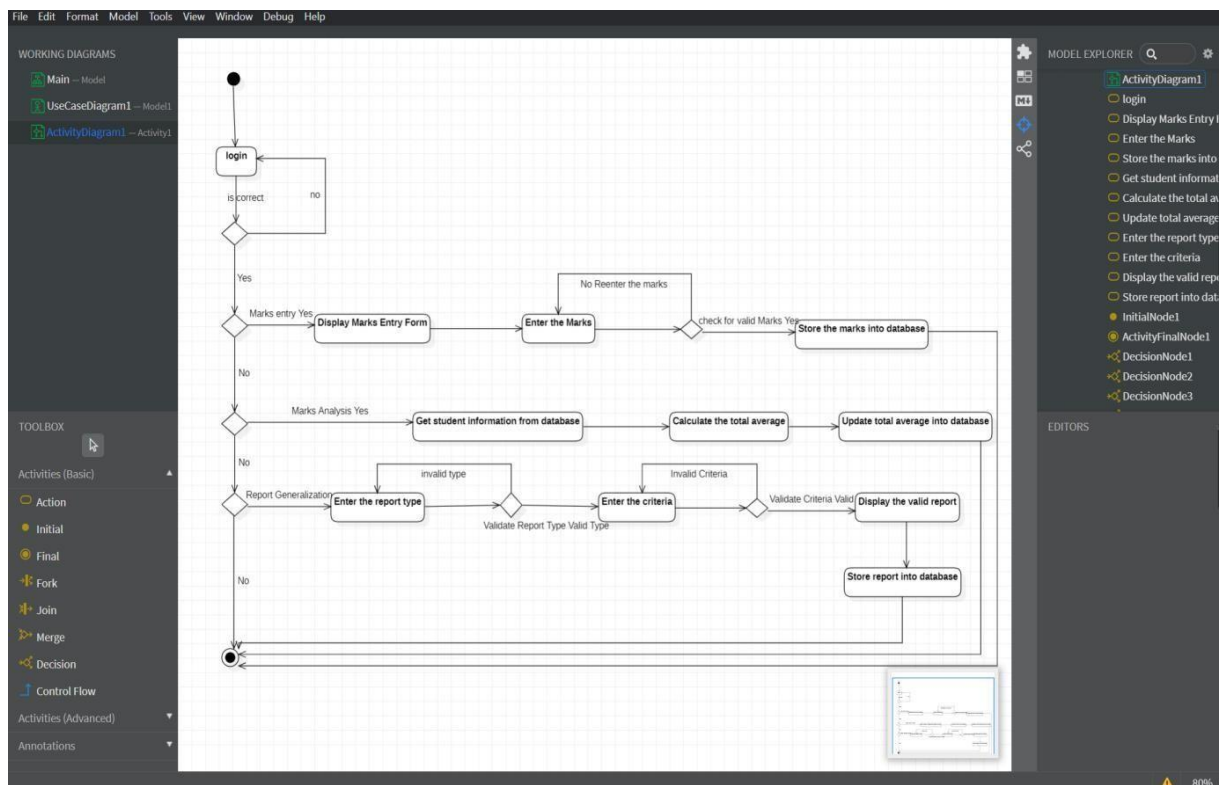
The system analyses the marks and generates the result reports. The marks and information about the students are stored in a database and the system works with the database. The faculty can enter the marks and student information through a visual environment. The updated details are stored in the database. The system generates the overall result by analyzing the marks. Mark analyzer monitors this process. One of the most important features of the system is creating reports based on the given criteria. The user can create the following reports.

STAR UML DIAGRAMS:

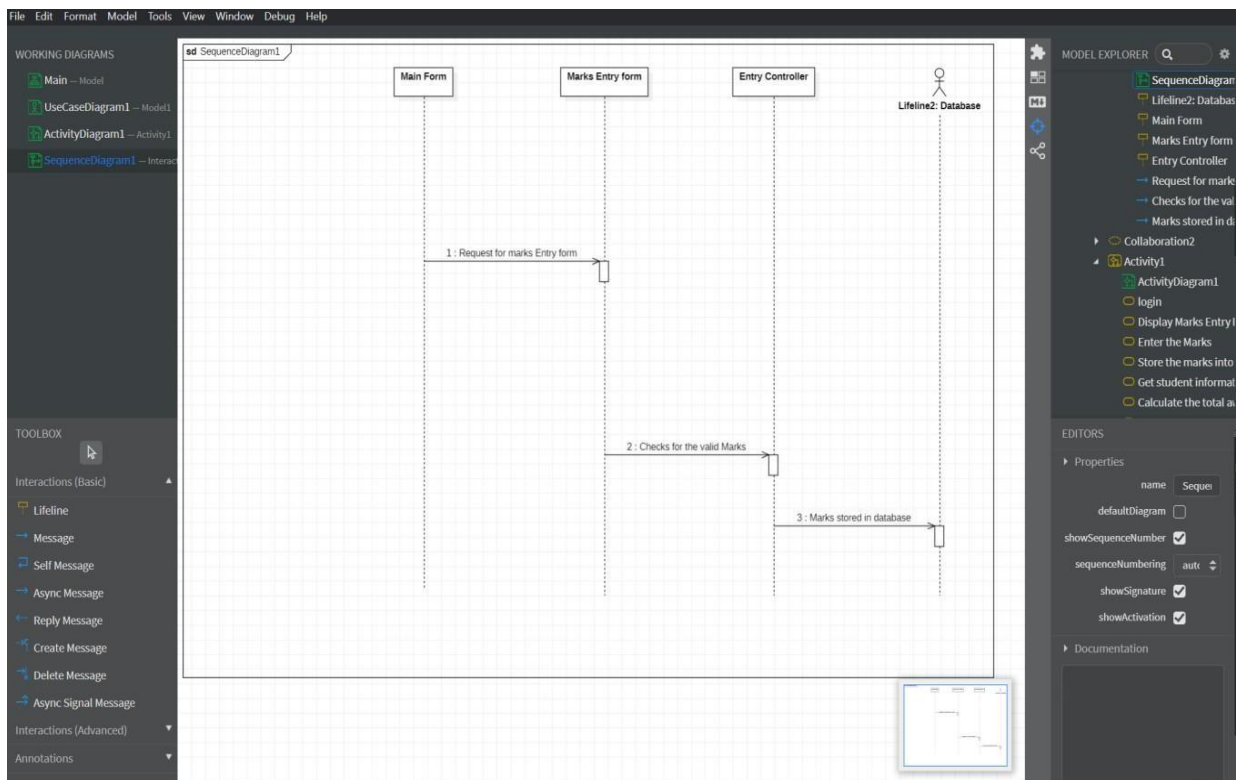
USE CASE DIAGRAM



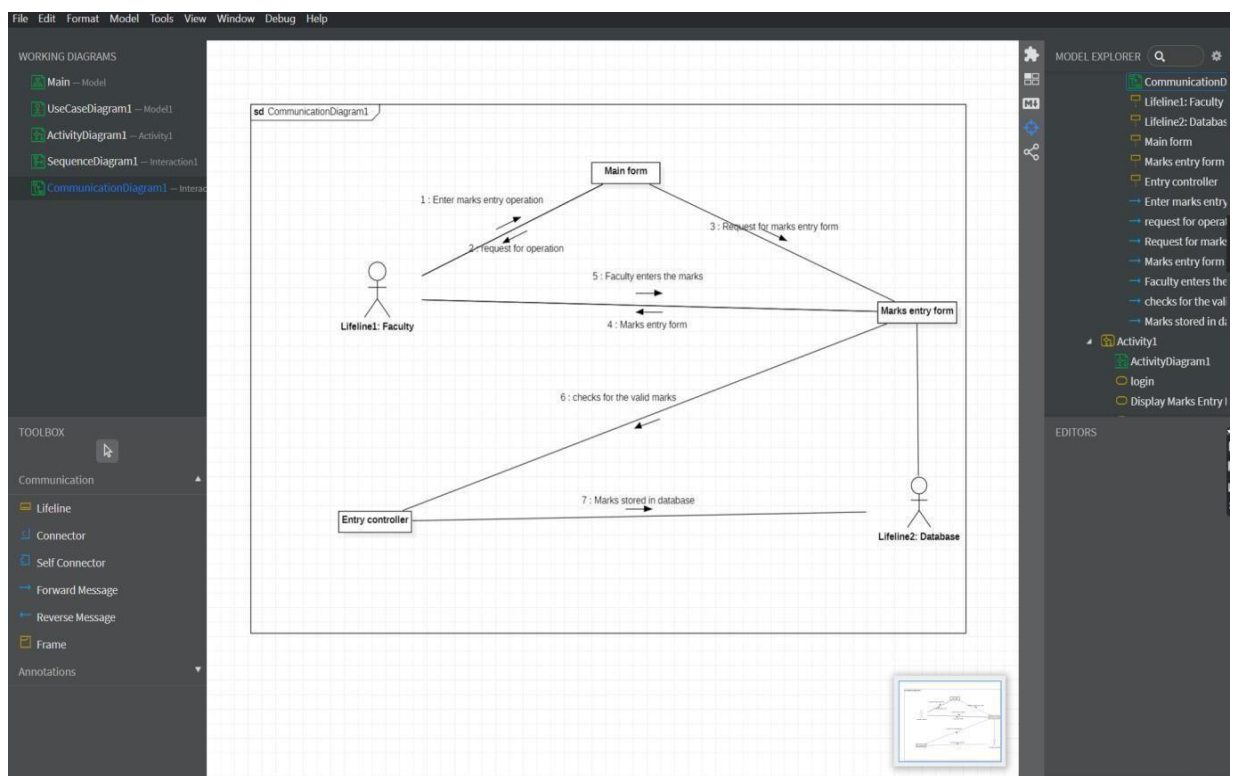
ACTIVITY DIAGRAM



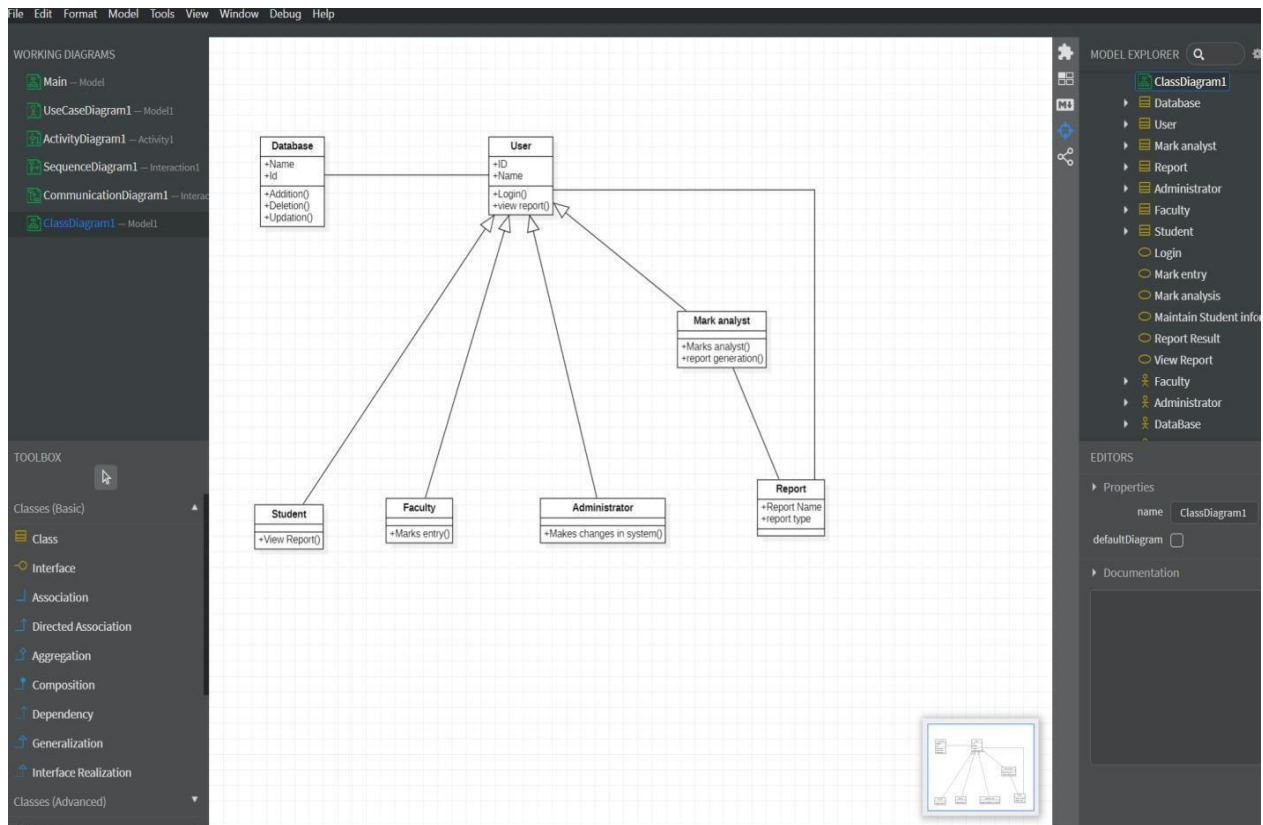
SEQUENCE DIAGRAM



COMMUNICATION DIAGRAM



CLASS DIAGRAM



STEPS TO GENERATE CODE IN STAR UML

1. Draw the class diagram and then save it.
2. Open the saved file with staruml.
3. Then go to tools at right corner.
4. Select the code language which we can prefer.
5. Then we have the code generator in tools so give access to it.
6. Then go to tools and select the language then it will take us to another window.
7. There select the class attribute to generate the code.
8. Then it will ask for the directory to store the code.
9. Select the directory and click on generate code.
10. Then code generated successfully.

OUTPUT:

> yoga oose > yoga oose exp3 > Model1

Name	Date modified	Type	Size
 Administrator.java	11-03-2023 10:07 AM	JAVA File	1 KB
 Database.java	11-03-2023 10:07 AM	JAVA File	1 KB
 Faculty.java	11-03-2023 10:07 AM	JAVA File	1 KB
 Mark analyst.java	11-03-2023 10:07 AM	JAVA File	1 KB
 Report.java	11-03-2023 10:07 AM	JAVA File	1 KB
 Student.java	11-03-2023 10:07 AM	JAVA File	1 KB
 User.java	11-03-2023 10:07 AM	JAVA File	1 KB

ADMINISTRATOR

```
> yoga oose > yoga oose exp3 > Model1
Administrator.java
File Edit View

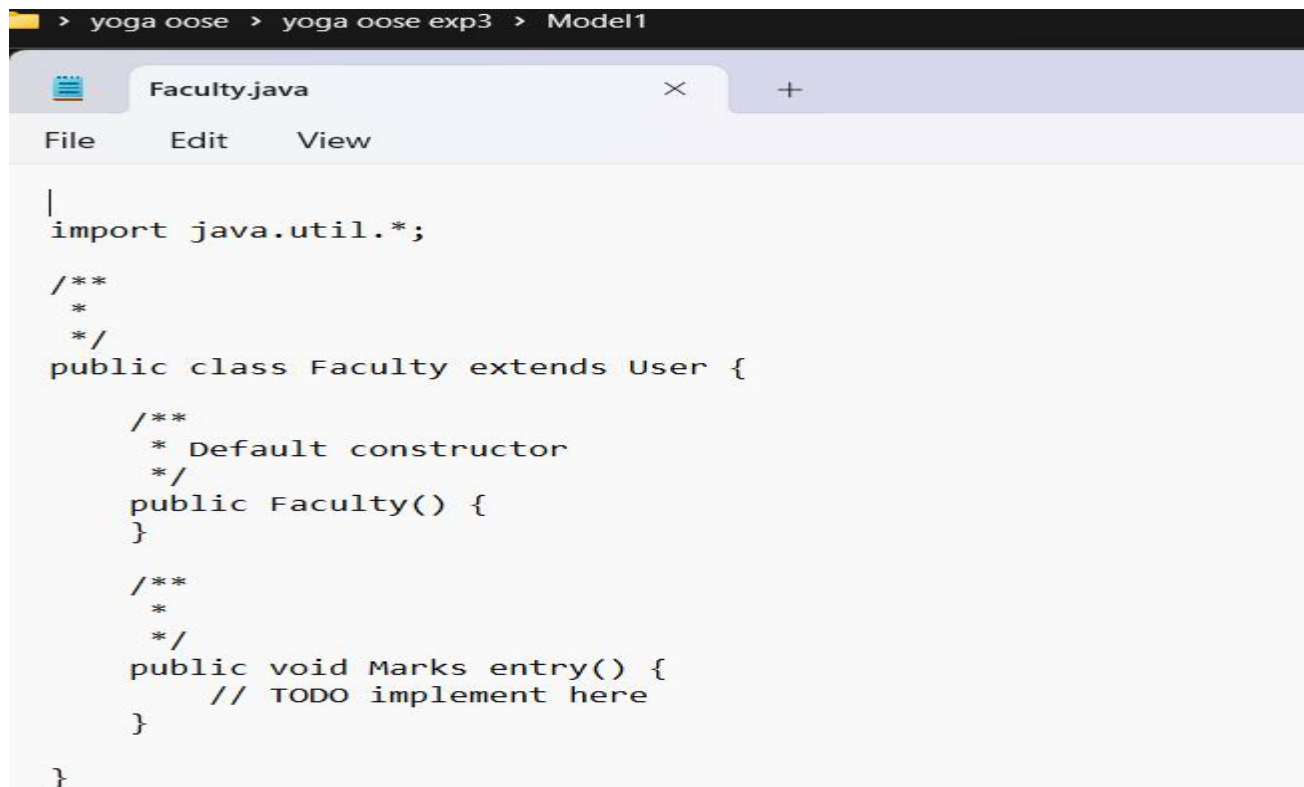
import java.util.*;
/**
 *
 */
public class Administrator extends User {
    /**
     * Default constructor
     */
    public Administrator() {
    }
}
```

DATA BASE

```
> yoga oose > yoga oose exp3 > Model1
Database.java
File Edit View

import java.util.*;
/**
 *
 */
public class Database {
    /**
     * Default constructor
     */
    public Database() {
    }
    /**
     *
     */
}
```

FACULTY



```
> yoga oose > yoga oose exp3 > Model1
Faculty.java
File Edit View

|
import java.util.*;

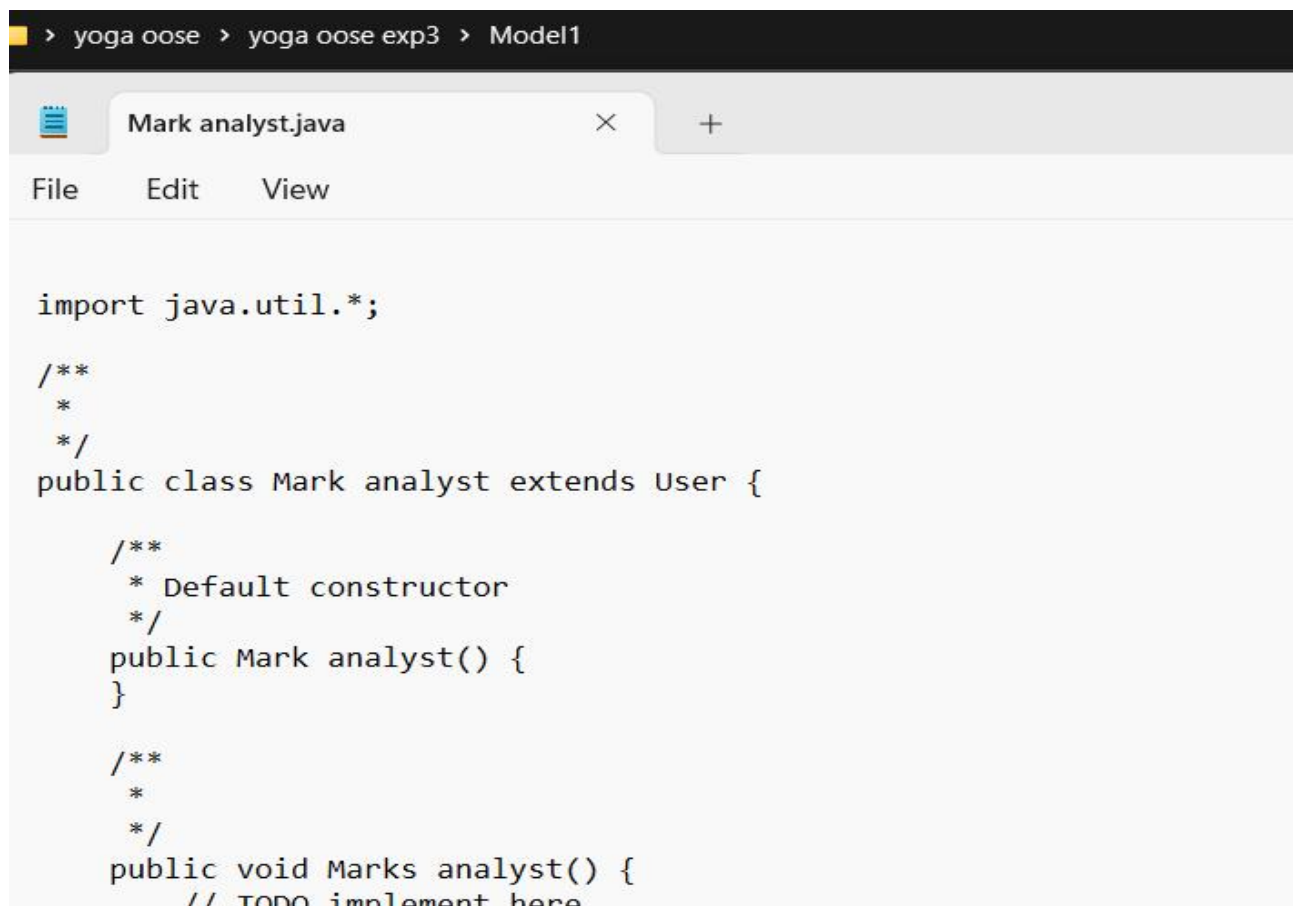
/**
 *
 */
public class Faculty extends User {

    /**
     * Default constructor
     */
    public Faculty() {
    }

    /**
     *
     */
    public void Marks entry() {
        // TODO implement here
    }

}
```

MARK ANALYST



```
> yoga oose > yoga oose exp3 > Model1
Mark analyst.java
File Edit View

import java.util.*;

/**
 *
 */
public class Mark analyst extends User {

    /**
     * Default constructor
     */
    public Mark analyst() {
    }

    /**
     *
     */
    public void Marks analyst() {
        // TODO implement here
    }

}
```

REPORT

```
> yoga oose > yoga oose exp3 > Model1

Report.java

import java.util.*;

/**
 *
 */
public class Report {

    /**
     * Default constructor
     */
    public Report() {
    }
}
```

STUDENT

```
> yoga oose > yoga oose exp3 > Model1

Student.java

import java.util.*;

/**
 *
 */
public class Student extends Student {

    /**
     * Default constructor
     */
    public Student() {
    }
}
```

USER

```
> yoga oose > yoga oose exp3 > Model1

User.java

import java.util.*;

/**
 *
 */
public class User {

    /**
     * Default constructor
     */
    public User() {
    }

    /**

```

Result:

Thus, the UML diagrams for Student Mark Analysis system in StarUML have been designed and java code template was generated successfully.

PAYROLL SYSTEM

AIM:

To develop the Payroll System using StarUML.

INTRODUCTION

This document deals with online Payroll Processing System. This document is designed in such a way that the reader understands it. The use case description and other documents are described in such a way that the system reaches the people easily.

OBJECTIVES

To develop a user-friendly online pay roll processing system which makes the work easy and faster.

SCOPE

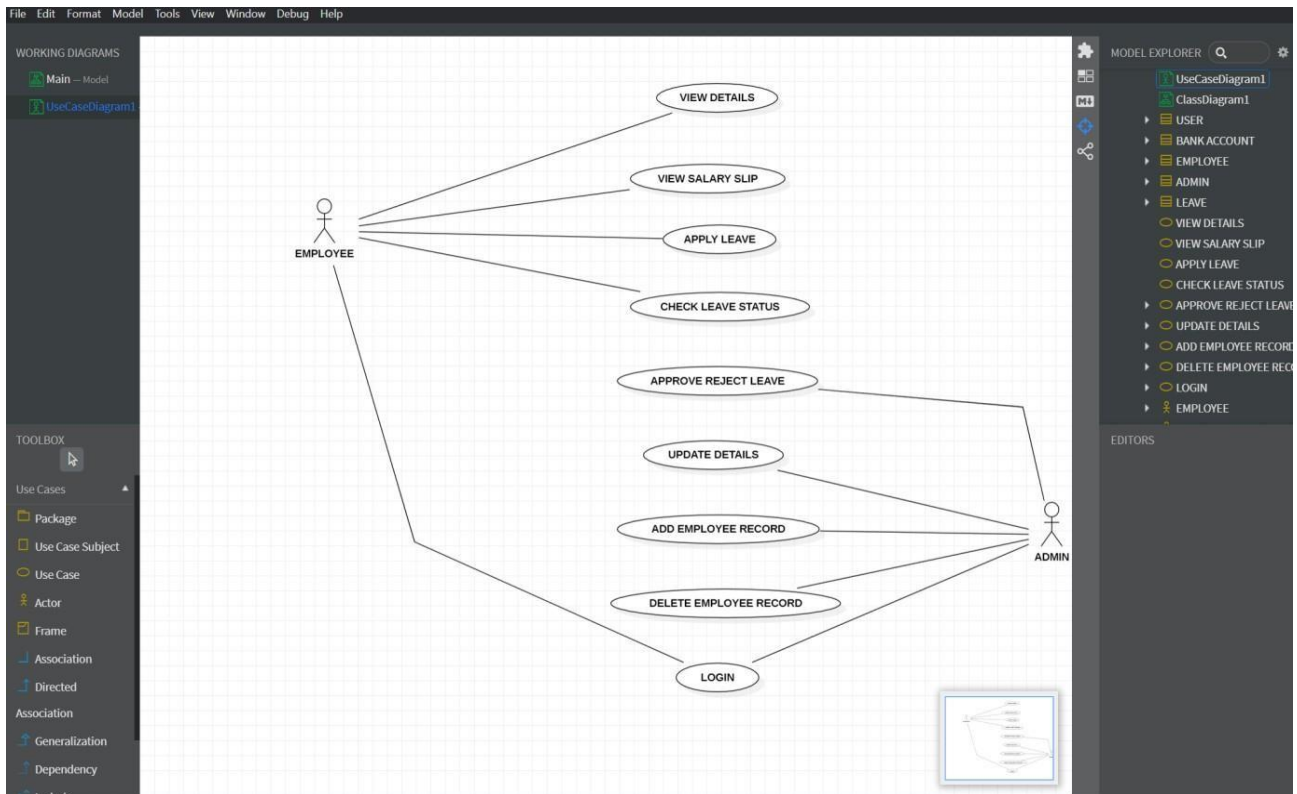
This project describes about the payroll system, there are two types of Users, normal users and Administrator. Users can log on the directory by a unique User ID that is provided to them and for every ID check there is a password. Administrator does authorization, he is the manager of that directory, and he can Search employee details with names and their employee id. This would help users to Create or find employee details easily.

PROBLEM STATEMENT

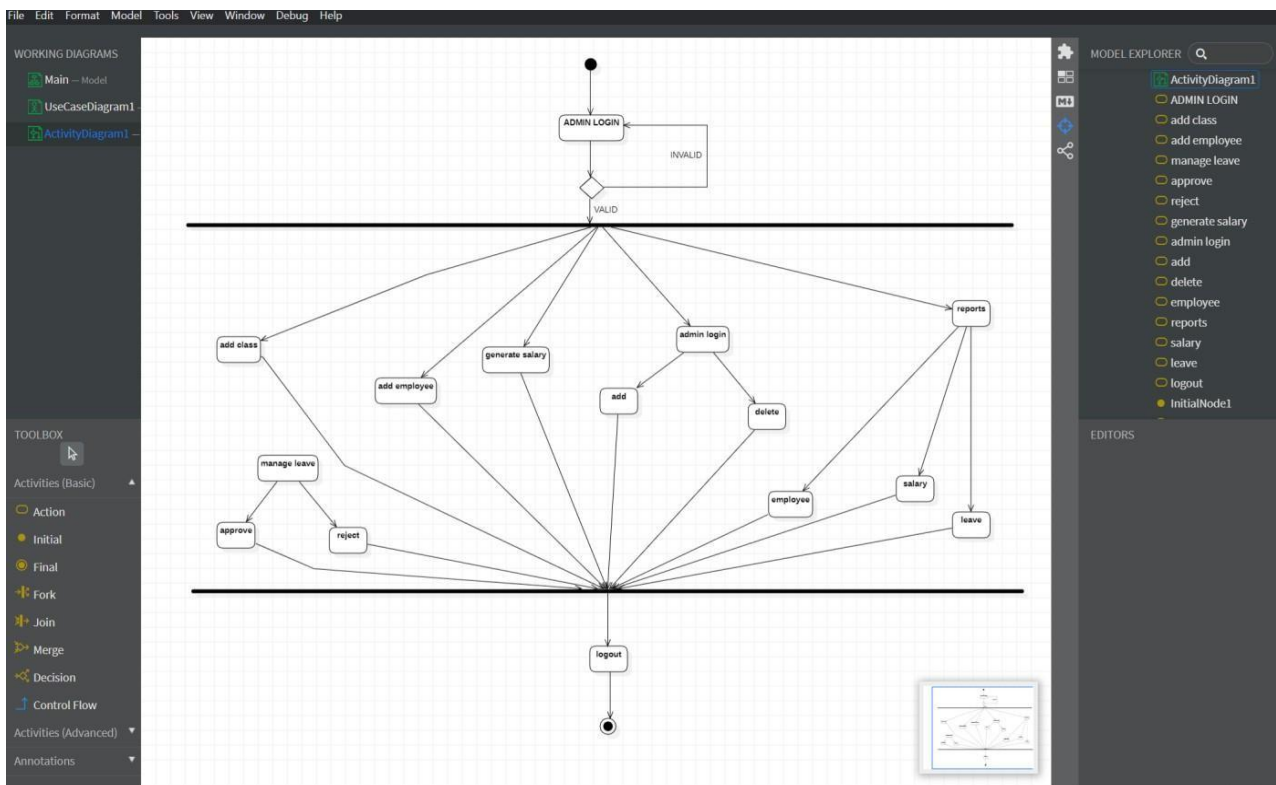
Designing a payroll system for users and administrator through Authentication Process, where only administrator can alter the data in Oracle database, Checking ID's and password of authorized users.

STAR UML DIAGRAMS :

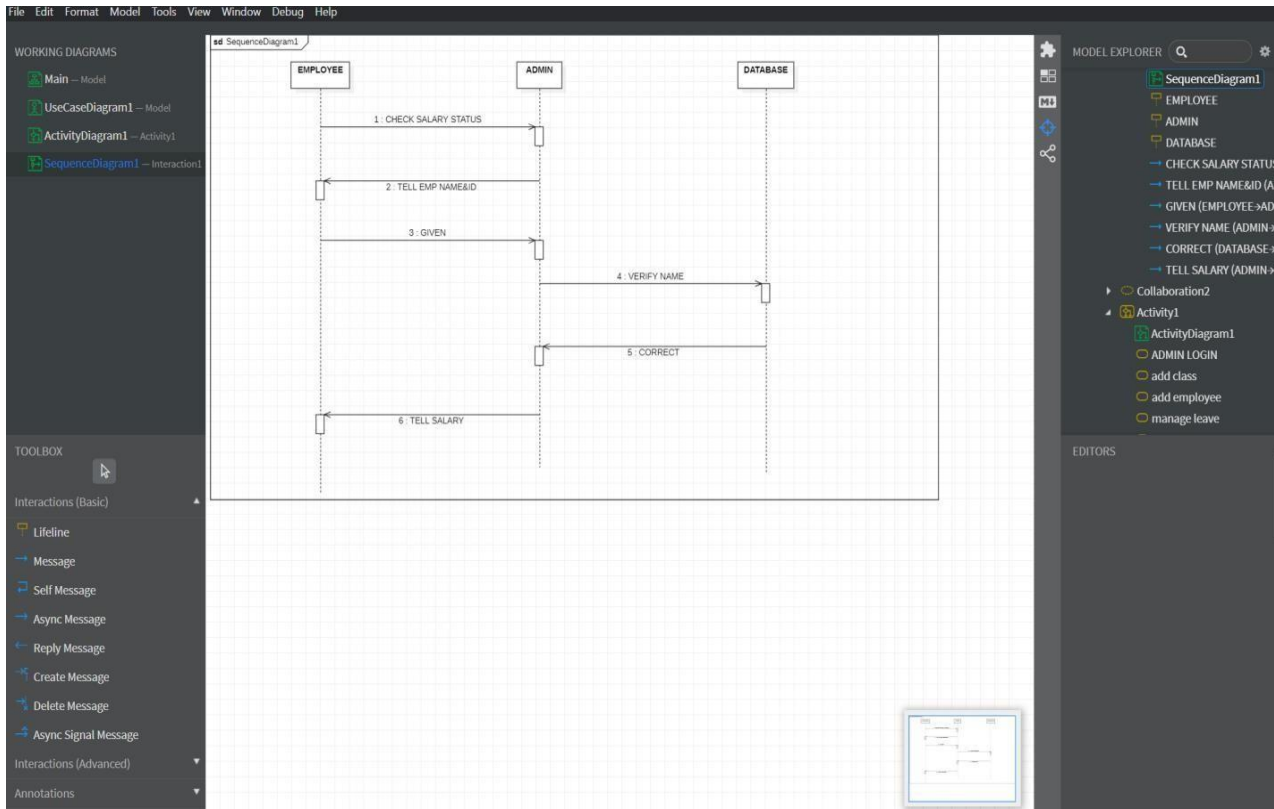
USE CASE DIAGRAM



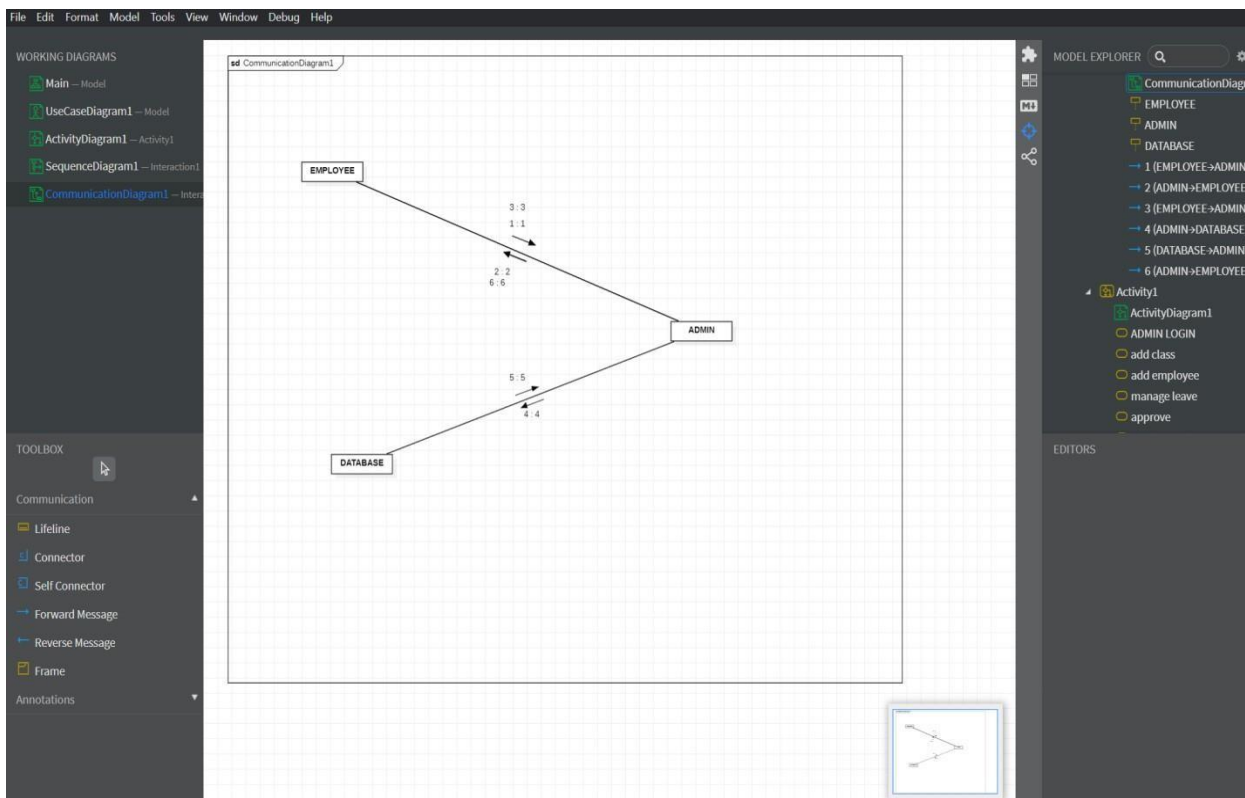
ACTIVITY DIAGRAM



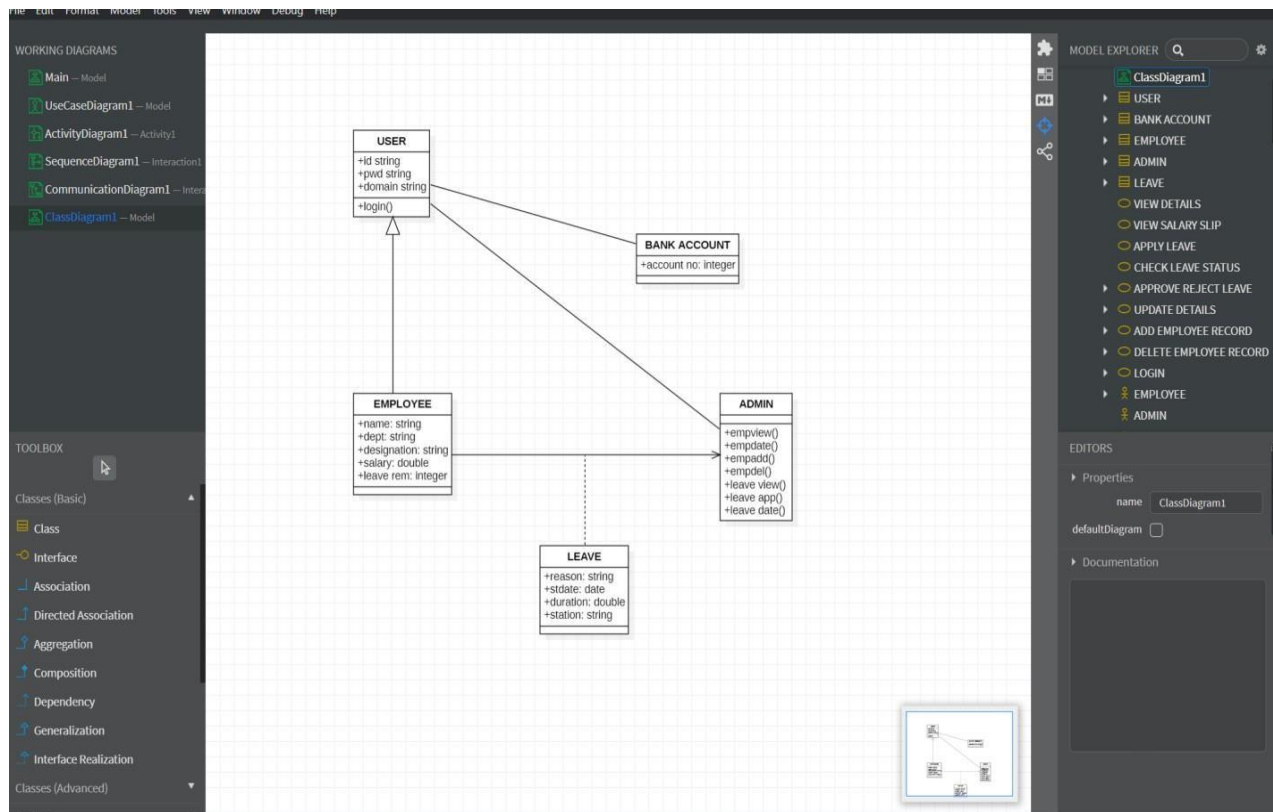
SEQUENCE DIAGRAM



COMMUNICATION DIAGRAM



CLASS DIAGRAM



STEPS TO GENERATE CODE IN STAR UML

1. Draw the class diagram and then save it.
2. Open the saved file with staruml.
3. Then go to tools at right corner.
4. Select the code language which we can prefer.
5. Then we have the code generator in tools so give access to it.
6. Then go to tools and select the language then it will take us to another window.
7. There select the class attribute to generate the code.
8. Then it will ask for the directory to store the code.
9. Select the directory and click on generate code.
10. Then code generated successfully.

OUTPUT:

> yoga oose > yoga oose exp4 > Model

Name	Date modified	Type	Size
ADMIN.java	11-03-2023 10:08 AM	JAVA File	1 KB
BANK ACCOUNT.java	11-03-2023 10:08 AM	JAVA File	1 KB
EMPLOYEE.java	11-03-2023 10:08 AM	JAVA File	1 KB
LEAVE.java	11-03-2023 10:08 AM	JAVA File	1 KB
USER.java	11-03-2023 10:08 AM	JAVA File	1 KB

ADMIN

```
> yoga oose > yoga oose exp4 > Model

ADMIN.java

File Edit View

import java.util.*;

/**
 *
 */
public class ADMIN {

    /**
     * Default constructor
     */
    public ADMIN() {
    }

    /**
     *
     */
}
```

BANK ACCOUNT

```
> yoga oose > yoga oose exp4 > Model

BANK ACCOUNT.java

File Edit View

import java.util.*;

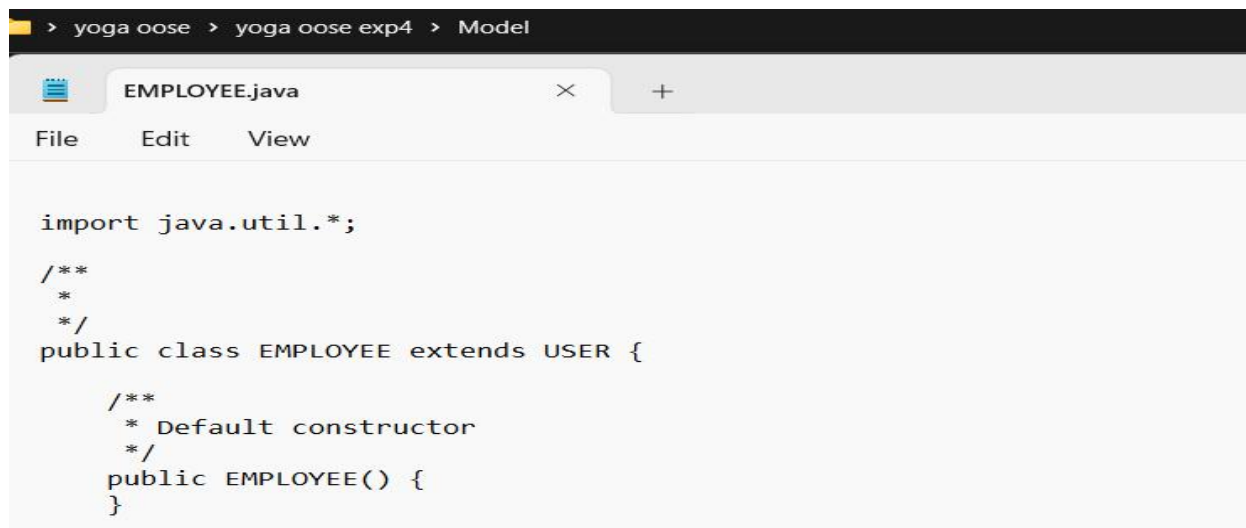
/**
 *
 */
public class BANK ACCOUNT {

    /**
     * Default constructor
     */
    public BANK ACCOUNT() {
    }

    /**
     *
     */
    public integer account no;

}
```

EMPLOYEE



```
> yoga oose > yoga oose exp4 > Model

EMPLOYEE.java

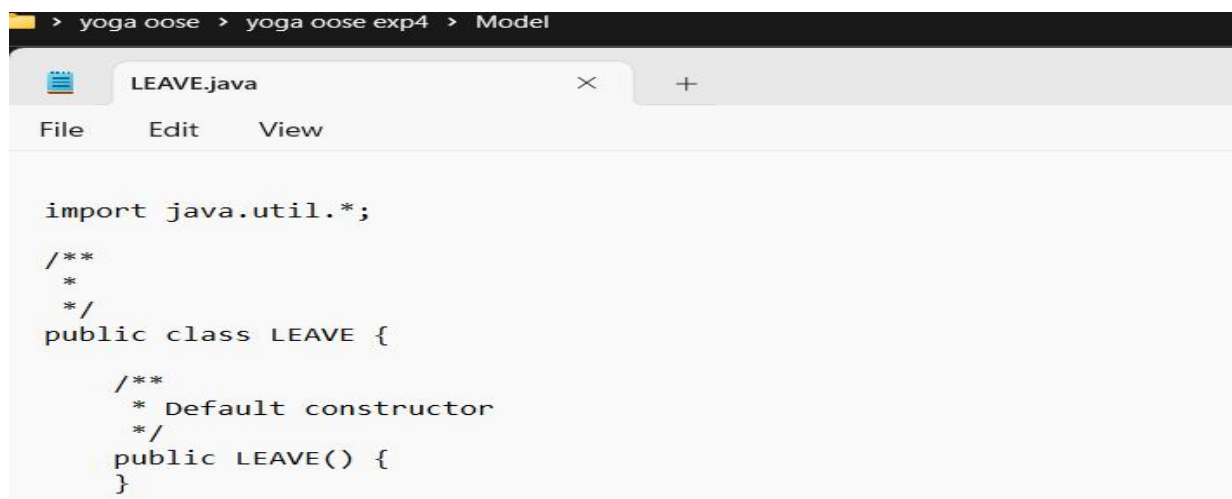
File Edit View

import java.util.*;

/**
 *
 */
public class EMPLOYEE extends USER {

    /**
     * Default constructor
     */
    public EMPLOYEE() {
    }
}
```

LEAVE



```
> yoga oose > yoga oose exp4 > Model

LEAVE.java

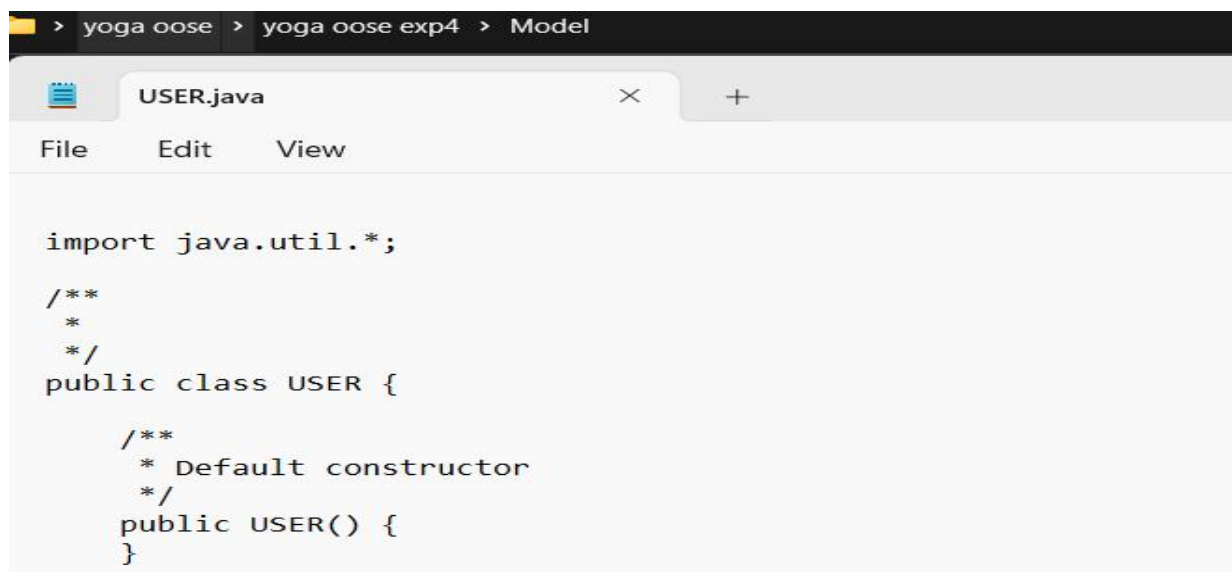
File Edit View

import java.util.*;

/**
 *
 */
public class LEAVE {

    /**
     * Default constructor
     */
    public LEAVE() {
    }
}
```

USER



```
> yoga oose > yoga oose exp4 > Model

USER.java

File Edit View

import java.util.*;

/**
 *
 */
public class USER {

    /**
     * Default constructor
     */
    public USER() {
    }
}
```

Result:

Thus, the UML diagrams for Payroll system in StarUML has been designed and java code the template was generated successfully.

INVENTORY SYSTEM

AIM

To develop the Inventory system using Star UML.

PROBLEM ANALYSIS AND PROJECT PLANNING

INTRODUCTION

The stock maintenance system is basically for the customers who access the information about the stock (here it is books in the book store) and retrieves the information.

OBJECTIVES

The purpose of the document is to define the requirements of the stock maintenance system. This supplementary specification lists the requirements that are not readily captured in the use cases of the use case model. The supplementary specification & the use case model together capture a complete set of requirement on the system.

SCOPE

This supplementary specification applies to the stock maintenance system. This specification defines the non-functional requirements of the system, such as reliability, usability, performance and supportability as well as functional requirements that are common across a number of use cases.

PROBLEM STATEMENT

A new stock maintenance system for a book store is to replace the existing maintenance system which is inefficient. The new stock maintenance system will allow the employee to record information of the books available in the book store and generate report based on the total amount of sales.

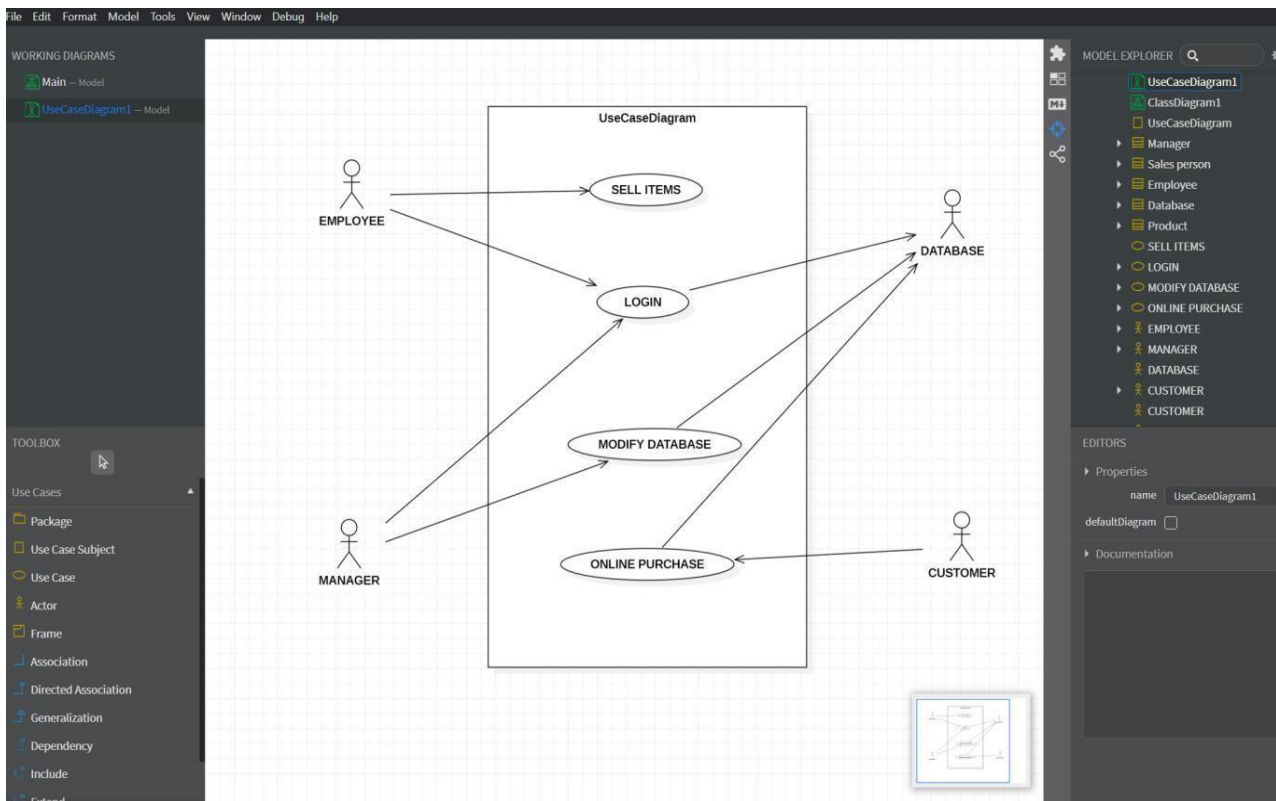
The new system will have a Windows based desktop interface to allow employees to enter the information of sale, purchase orders, change employee preferences and create reports. Employees can only access the information and purchase orders for security purpose.

The system retains information on all the books in the shop. The system retains the records of the cost, edition, author, publication of the books. The employee maintains the information of the sale of books. He can add the books at right time and update the database.

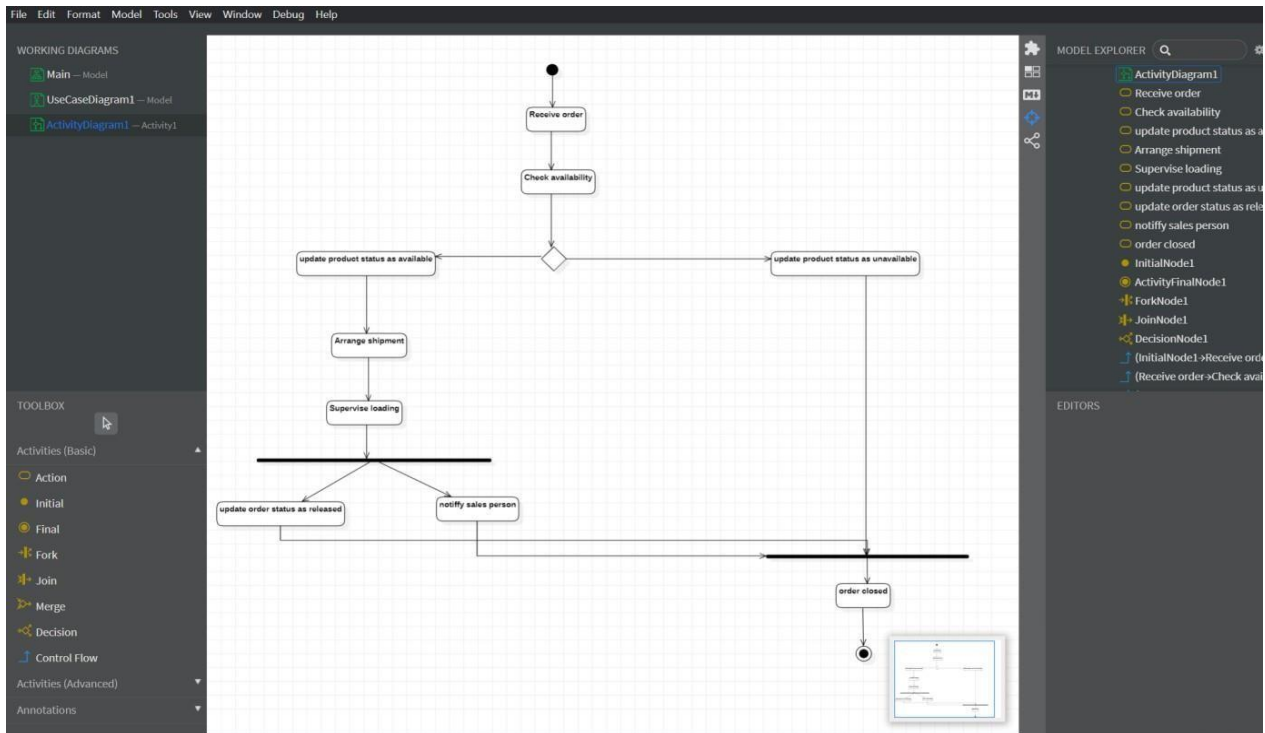
The customer can view the availability of the required books and the price of the books. The customer can just view them but cannot make any changes.

STAR UML DIAGRAMS :

USE CASE DIAGRAM

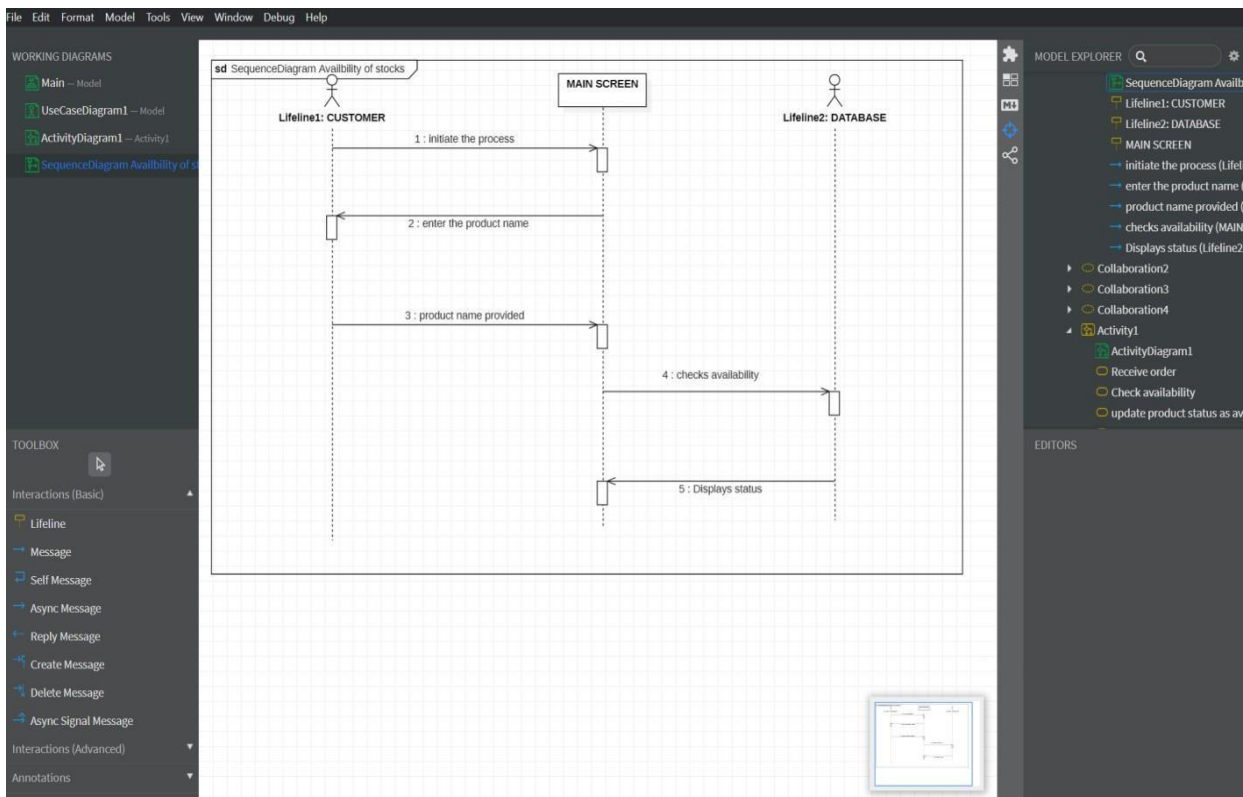


ACTIVITY DIAGRAM

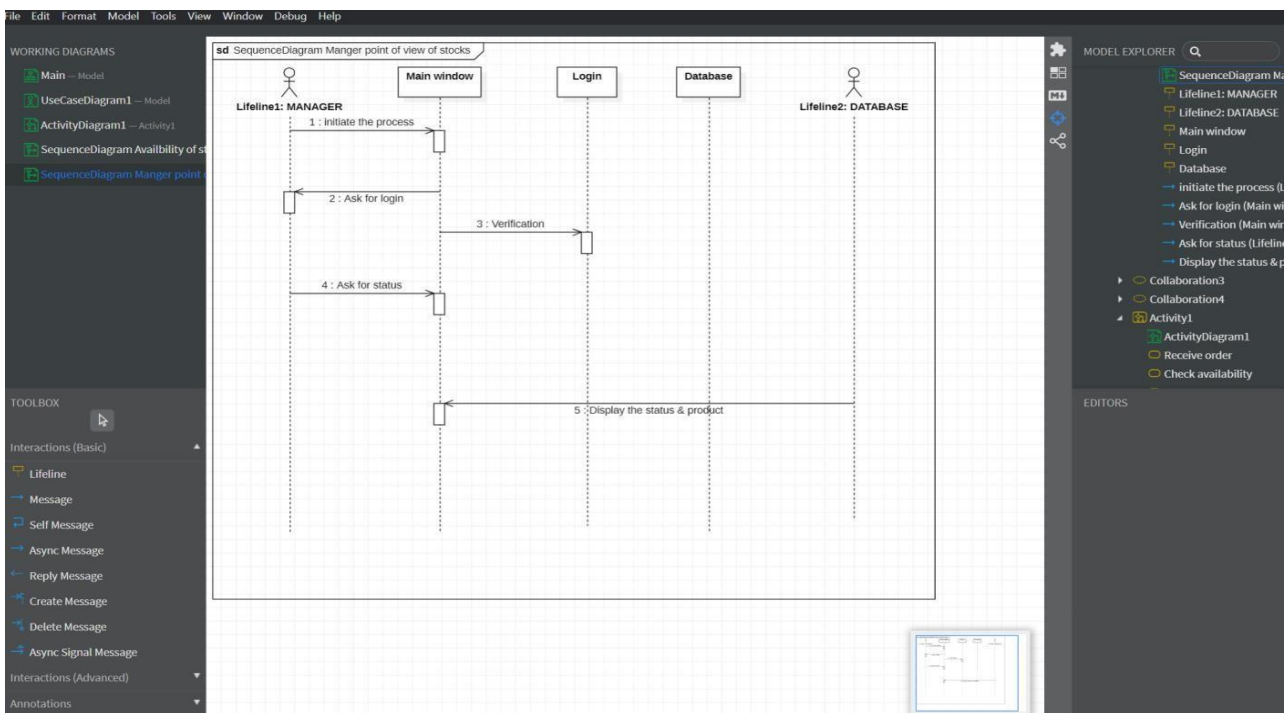


SEQUENCE DIAGRAM

AVAILABILITY OF STOCKS

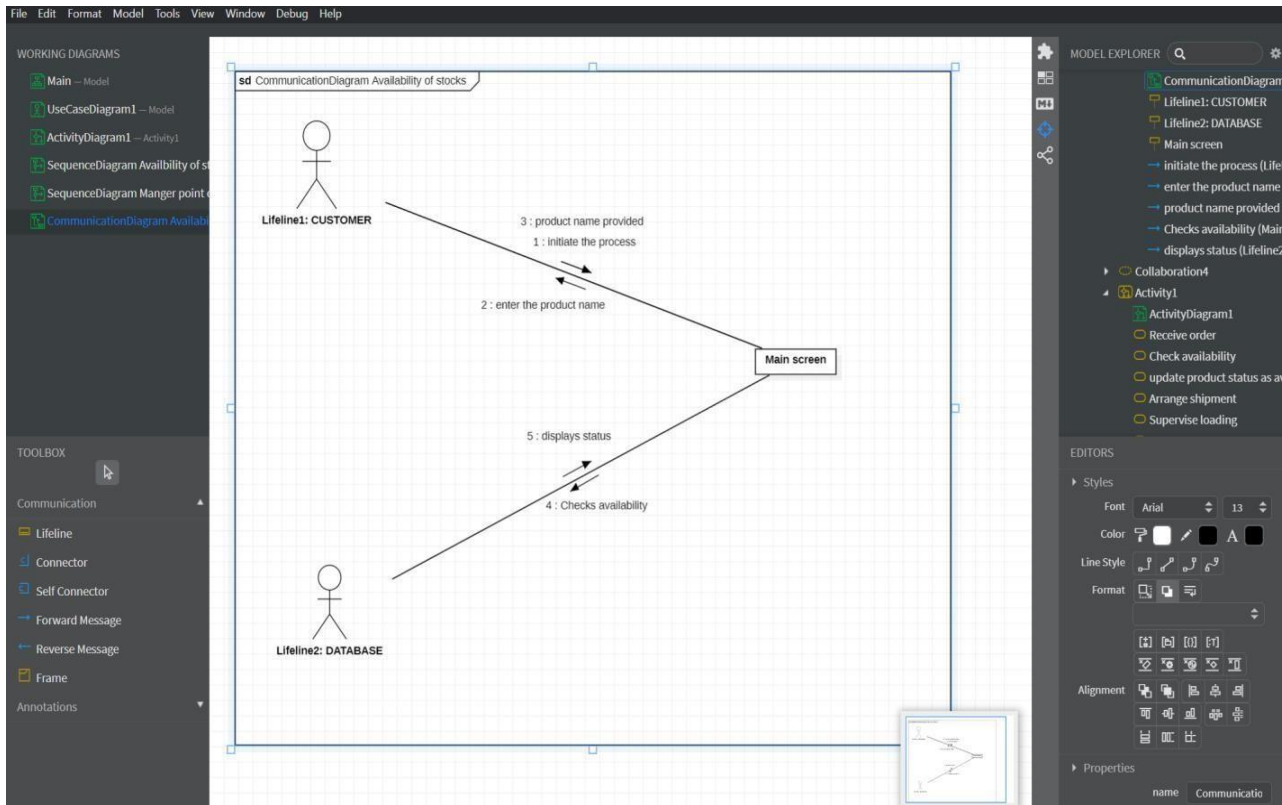


MANAGER POINT OF VIEW

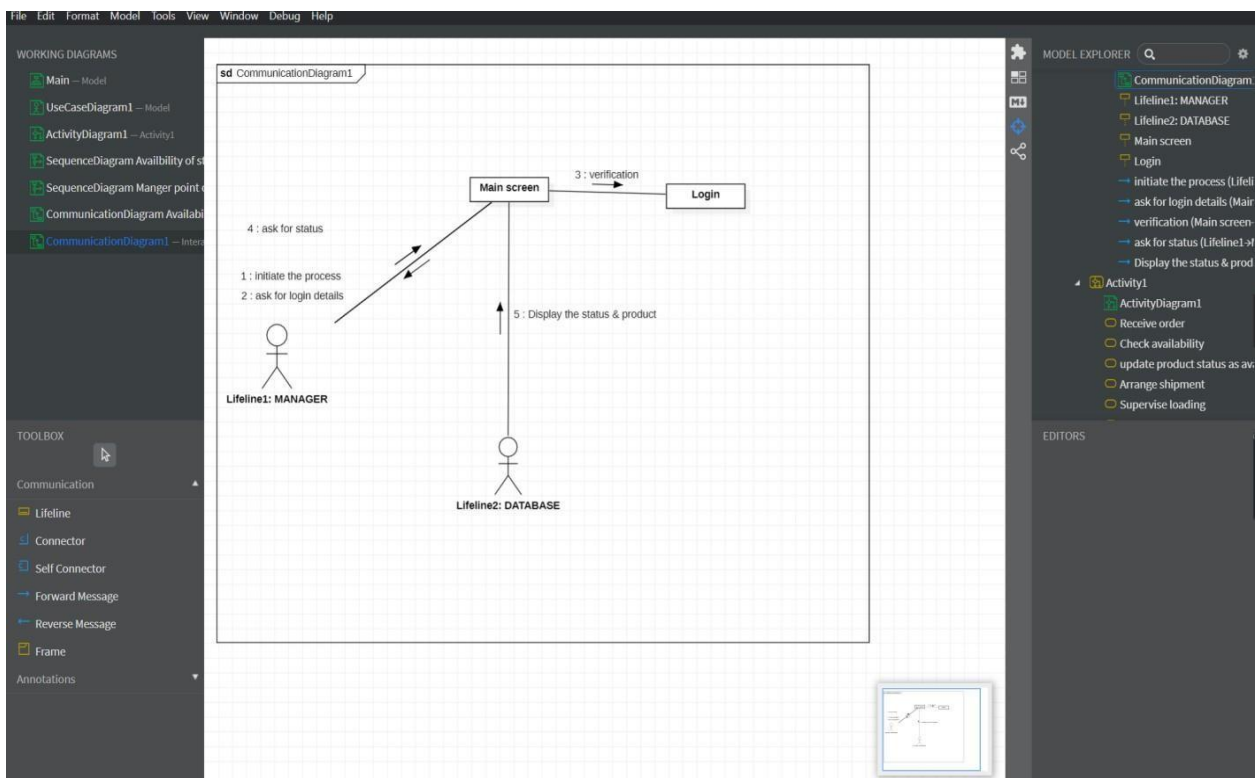


COMMUNICATION DIAGRAM

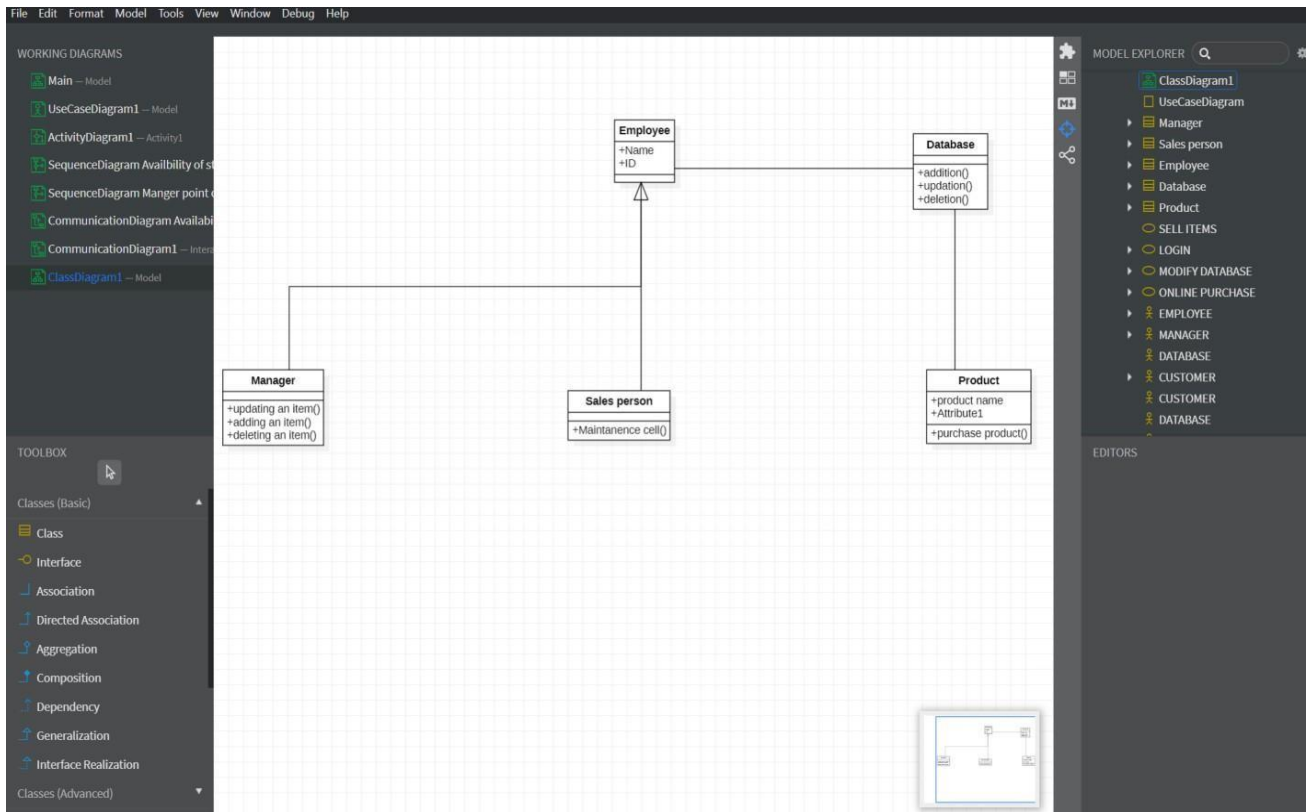
AVAILABILITY OF STOCKS



MANAGER POINT OF VIEW



CLASS DIAGRAM



STEPS TO GENERATE CODE IN STAR UML

1. Draw the class diagram and then save it.
2. Open the saved file with star uml.
3. Then go to tools at right corner.
4. Select the code language which we can prefer.
5. Then we have the code generator in tools so give access to it.
6. Then go to tools and select the language then it will take us to another window.
7. There select the class attribute to generate the code.
8. Then it will ask for the directory to store the code.
9. Select the directory and click on generate code.
10. Then code generated successfully.

OUTPUT:

> yoga oose > yoga oose exp5 > Model

Name	Date modified	Type	Size
 Database.java	11-03-2023 10:09 AM	JAVA File	1 KB
 Employee.java	11-03-2023 10:09 AM	JAVA File	1 KB
 Manager.java	11-03-2023 10:09 AM	JAVA File	1 KB
 Product.java	11-03-2023 10:09 AM	JAVA File	1 KB
 Sales person.java	11-03-2023 10:09 AM	JAVA File	1 KB

DATA BASE

```
> yoga oose > yoga oose exp5 > Model

Database.java
File Edit View

import java.util.*;
/**
 *
 */
public class Database {
    /**
     * Default constructor
     */
    public Database() {
    }
}
```

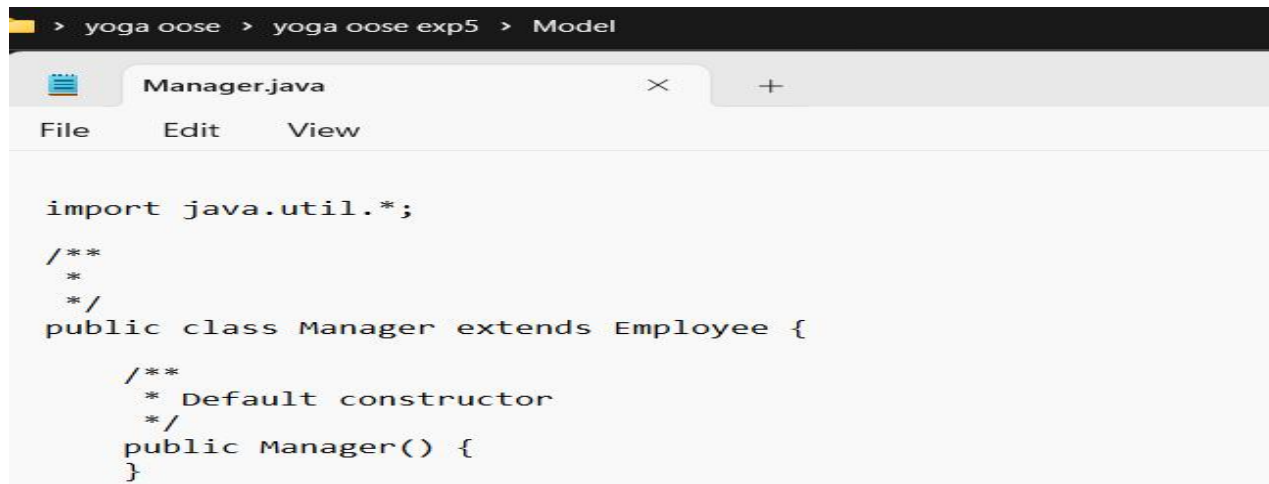
EMPLOYEE

```
> yoga oose > yoga oose exp5 > Model

Employee.java
File Edit View

import java.util.*;
/**
 *
 */
public class Employee {
    /**
     * Default constructor
     */
    public Employee() {
    }
    /**
     *
     */
    public void Name;
```

MANAGER



```
> yoga oose > yoga oose exp5 > Model
Manager.java
File Edit View

import java.util.*;

/**
 *
 */
public class Manager extends Employee {

    /**
     * Default constructor
     */
    public Manager() {
    }
}
```

PRODUCT



```
> yoga oose > yoga oose exp5 > Model
Product.java
File Edit View

import java.util.*;

/**
 *
 */
public class Product {

    /**
     * Default constructor
     */
    public Product() {
    }

    /**
     *
     */
    public void product name();
}
```

SALES PERSON



```
> yoga oose > yoga oose exp5 > Model
Sales person.java
File Edit View

import java.util.*;

/**
 *
 */
public class Sales person {

    /**
     * Default constructor
     */
    public Sales person() {
    }

    /**
     *
     */
    public void Maintenance cell() {
        // TODO implement here
    }
}
```

Result:

Thus, the UML diagrams for Inventory system in StarUML has been designed and java code template was generated successfully.

AUTOMATING THE BANKING SYSTEM

AIM

To develop the Automating the banking System using StarUML.

PROBLEM ANALYSIS AND PROJECT PLANNING

INTRODUCTION

Banking is one of the common day to day activities of life. Nowadays it is totally different from what existed a few years back, today banking has become completely computerized, new facilities such as credit cards, debit cards & ATM has been introduced. ATM stands for Automatic Teller Machine which is basically used to withdraw money from an account.

OBJECTIVES

The objective of this software is similar to the ATM software installed in an ATM center. It first validates the pin of the ATM card, then the type of transaction is enquired and the information from the customer is validated. If it is a withdrawal the amount is asked. After the money is delivered the transaction details are updated in the database where the customer's information is stored.

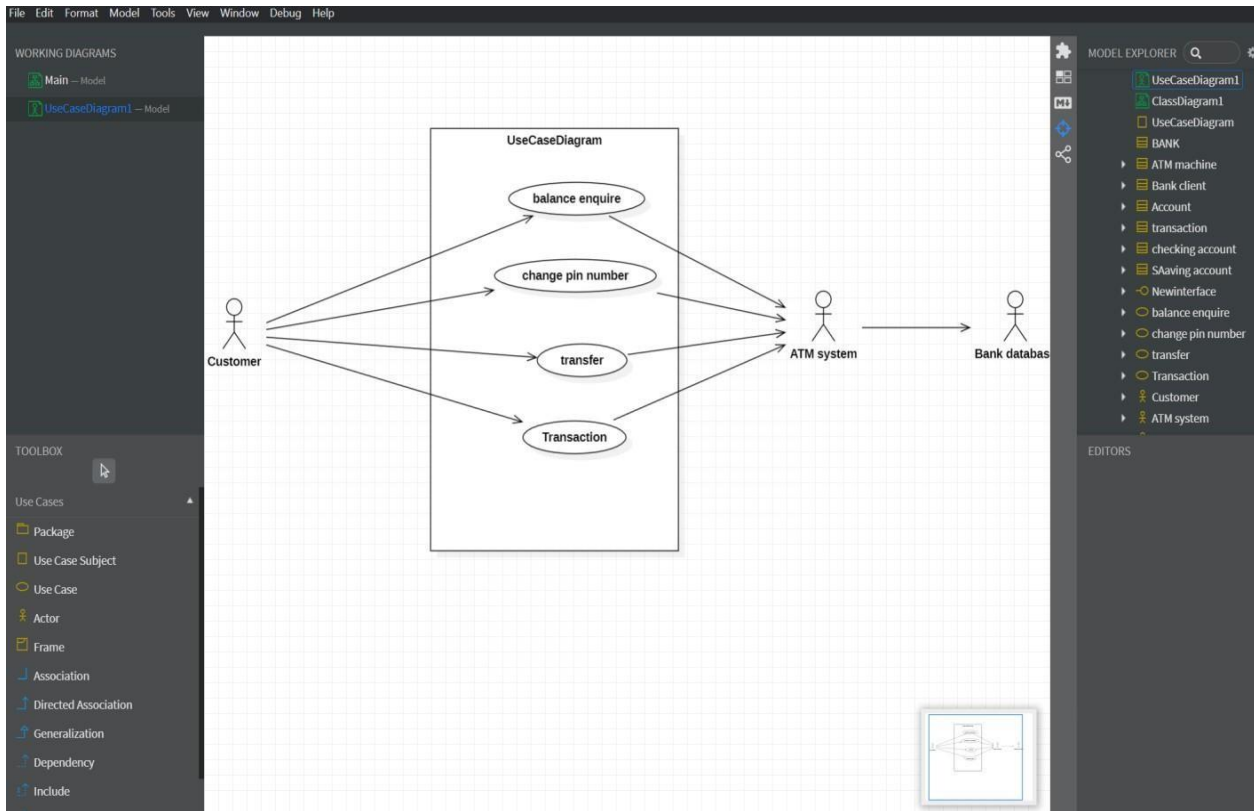
SCOPE

The scope of the project is to design an ATM system that will help in completely automatic banking, this software is going to be designed for withdrawal and deposit and also register the transaction in the database where the customer's information is stored.

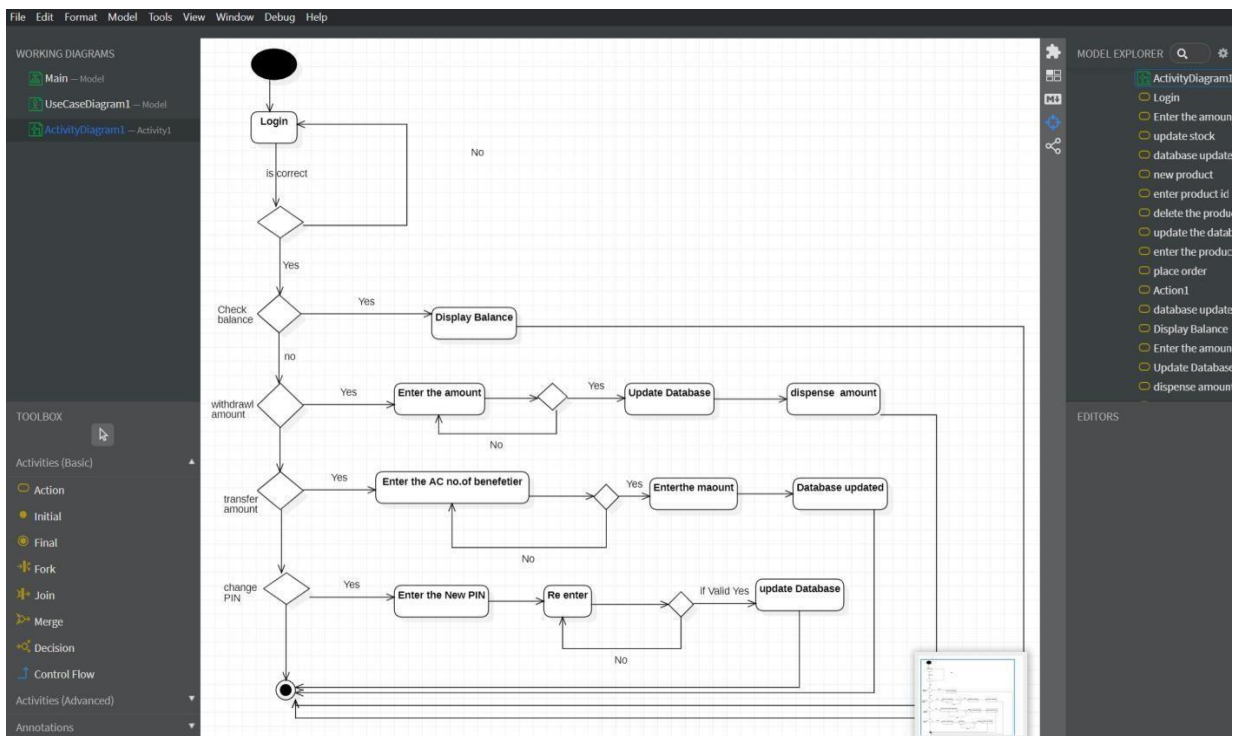
PROBLEM STATEMENT

ATM banking is another type of banking where the most frequent type of transaction made is withdrawal. A user may withdraw as much as he wants until his account holds a sum greater than his withdrawal amount. ATM is completely automated and there is no necessity of the ATM center being placed at the bank itself. It can be placed in the shopping malls, airports, railway stations etc. This ATM system can use any kind of interface. But it should be user friendly and not confusing. Help manuals should be provided in case any customer has problem working with the software.

STAR UML DIAGRAMS : USE CASE DIAGRAM

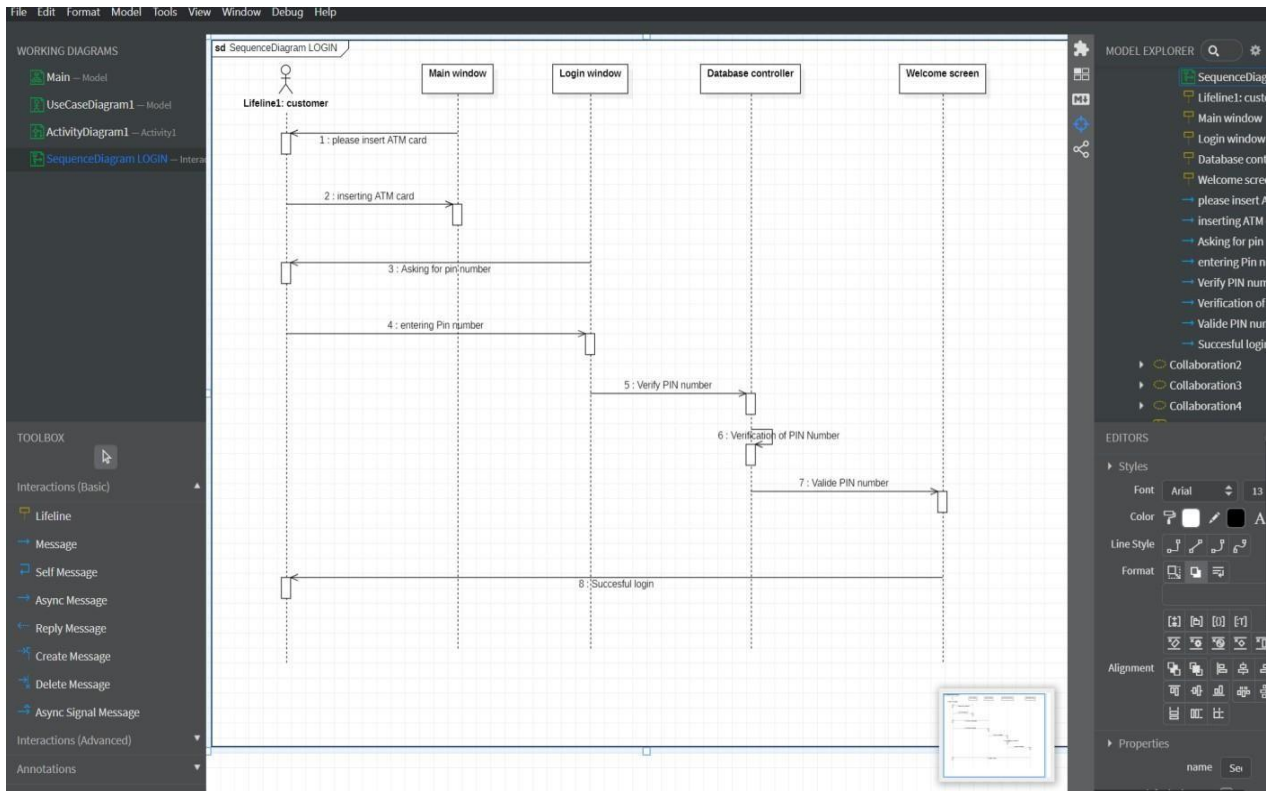


ACTIVITY DIAGRAM

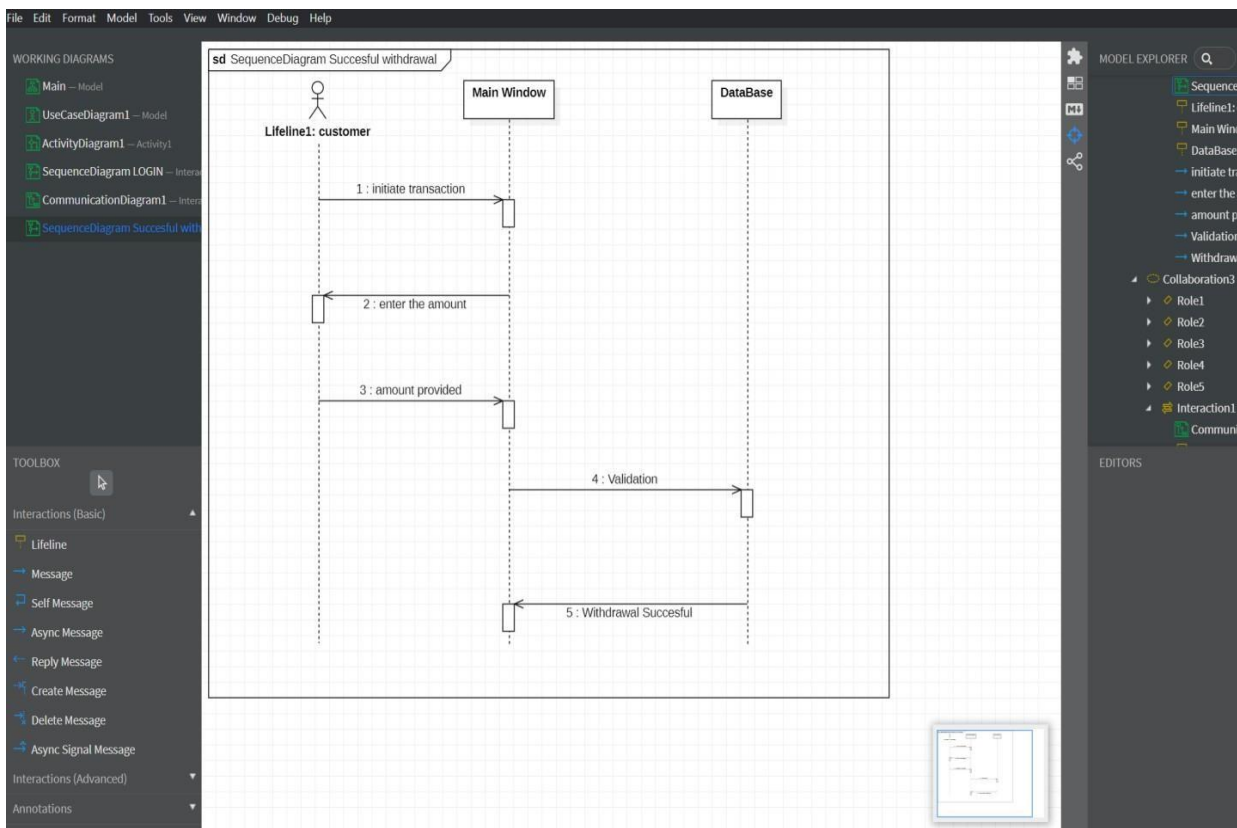


SEQUENCE DIAGRAM

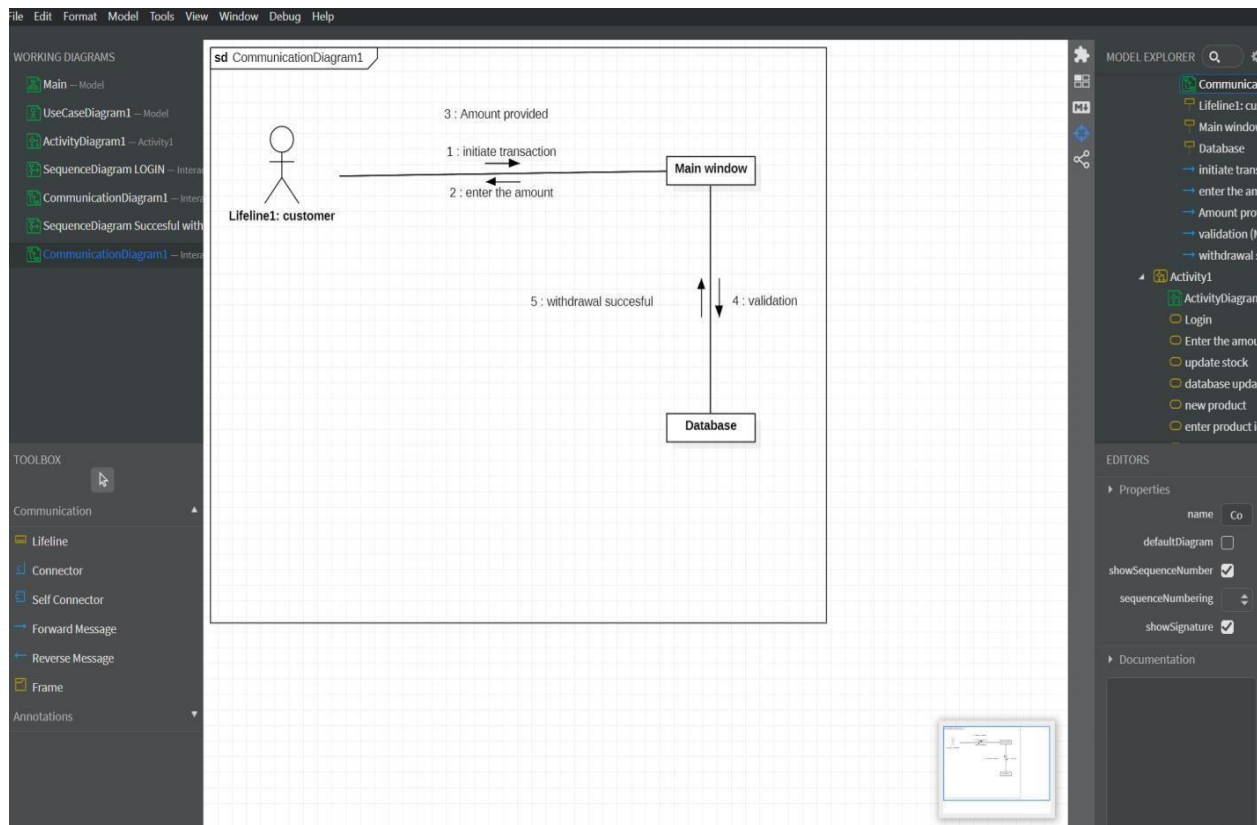
LOGIN



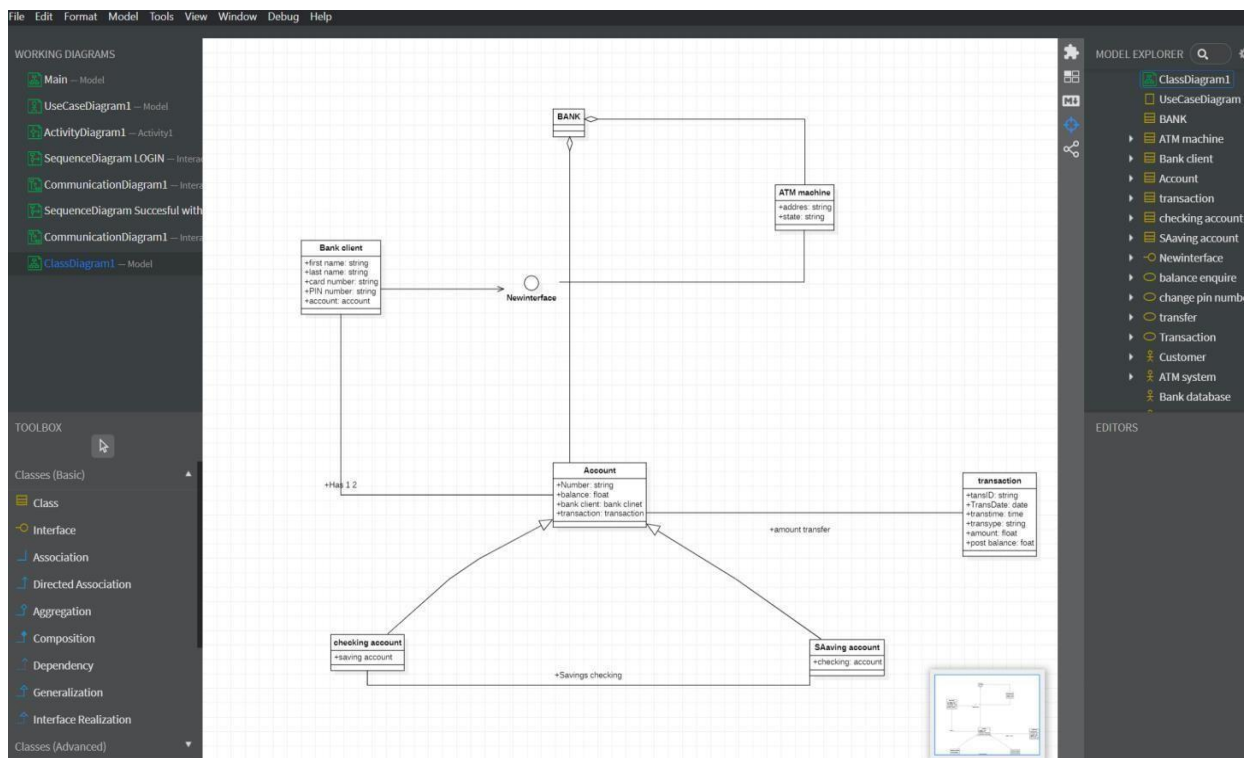
SUCCESSFUL WITHDRAWAL



COMMUNICATION DIAGRAM



CLASS DIAGRAM



STEPS TO GENERATE CODE IN STAR UML

1. Draw the class diagram and then save it.
2. Open the saved file with star uml.
3. Then go to tools at right corner.
4. Select the code language which we can prefer.
5. Then we have the code generator in tools so give access to it.
6. Then go to tools and select the language then it will take us to another window.
7. There select the class attribute to generate the code.
8. Then it will ask for the directory to store the code.
9. Select the directory and click on generate code.
10. Then code generated successfully.

OUTPUT:

> yoga oose > yoga oose exp6 > Model

Name	Date modified	Type	Size
Account.java	11-03-2023 10:10 AM	JAVA File	1 KB
ATM machine.java	11-03-2023 10:10 AM	JAVA File	1 KB
Bank client.java	11-03-2023 10:10 AM	JAVA File	1 KB
BANK.java	11-03-2023 10:10 AM	JAVA File	1 KB
checking account.java	11-03-2023 10:10 AM	JAVA File	1 KB
Newinterface.java	11-03-2023 10:10 AM	JAVA File	1 KB
SAaving account.java	11-03-2023 10:10 AM	JAVA File	1 KB
transaction.java	11-03-2023 10:10 AM	JAVA File	1 KB

ACCOUNT

```
> yoga oose > yoga oose exp6 > Model

Account.java

File Edit View

import java.util.*;

/**
 *
 */
public class Account {

    /**
     * Default constructor
     */
    public Account() {
    }
```

ATM MACHINE

```
> yoga oose > yoga oose exp6 > Model

ATM machine.java

File Edit View

import java.util.*;

/**
 *
 */
public class ATM machine {

    /**
     * Default constructor
     */
    public ATM machine() {
    }
```

BANK CLIENT



The screenshot shows an IDE window titled 'yoga oose > yoga oose exp6 > Model'. The active file is 'Bank client.java'. The code defines a public class 'Bank client' with a default constructor.

```
import java.util.*;

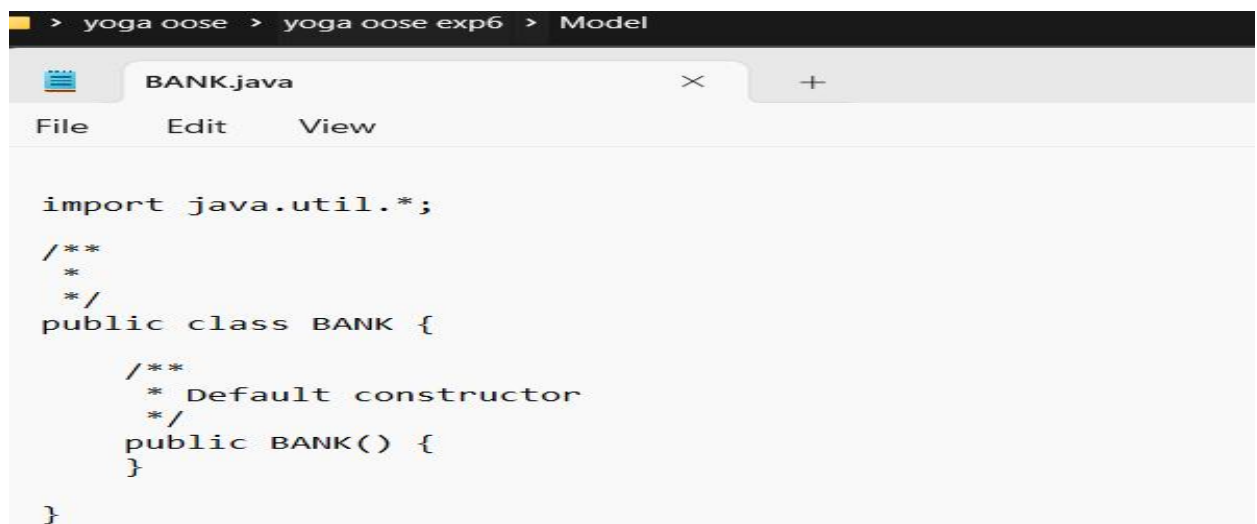
/**
 *
 */
public class Bank client {

    /**
     * Default constructor
     */
    public Bank client() {

    }

}
```

BANK



The screenshot shows an IDE window titled 'yoga oose > yoga oose exp6 > Model'. The active file is 'BANK.java'. The code defines a public class 'BANK' with a default constructor.

```
import java.util.*;

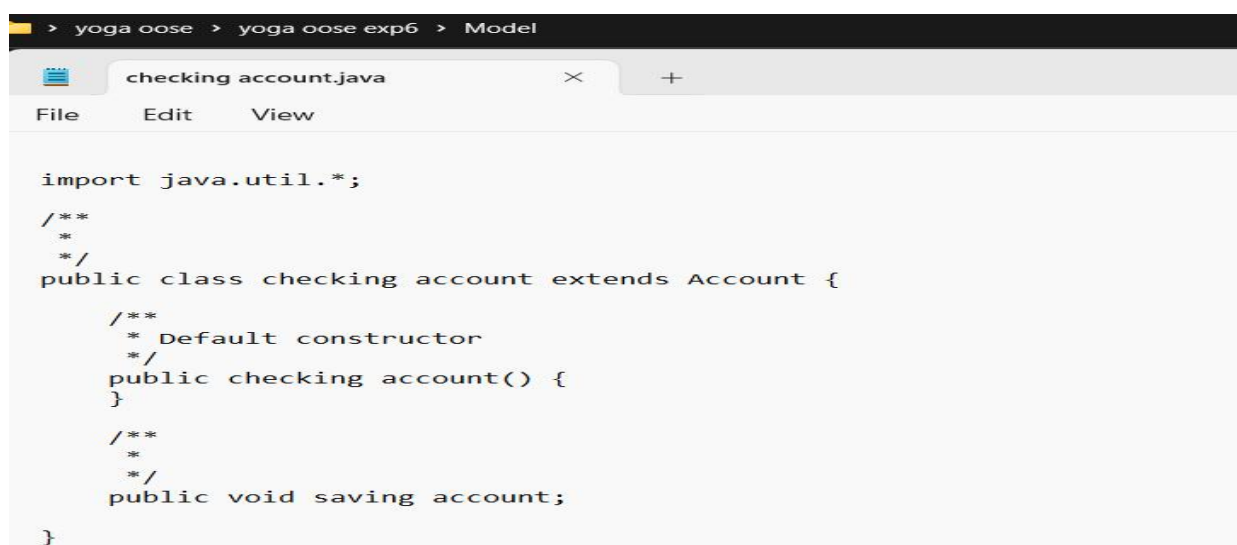
/**
 *
 */
public class BANK {

    /**
     * Default constructor
     */
    public BANK() {

    }

}
```

CHECKING ACCOUNT



The screenshot shows an IDE window titled 'yoga oose > yoga oose exp6 > Model'. The active file is 'checking account.java'. The code defines a public class 'checking account' that extends 'Account'. It has a default constructor and a method 'saving account'.

```
import java.util.*;

/**
 *
 */
public class checking account extends Account {

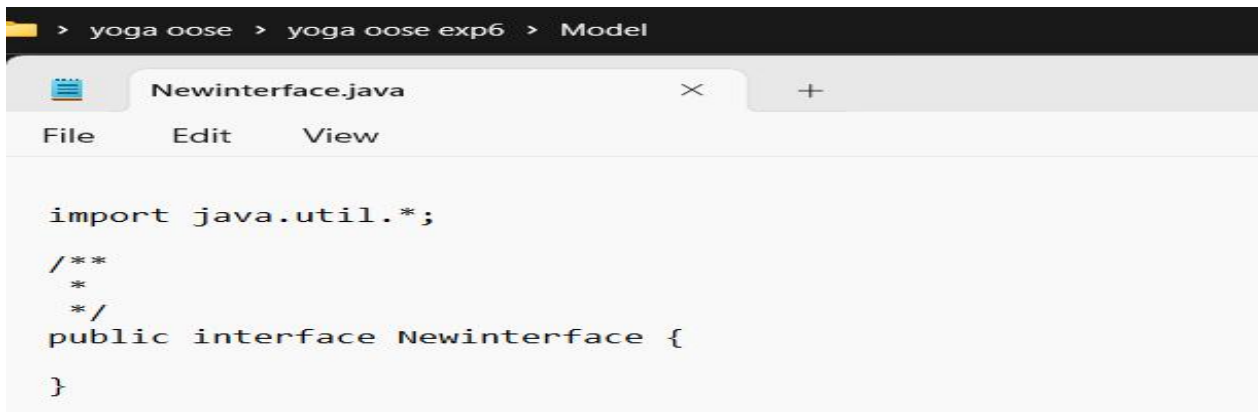
    /**
     * Default constructor
     */
    public checking account() {

    }

    /**
     *
     */
    public void saving account;

}
```

NEW INTERFACE



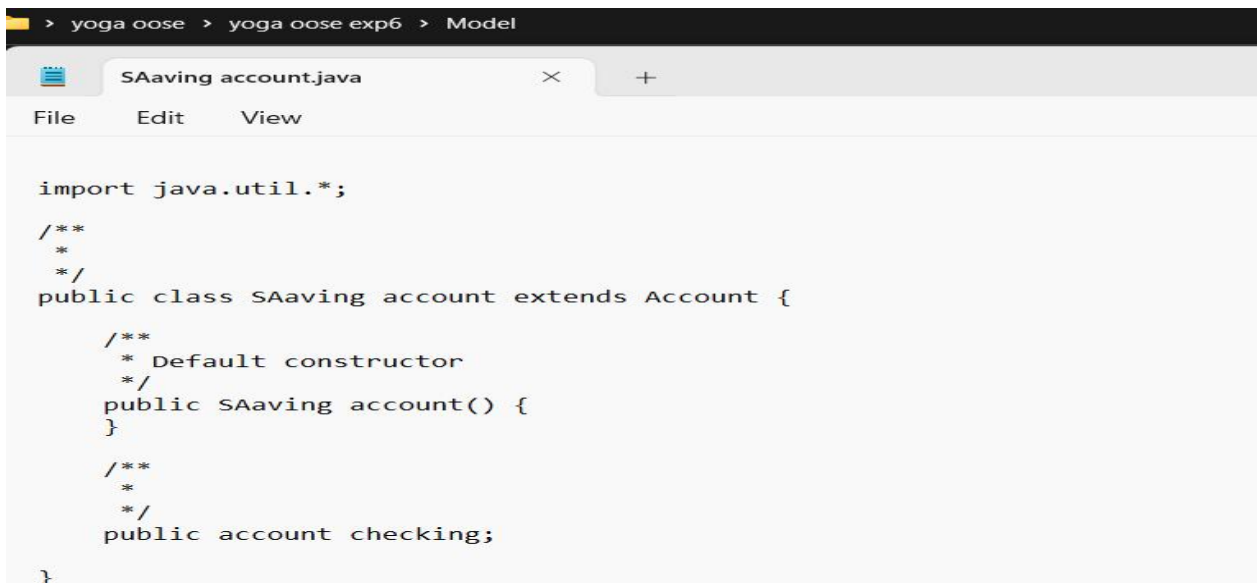
The screenshot shows an IDE window with the file path > yoga oose > yoga oose exp6 > Model. The active file is Newinterface.java. The code defines a public interface named Newinterface that imports java.util.* and contains a single empty method signature.

```
import java.util.*;

/**
 *
 */
public interface Newinterface {

}
```

SAVING ACCOUNT



The screenshot shows an IDE window with the file path > yoga oose > yoga oose exp6 > Model. The active file is SAaving account.java. The code defines a public class named SAaving account that extends Account. It includes a default constructor and a public attribute named account checking.

```
import java.util.*;

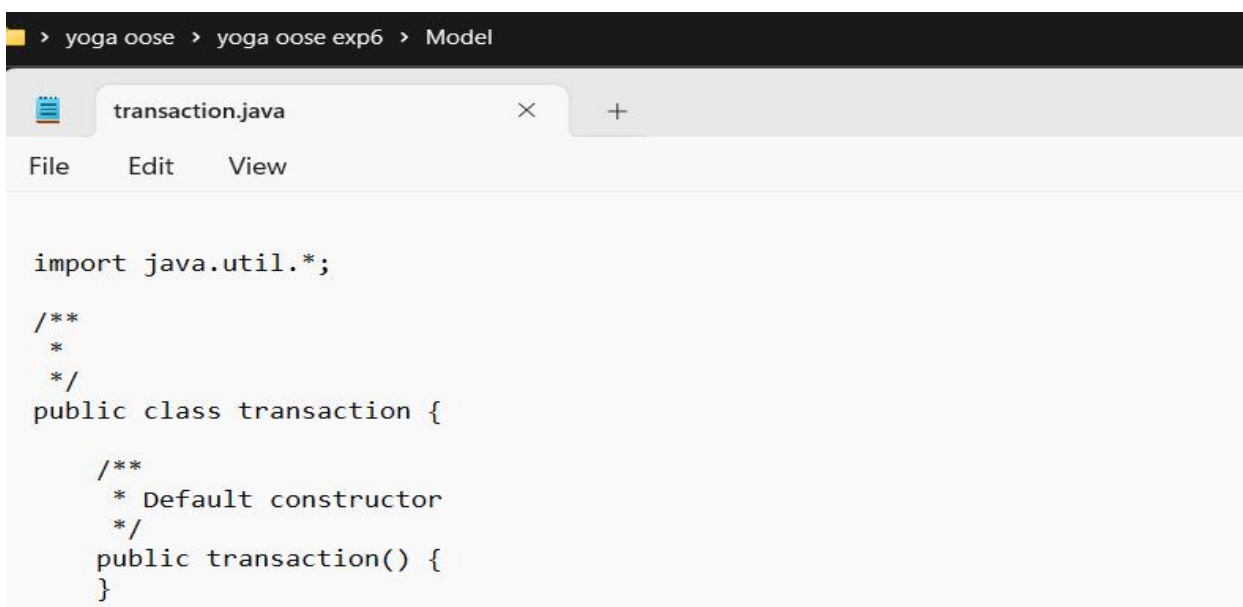
/**
 *
 */
public class SAaving account extends Account {

    /**
     * Default constructor
     */
    public SAaving account() {
    }

    /**
     *
     */
    public account checking;

}
```

TRANSACTION



The screenshot shows an IDE window with the file path > yoga oose > yoga oose exp6 > Model. The active file is transaction.java. The code defines a public class named transaction that imports java.util.* and contains a single empty method signature.

```
import java.util.*;

/**
 *
 */
public class transaction {

    /**
     * Default constructor
     */
    public transaction() {
    }

}
```

Result:

Thus,the UML diagrams for Banking system in StarUML has been designed and java code template was generated successfully.

COURSE REGISTRATION SYSTEM

AIM

To develop the Course Registration System using Star UML.

PROBLEM ANALYSIS AND PROJECT PLANNING

INTRODUCTION

This software is designed in such a way that it receives the name and other particulars from the student. Based on marks the student has scored the list of possible branches that will accommodate for the student will be displayed. Only work for the student is what he has to fill in the form and submit it.

OBJECTIVES

The ultimate objective of this software is to eliminate hassles that the student overcomes while registering himself. This software will reduce the paper work. This also reduces the time delay.

SCOPE

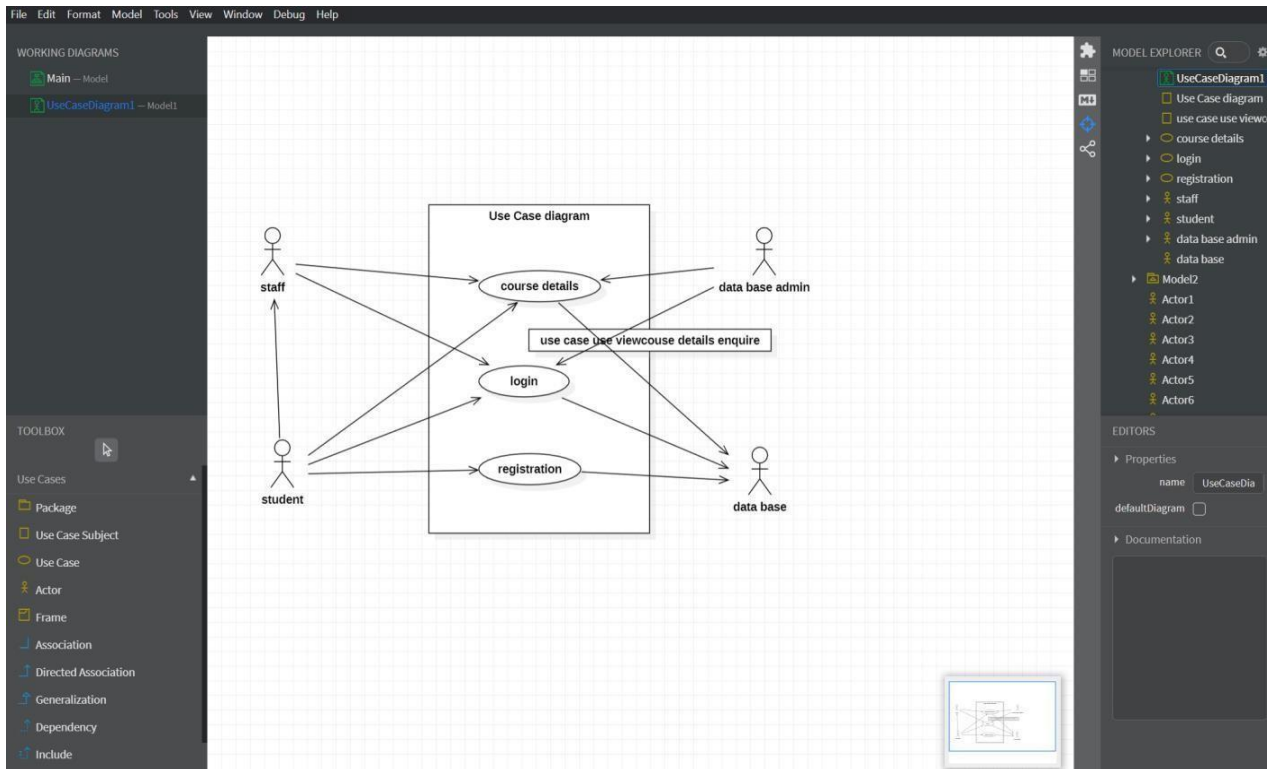
The student is first requested to fill the form. This form will contain important particulars of the student like his name, DOB, preferred branch, his marks. Once the student fills it, a unique id number will be generated. An important thing within this is to decide what mode the student opts to make the payment. It may either be by demand draft or credit card. As soon as a student is registered then the number of seats available is displayed.

PROBLEM STATEMENT

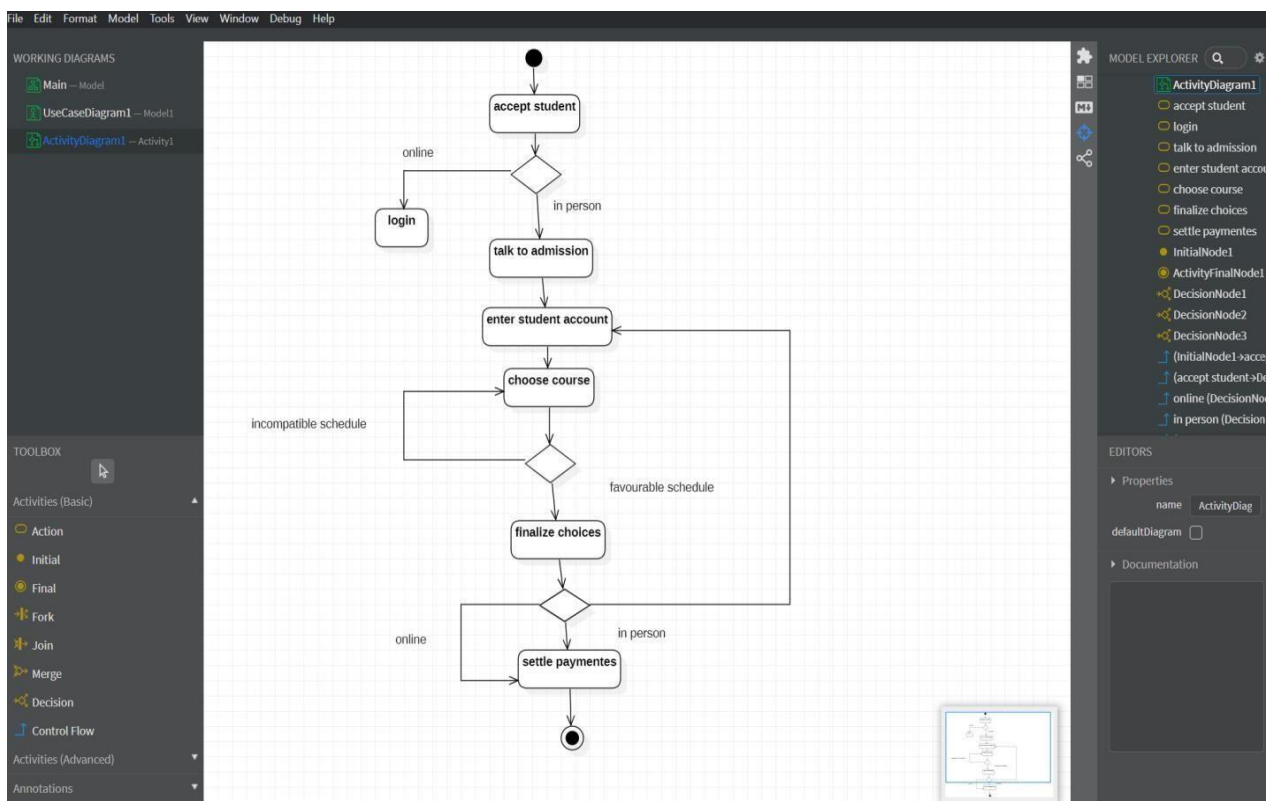
As project developers we developed a new course registration system to replace the existing manual registration since manual system are prone to errors and take more time. The system made is user friendly and reduces the burden of users. Our system can be made available even in the website of a college.

Students can easily register the course in our system without any difficulty and can easily understand and also the time taken for registration is less when compared to manual registration. Options are given to the student to select their electives and also it shows the number of papers available along with the number of students who have registered and also the number of days for particular elective per semester is also displayed at the side. This makes their work easier as compared to manual registration since they need to make a copy of HOD, staff separately and even if one is missed out the whole process has to be redone. Their information will be stored as soon as they are registered.

STAR UML DIAGRAMS: USE CASE DIAGRAM

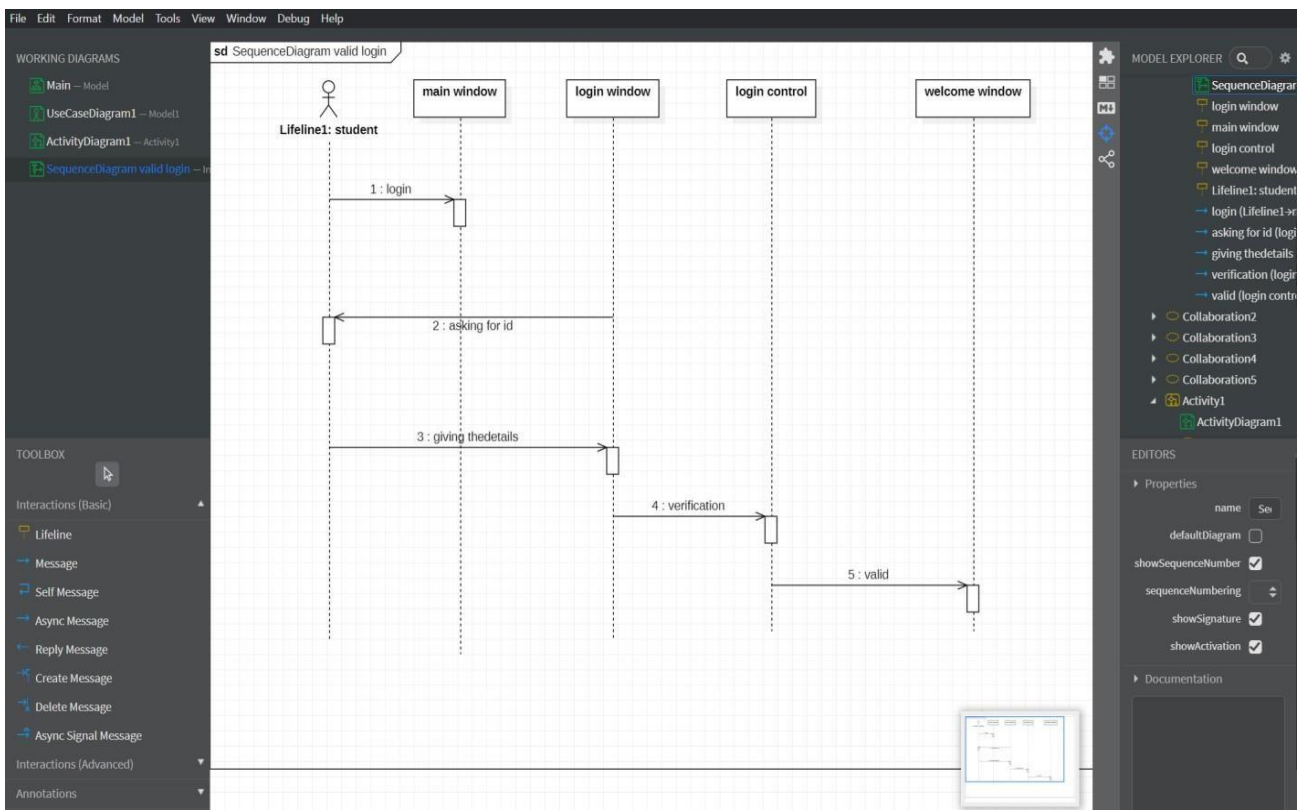


ACTIVITY DIAGRAM

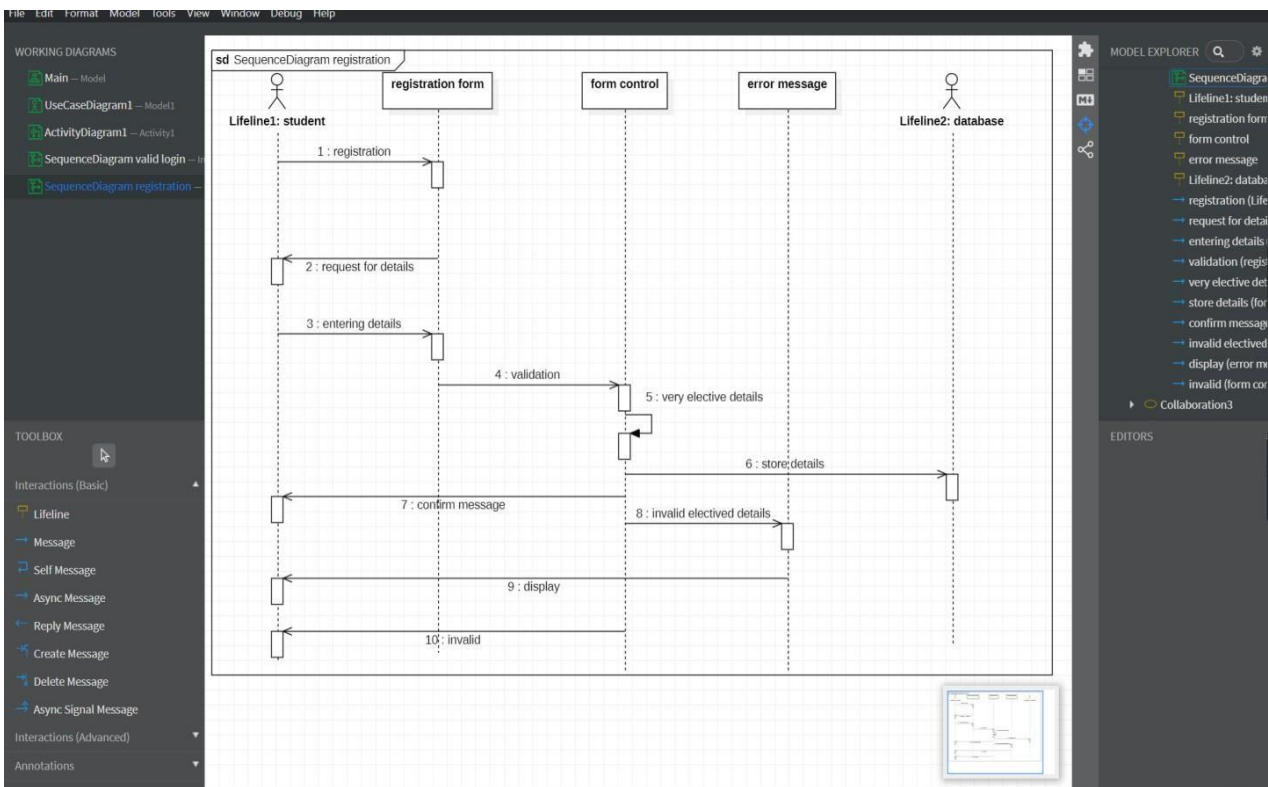


SEQUENCE DIAGRAM

VALID LOGIN

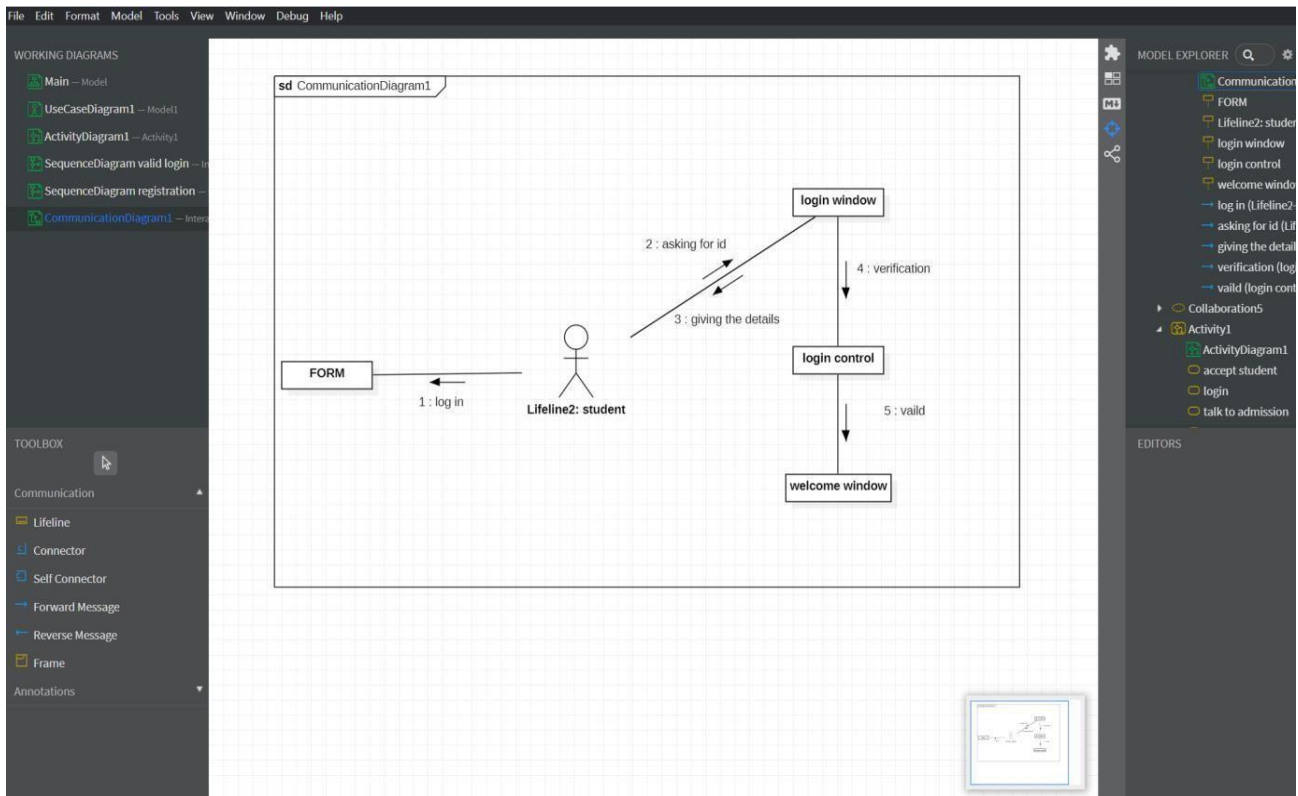


REGISTRATION

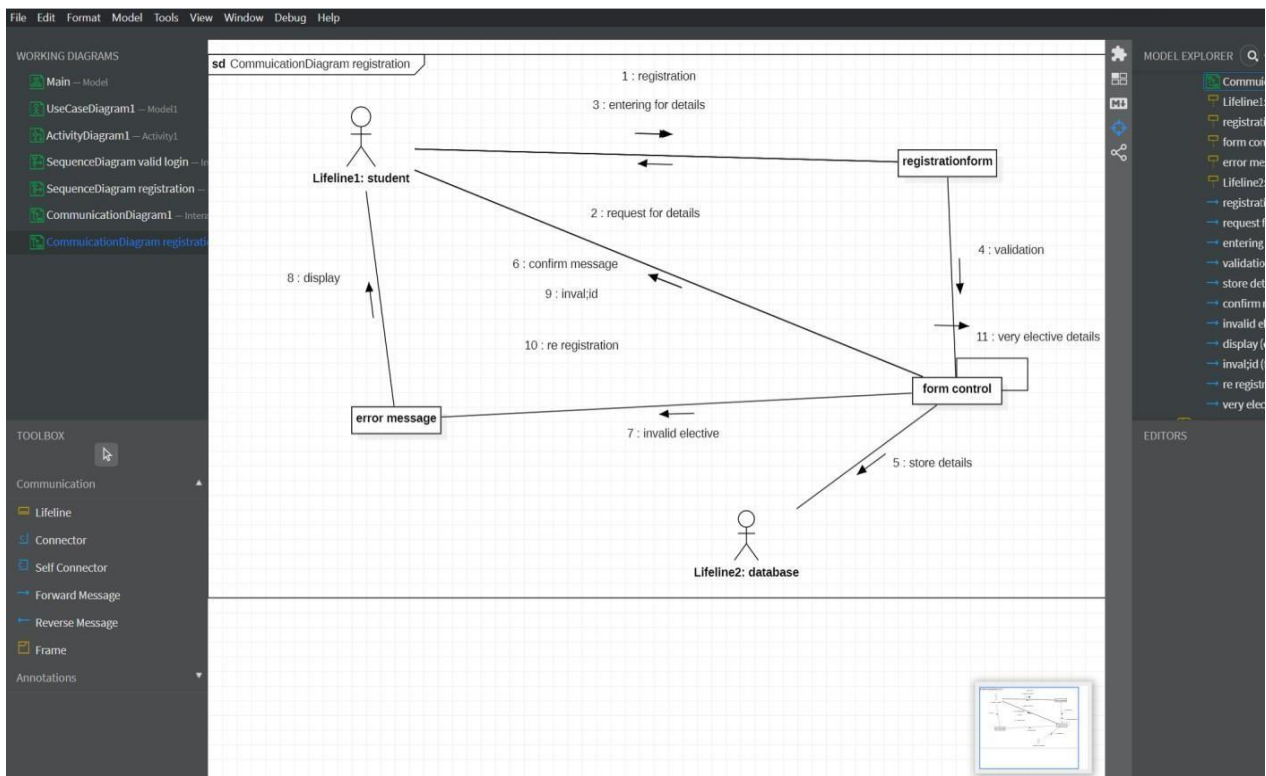


COMMUNICATION DIAGRAM

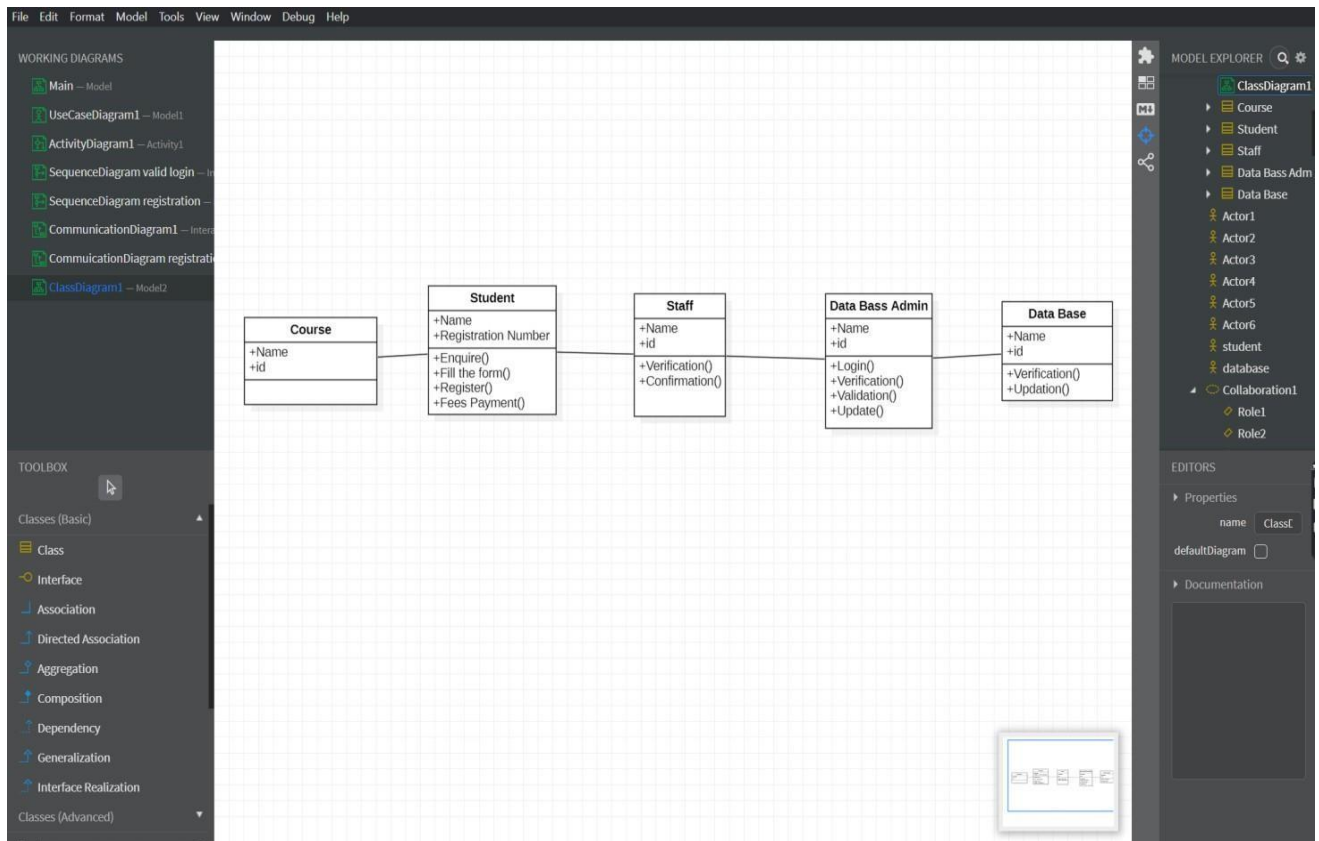
LOGIN



REGISTRATION



CLASS DIAGRAM



STEPS TO GENERATE CODE IN STAR UML

1. Draw the class diagram and then save it.
2. Open the saved file with star uml.
3. Then go to tools at right corner.
4. Select the code language which we can prefer.
5. Then we have the code generator in tools so give access to it.
6. Then go to tools and select the language then it will take us to another window.
7. There select the class attribute to generate the code.
8. Then it will ask for the directory to store the code.
9. Select the directory and click on generate code.
10. Then code generated successfully.

OUTPUT:

> yoga oose > yoga oose exp7 > Model2

Name	Date modified	Type	Size
Course.java	11-03-2023 10:10 AM	JAVA File	1 KB
Data Base.java	11-03-2023 10:10 AM	JAVA File	1 KB
Data Bass Admin.java	11-03-2023 10:10 AM	JAVA File	1 KB
Staff.java	11-03-2023 10:10 AM	JAVA File	1 KB
Student.java	11-03-2023 10:10 AM	JAVA File	1 KB

COURSE

```
> yoga oose > yoga oose exp7 > Model2

Course.java

import java.util.*;

/**
 *
 */
public class Course {

    /**
     * Default constructor
     */
    public Course() {
    }
}
```

DATA BASE

```
> yoga oose > yoga oose exp7 > Model2

Data Base.java

import java.util.*;

/**
 *
 */
public class Data Base {

    /**
     * Default constructor
     */
    public Data Base() {
    }
}
```

DATA BASE ADMIN

```
> yoga oose > yoga oose exp7 > Model2

Data Bass Admin.java

File Edit View

import java.util.*;

/**
 *
 */
public class Data Bass Admin {

    /**
     * Default constructor
     */
    public Data Bass Admin() {
    }
}
```

STAFF

```
> yoga oose > yoga oose exp7 > Model2

Staff.java

File Edit View

import java.util.*;

/**
 *
 */
public class Staff {

    /**
     * Default constructor
     */
    public Staff() {
    }
}
```

STUDENT

```
> yoga oose > yoga oose exp7 > Model2

Student.java

File Edit View

import java.util.*;

/**
 *
 */
public class Student {

    /**
     * Default constructor
     */
    public Student() {
    }

    /**
     *
     */
    public void Name;
}
```

Result:

Thus, the UML diagrams for Course registration system in StarUML has been designed and java code template was generated successfully.

LIBRARY MANAGEMENT SYSTEM

AIM

To develop the Library Management System using Star UML.

INTRODUCTION

To design a library management system using IBM Rational Rose. Implement this system using Visual Basic Language with Oracle/SQL.

OBJECTIVE

Recently the importance of library management system within the service context of libraries has been unchallenged. Students check the list of books available and borrow the books if the book is a borrow book otherwise it is of waste for the student to come to the library to come to check for the books if the student doesn't get the book. Then the librarian checks the student id and allows the member to check out the book and the librarian then updates the member database and also the books database.

SCOPE

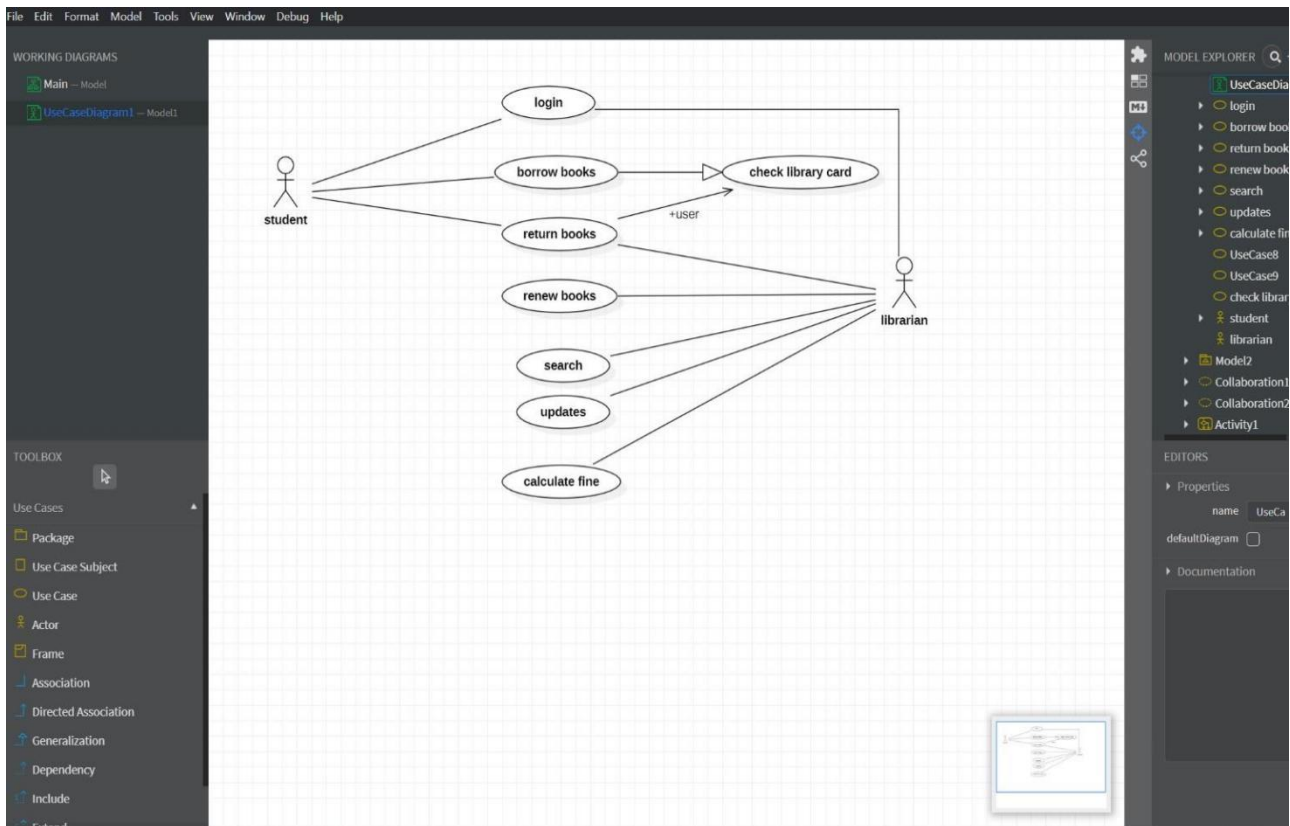
This system would be used by members who may be students or professors of that University to check the availability of the books and borrow the books, and by the librarian to update the databases. The purpose of this document is to analyze and elaborate on the high level needs and features of the Online Library System. It focuses on the capabilities and facilities provided by a Library.

PROBLEM STATEMENT

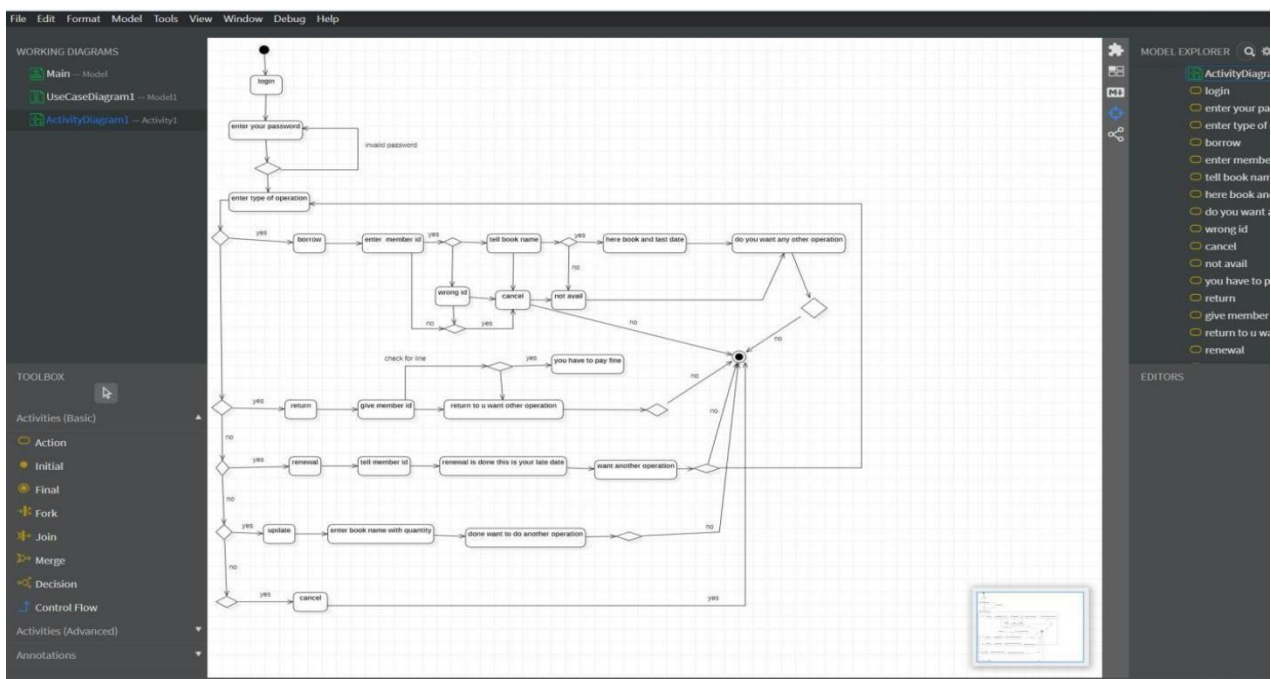
The Library Management System is designed & developed for a receipt and issuance of books in the library along with the student's details. A librarian can add, delete and update book status and search from the database. A user can borrow, return books, reserve books and search for books. He can also renew his loan period. If a book is overdue, the user will be fined Rs.5 each day over the due. If a book is reported lost, the user will pay the full cost of the book. The books received in the library are entered in Books Entry form and the new student is entered in the student entry form. When the student wants to get the desired book the same is issued on the availability basis to the student.

STAR UML DIAGRAMS:

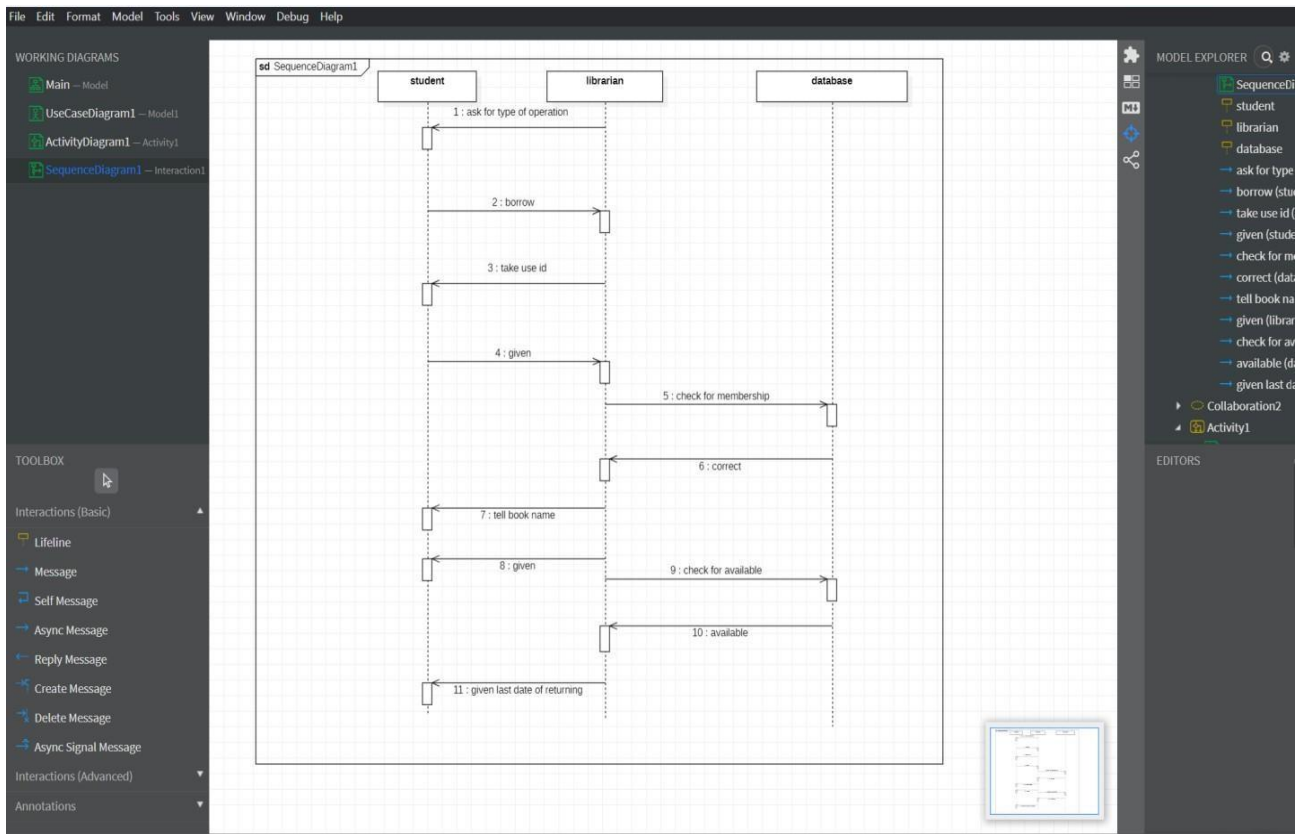
USE CASE DIAGRAM



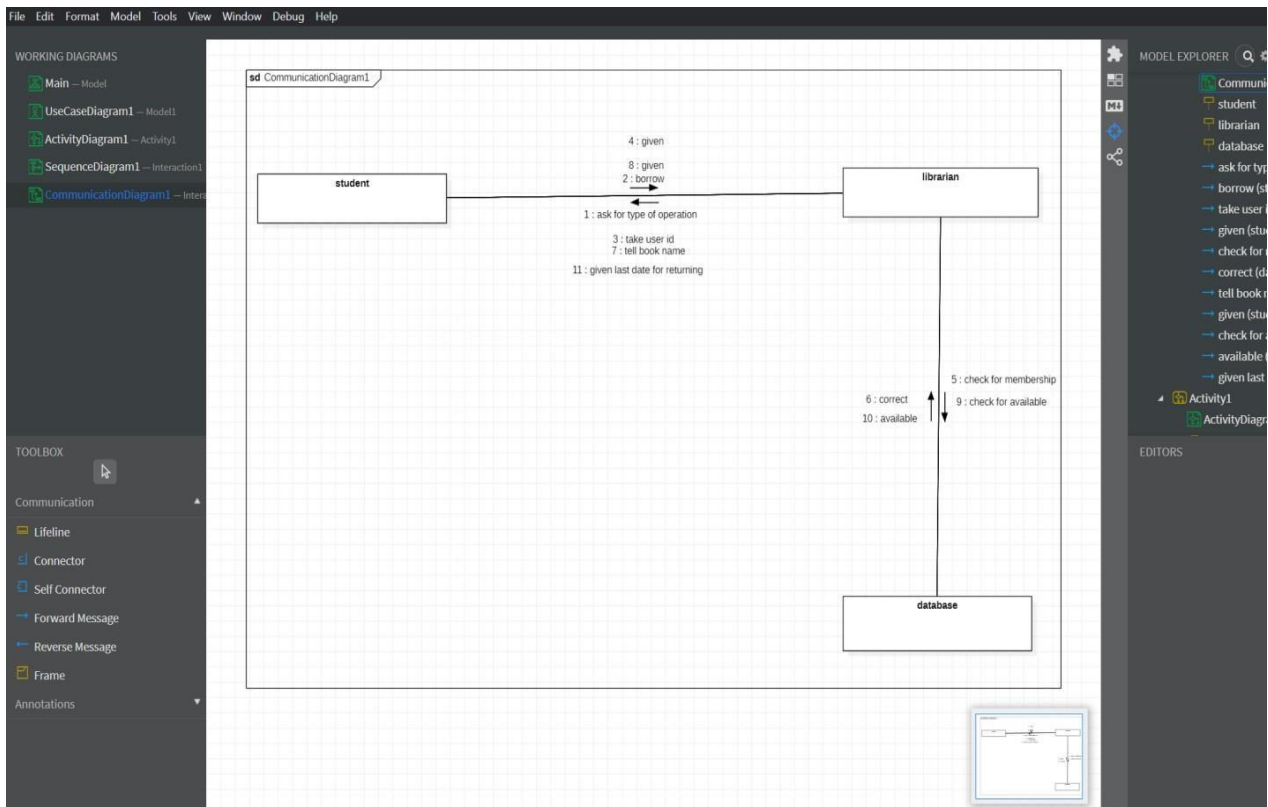
ACTIVITY DIAGRAM



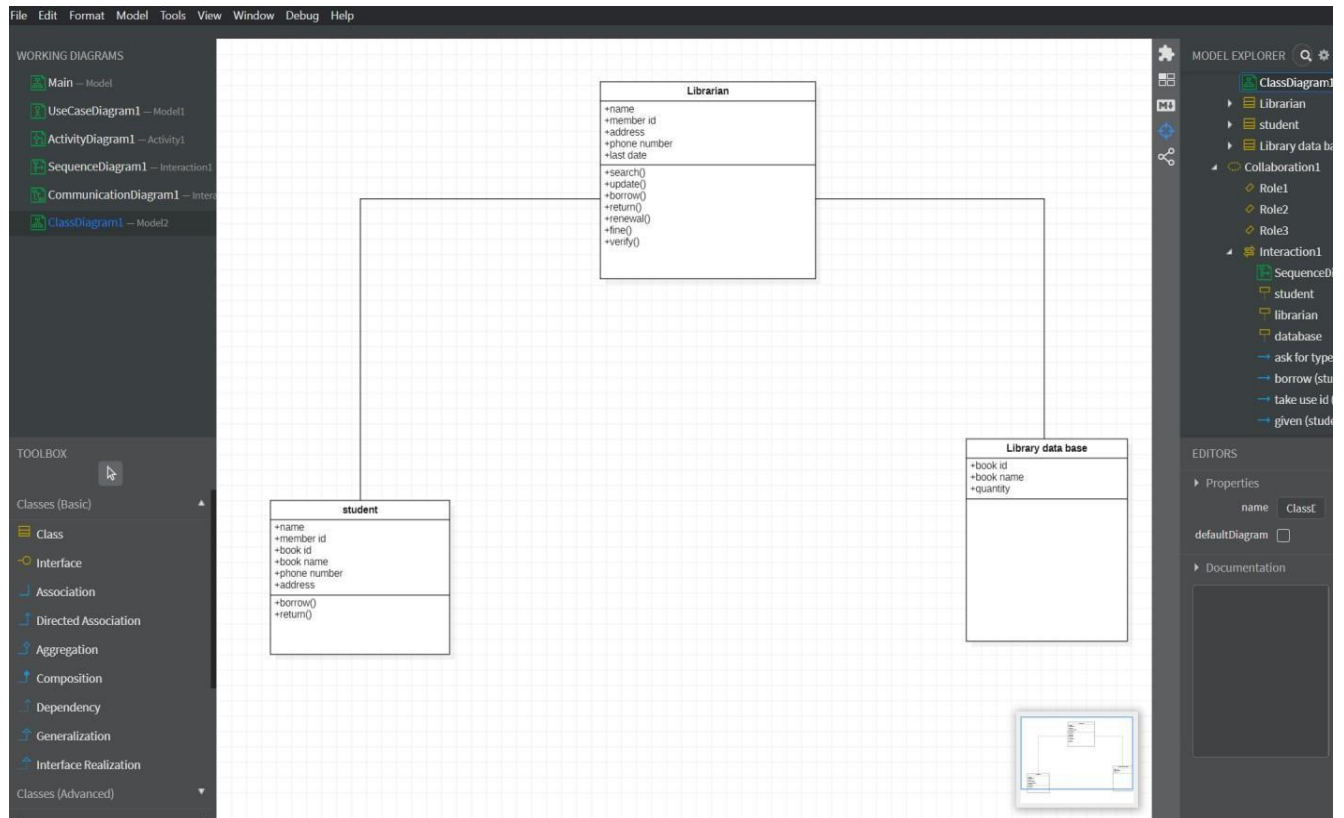
SEQUENCE DIAGRAM



COMMUNICATION DIAGRAM



CLASS DIAGRAM



STEPS TO GENERATE CODE IN STAR UML

1. Draw the class diagram and then save it.
2. Open the saved file with star uml.
3. Then go to tools at right corner.
4. Select the code language which we can prefer.
5. Then we have the code generator in tools so give access to it.
6. Then go to tools and select the language then it will take us to another window.
7. There select the class attribute to generate the code.
8. Then it will ask for the directory to store the code.
9. Select the directory and click on generate code.
10. Then code generated successfully.

OUTPUT:

➤ yoga oose ➤ yoga oose exp8 ➤ Model2

Name	Date modified	Type	Size
 Librarian.java	11-03-2023 10:11 AM	JAVA File	1 KB
 Library data base.java	11-03-2023 10:11 AM	JAVA File	1 KB
 student.java	11-03-2023 10:11 AM	JAVA File	1 KB

LIBRARIAN

```
➤ yoga oose ➤ yoga oose exp8 ➤ Model2

Librarian.java

File Edit View

import java.util.*;

/**
 *
 */
public class Librarian {

    /**
     * Default constructor
     */
    public Librarian() {
    }
}
```

LIBRARY DATA BASE

```
➤ yoga oose ➤ yoga oose exp8 ➤ Model2

Library data base.java

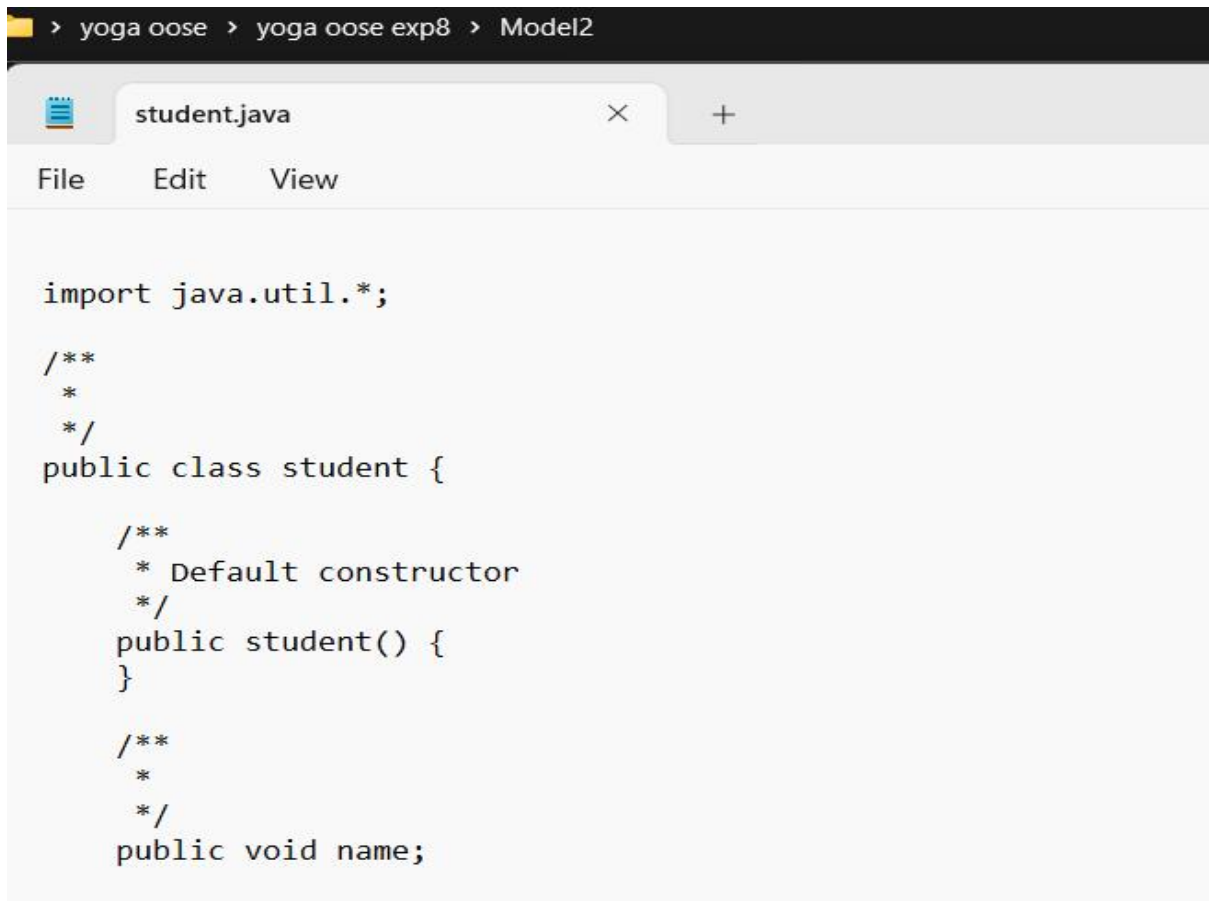
File Edit View

import java.util.*;

/**
 *
 */
public class Library data base {

    /**
     * Default constructor
     */
    public Library data base() {
    }
}
```

STUDENT



The screenshot shows a code editor window with the following elements:

- Path bar: > yoga oose > yoga oose exp8 > Model2
- Tab: student.java
- Menu bar: File, Edit, View
- Code content:

```
import java.util.*;

/**
 *
 */
public class student {

    /**
     * Default constructor
     */
    public student() {
    }

    /**
     *
     */
    public void name;
```

RESULT:

Thus, the UML diagrams for Library Management system in StarUML has been designed and java code template was generated successfully.

PASSPORT AUTOMATION SYSTEM

AIM

To develop the Passport Automation System using StarUML

OBJECTIVE

To develop the passport automation system software using UML language. It is the interface between applicant and authority responsible for issue the passport. It aims at improving efficiency and reducing complexities.

SCOPE

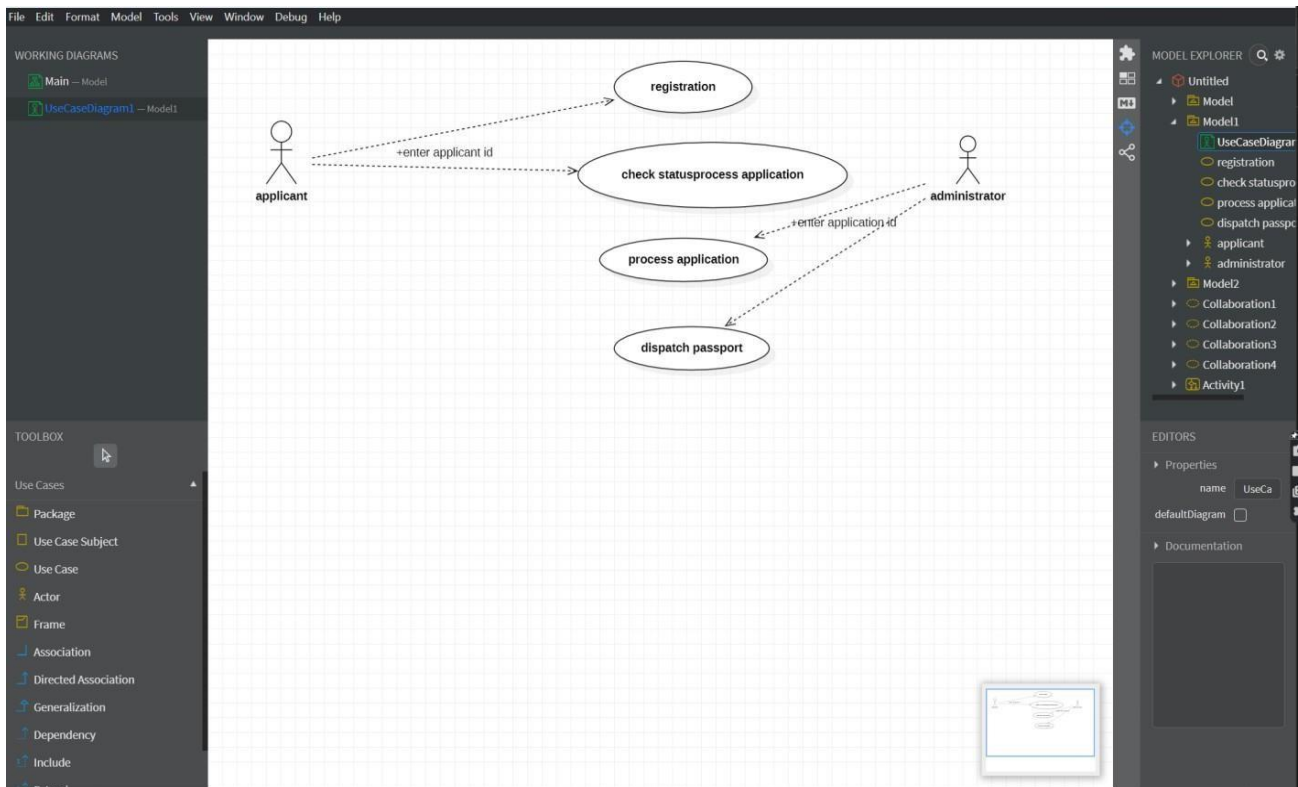
The system provides the online interface to the user where we can fill their form and personal detail with necessary proof. The authority concerned with the issue of passport can use this system to reduce its workload and process it in a speedy manner. It provides a communication platform between administrator and applicant. To transfer the data between passport authority and local police verifying the applicant's information.

PROBLEM STATEMENT

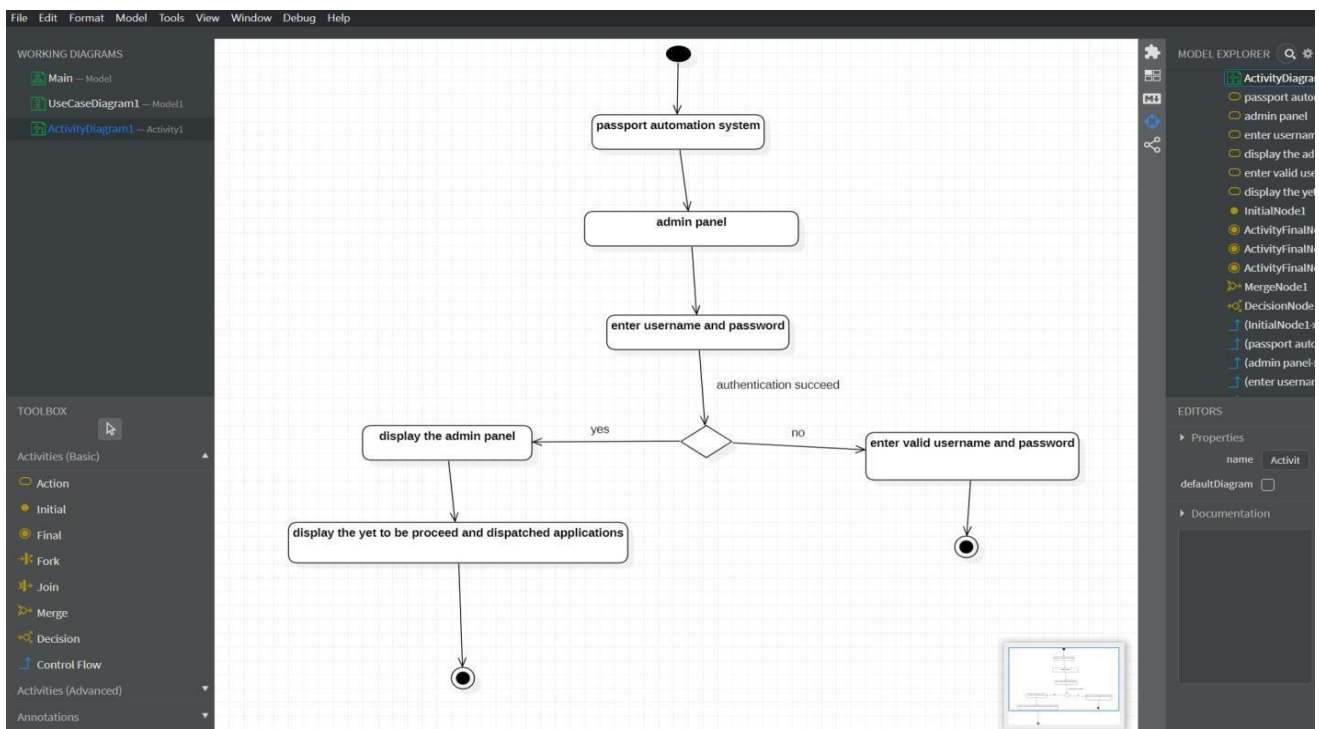
My project title is passport automation system. In this project we can develop the modules such as login, fill the form with necessary proof, verifying the applicant's information, validity checking and issue the passport for that particular applicant. In this login module, we can perform that that perform that enters into the login website for the different actors, and then fill for the can be done by the primary and verification, checking and issuing the passport can be done for the supporting actor.

STAR UML DIAGRAMS :

USE CASE DIAGRAM

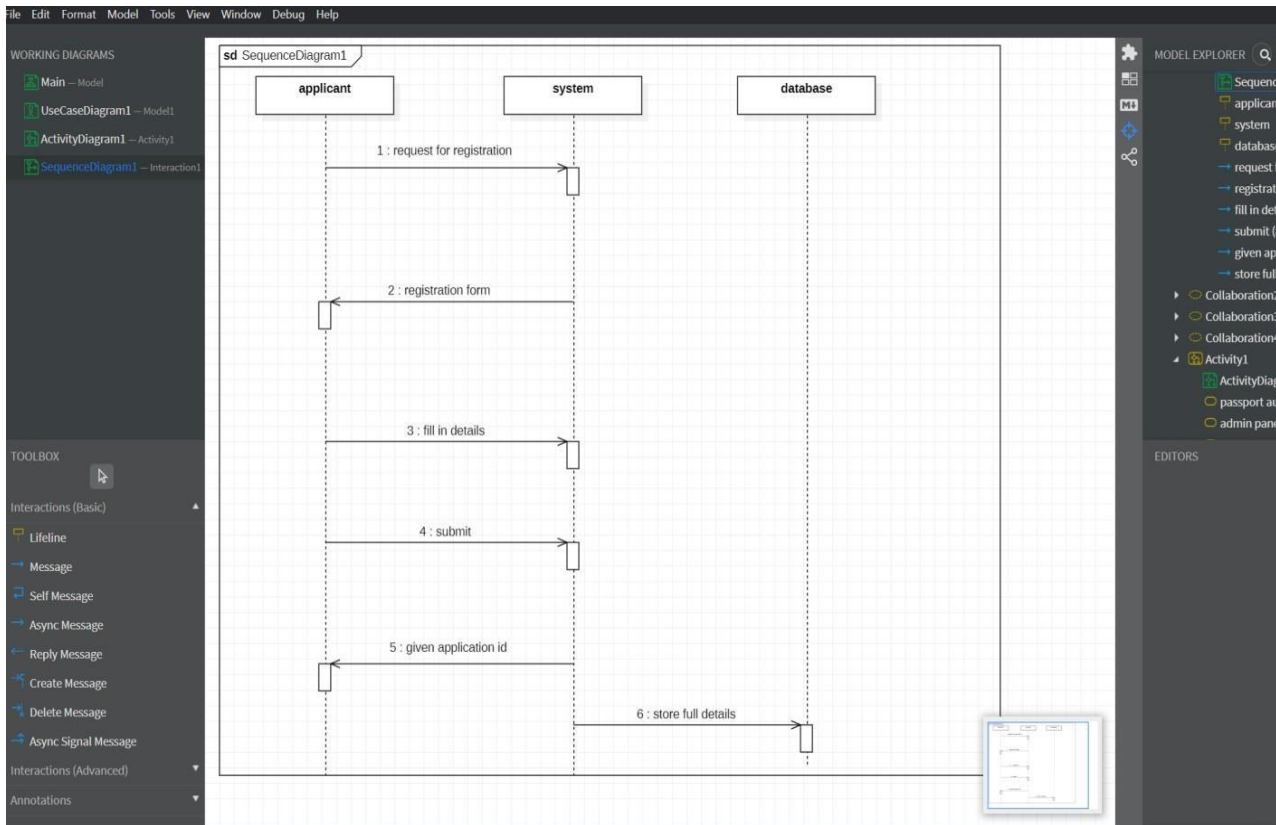


ACTIVITY DIAGRAM

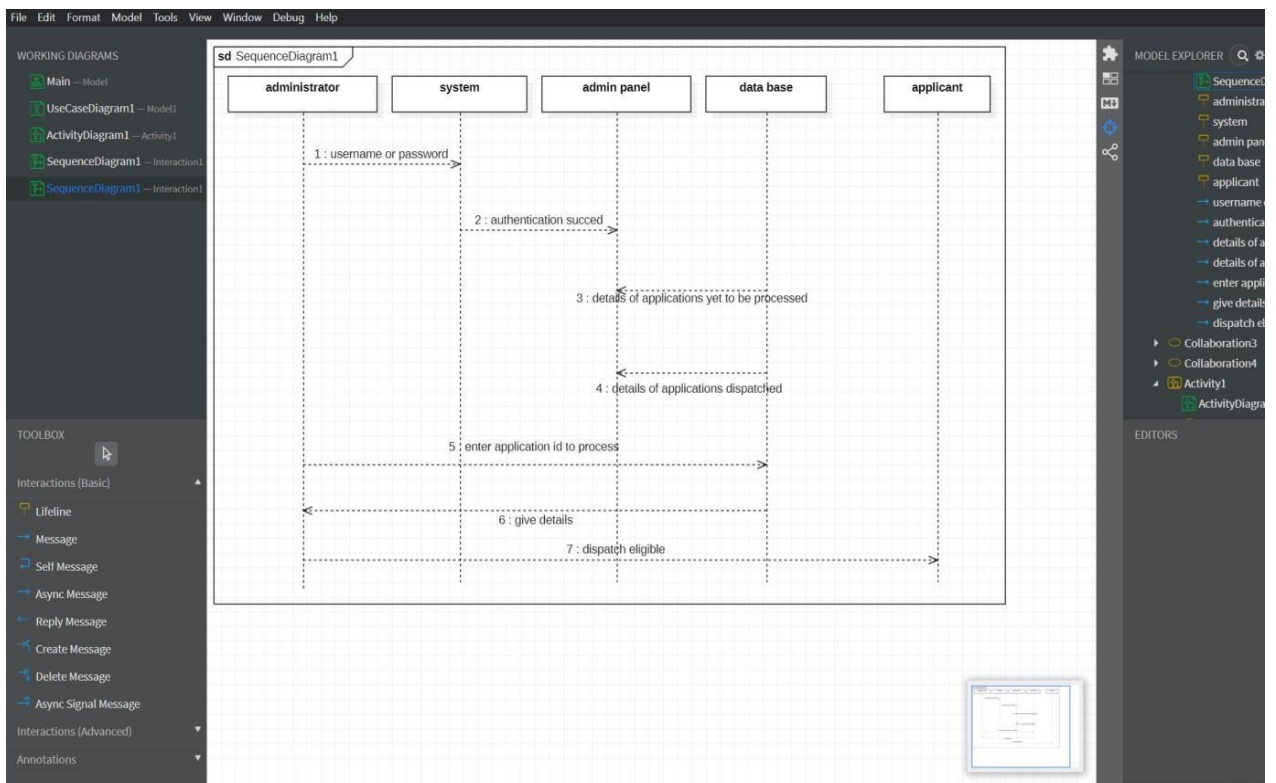


SEQUENCE DIAGRAM

NEW REGISTRATION

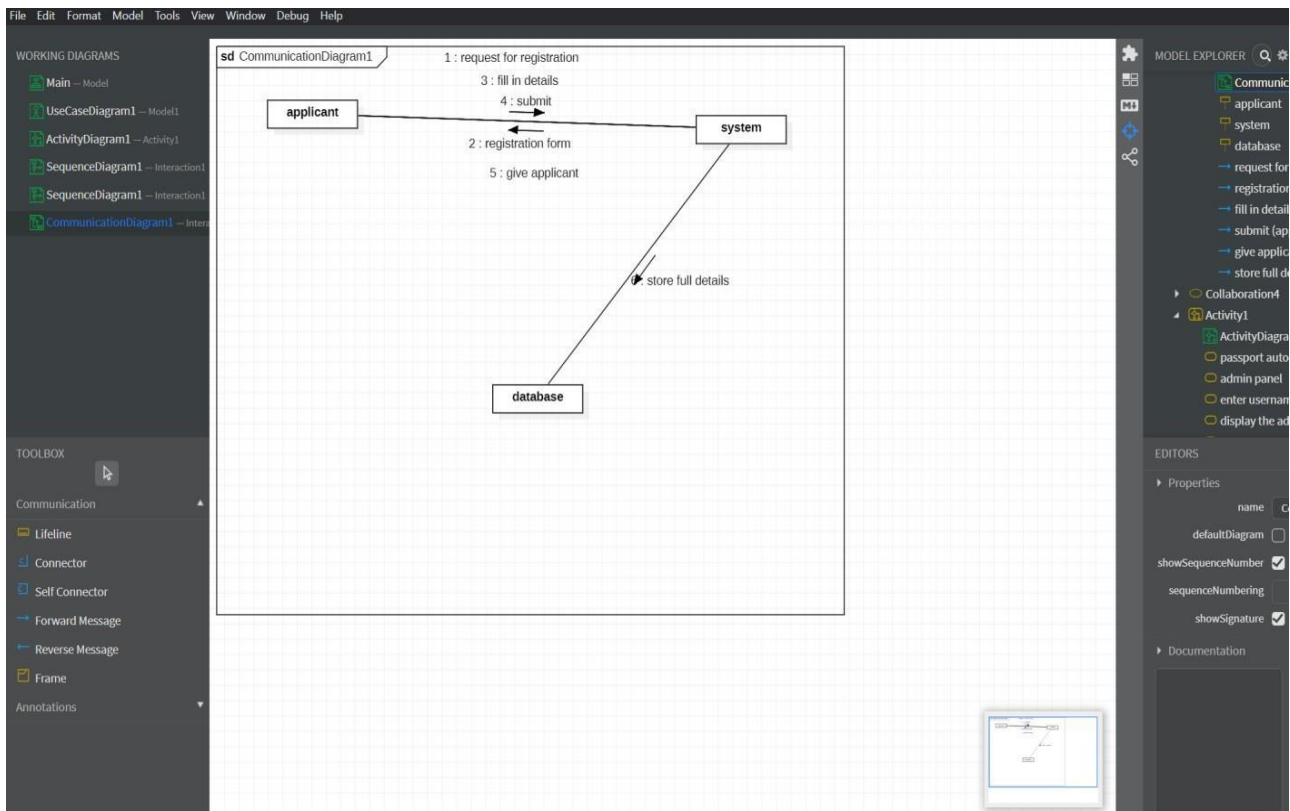


ADMIN PANEL

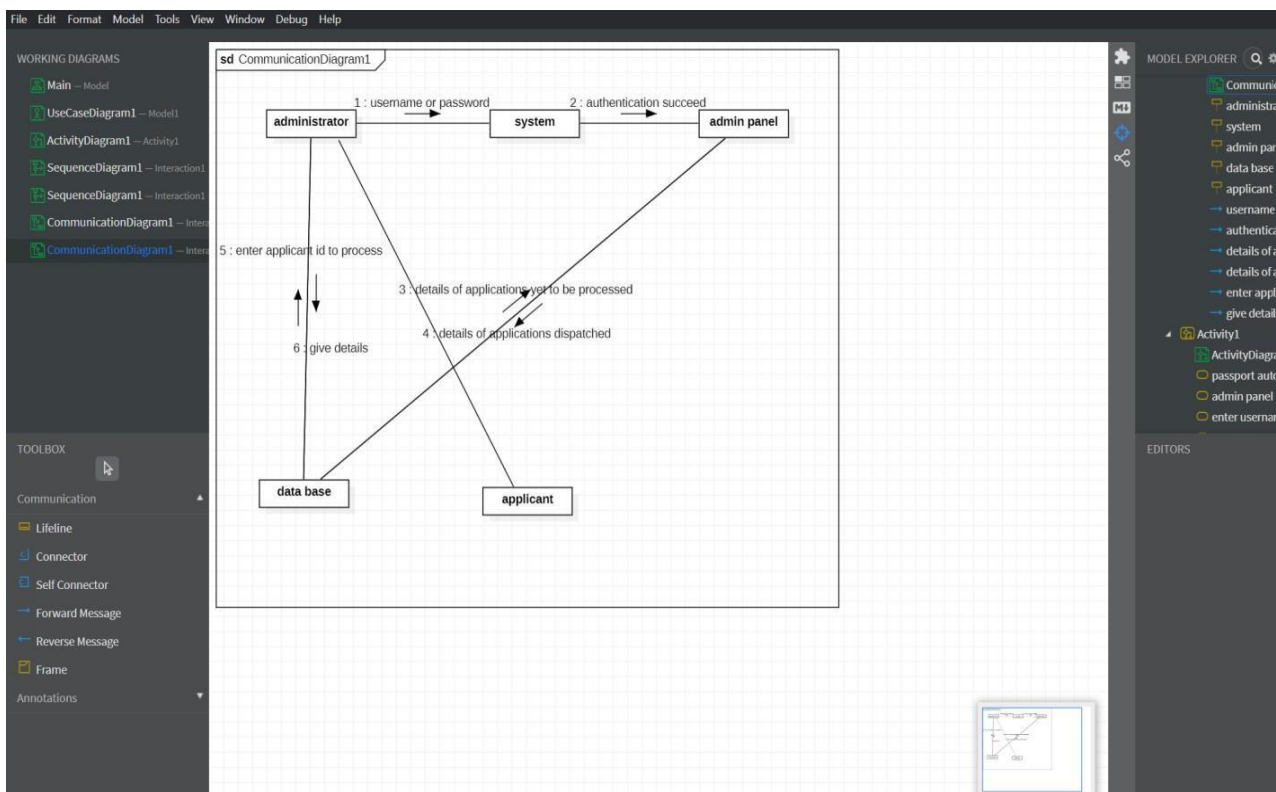


COMMUNICATION DIAGRAM

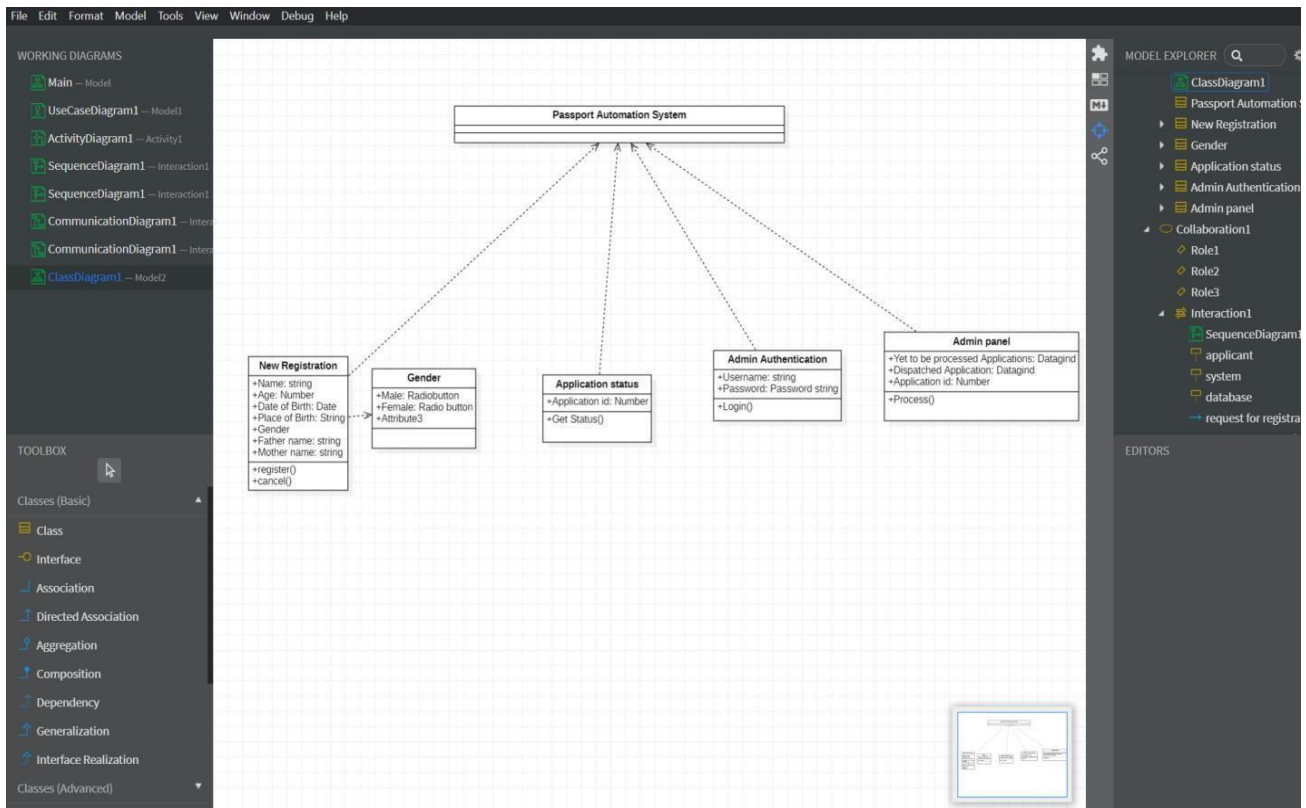
NEW REGISTRATION



ADMIN PANEL



CLASS DIAGRAM



STEPS TO GENERATE CODE IN STAR UML

1. Draw the class diagram and then save it.
2. Open the saved file with staruml.
3. Then go to tools at right corner.
4. Select the code language which we can prefer.
5. Then we have the code generator in tools so give access to it.
6. Then go to tools and select the language then it will take us to another window.
7. There select the class attribute to generate the code.
8. Then it will ask for the directory to store the code.
9. Select the directory and click on generate code.
10. Then code generated successfully.

OUTPUT:

> yoga oose > yoga oose exp9 > Model2

Name	Date modified	Type	Size
 Admin Authentication.java	11-03-2023 10:11 AM	JAVA File	1 KB
 Admin panel.java	11-03-2023 10:11 AM	JAVA File	1 KB
 Application status.java	11-03-2023 10:11 AM	JAVA File	1 KB
 Gender.java	11-03-2023 10:11 AM	JAVA File	1 KB
 New Registration.java	11-03-2023 10:11 AM	JAVA File	1 KB
 Passport Automation System.java	11-03-2023 10:11 AM	JAVA File	1 KB

ADMIN AUTHENTICATION

```
> yoga oose > yoga oose exp9 > Model2

Admin Authentication.java

File Edit View

import java.util.*;

/**
 *
 */
public class Admin Authentication {

    /**
     * Default constructor
     */
    public Admin Authentication() {
    }
```

ADMIN PANEL

```
> yoga oose > yoga oose exp9 > Model2

Admin panel.java

File Edit View

import java.util.*;

/**
 *
 */
public class Admin panel {

    /**
     * Default constructor
     */
    public Admin panel() {
    }
```

APPLICATION STATUS

```
> yoga oose > yoga oose exp9 > Model2

Application status.java

File Edit View

import java.util.*;

/**
 *
 */
public class Application status {

    /**
     * Default constructor
     */
    public Application status() {
    }
}
```

GENDER

```
> yoga oose > yoga oose exp9 > Model2

Gender.java

File Edit View

import java.util.*;

/**
 *
 */
public class Gender {

    /**
     * Default constructor
     */
    public Gender() {
    }
}
```

NEW REGISTRATION

```
> yoga oose > yoga oose exp9 > Model2

New Registration.java

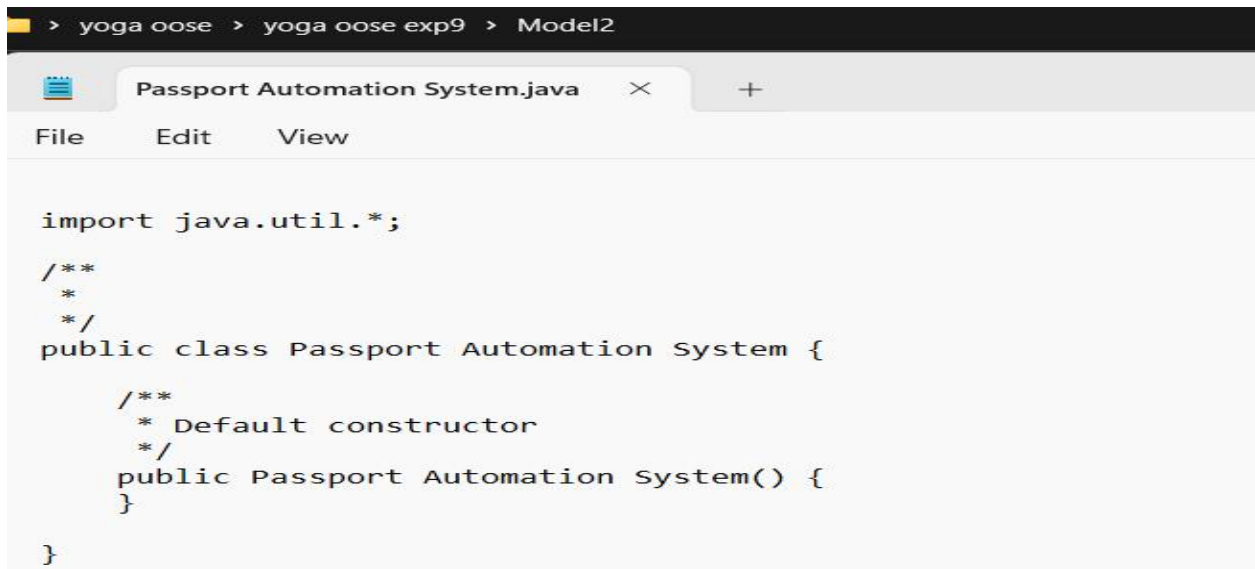
File Edit View

import java.util.*;

/**
 *
 */
public class New Registration {

    /**
     * Default constructor
     */
    public New Registration() {
    }
}
```

PASSPORT AUTOMATION SYSTEM



The screenshot shows a Java IDE window titled "Passport Automation System.java". The breadcrumb navigation at the top indicates the file path: > yoga oose > yoga oose exp9 > Model2. The menu bar includes "File", "Edit", and "View". The code editor contains the following Java code:

```
import java.util.*;

/**
 *
 */
public class Passport Automation System {

    /**
     * Default constructor
     */
    public Passport Automation System() {
    }

}
```

Result:

Thus,the UML diagrams for Passport Automation system in StarUML has been designed and java code template was generated successfully.

